Graph Neural Networks and Graph Foundation Models for Scientific Discovery: Architectures, Paradigms, and Frontiers

1 Introduction and the Paradigm Shift to Graph Foundation Models

1.1 From Task-Specific GNNs to Universal Graph Foundation Models

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The trajectory of graph machine learning has witnessed a profound metamorphosis over the last decade, evolving from the development of specialized, single-task architectures toward the emergence of versatile, universal Graph Foundation Models (GFMs). In the nascent stages of this field, Graph Neural Networks (GNNs) were primarily conceptualized as powerful tools for learning on non-Euclidean data, adept at capturing complex dependencies and interrelationships that traditional deep learning models in Euclidean spaces could not address [1]. These early iterations were predominantly designed with a narrow focus, optimized end-to-end for specific predictive tasks such as node classification, link prediction, or graph classification within isolated datasets. While these task-specific GNNs achieved state-of-the-art performance in their respective domains, they inherently suffered from a lack of generalizability, often failing to transfer learned representations to unseen graphs or disparate domains without extensive retraining [2].

This traditional paradigm relied heavily on the assumption that training and testing distributions were identical, a constraint that severely limited the applicability of GNNs in real-world scientific discovery where data scarcity and distribution shifts are commonplace. Researchers observed that embeddings learned through existing GNNs were highly task-dependent, rendering them ineffective for sharing across different datasets or adapting to new challenges [3]. This specialization created a fragmented landscape where a model trained on molecular graphs could not easily be adapted to social network analysis or materials science, necessitating the design of unique architectures for every new problem. Furthermore, the efficacy of these models was tightly coupled with the availability of abundant, high-quality labeled data, a luxury rarely afforded in high-stakes scientific domains like drug discovery or protein folding [1].

Recognizing these limitations, the research community initially explored strategies to enhance generalization through self-supervised learning (SSL) to leverage vast amounts of unlabeled graph data. However, conventional SSL frameworks often relied on a single pretext task philosophy, such as mutual information maximization or generative reconstruction, which proved insufficient for capturing the diverse knowledge required for robust multi-task generalization [4]. The realization that a single learning objective could not span the extensive feature space of diverse graphs prompted a shift toward multi-task and meta-learning approaches. These methods aimed to produce node embeddings capable of performing multiple tasks with performance comparable to specialized single-task models, effectively learning to adapt quickly to new tasks with only a few steps of gradient descent [5]. Despite these advancements, the field still lacked a unified framework that could truly operate as a “foundation” model, capable of zero-shot adaptation across heterogeneous domains.

The paradigm shift toward Graph Foundation Models represents a fundamental reimagining of graph learning, inspired by the success of foundation models in natural language processing and computer vision. This new vision envisions models that are pre-trained on extensive, diverse graph corpora and can be adapted to a wide array of downstream tasks with minimal fine-tuning [6]. Unlike their task-specific predecessors, GFMs are designed to transcend the boundaries of individual datasets, learning invariant structural patterns and universal representations that hold across different domains. A critical enabler of this transition is the unification of graph data representations. Recent frameworks have proposed leveraging Text-Attributed Graphs (TAGs) to describe nodes and edges using natural language, thereby allowing Large Language Models (LLMs) to encode diverse attributes into a shared embedding space [7]. This approach directly addresses the historical challenge of distinct attribute spaces and label distributions that previously hindered cross-domain transfer, setting the stage for a more rigorous definition of what constitutes a foundational architecture.

The emergence of GFMs also introduces novel mechanisms for adaptation, moving beyond rigid fine-tuning to flexible prompting strategies. Innovations such as graph instruction tuning enable models to understand complex topological features through natural language prompts, facilitating zero-shot prediction abilities on unseen graphs [8]. By aligning graph structures with the semantic reasoning capabilities of LLMs, these models can bridge the gap between structural data and task instructions, a capability absent in traditional GNNs [9]. Furthermore, the development of unified graph tokenizers and scalable graph transformers has allowed these foundation models to capture node-wise dependencies within a global topological context, ensuring robustness even when underlying graph properties differ significantly from training instances [10].

This evolution from specialized tools to universal foundations marks a critical juncture in scientific AI. Where once researchers were confined to building custom models for every specific scientific query, the GFM paradigm offers a scalable, adaptable infrastructure capable of accelerating discovery across physics, biology, and chemistry. By synthesizing knowledge from diverse graph domains and leveraging the power of self-supervised pre-training and instruction tuning, Graph Foundation Models are poised to overcome the data scarcity and generalization bottlenecks that have long plagued task-specific approaches, ushering in an era of more autonomous and versatile scientific exploration [6]. The transition is not merely an incremental improvement but a structural reorientation of how machine learning interacts with relational data. However, realizing this vision requires addressing the inherent heterogeneity of graph data through a unifying theoretical framework, leading to the concept of a “graph vocabulary” that defines the core characteristics of these foundational models.

1.2 Defining Core Characteristics and the “Graph Vocabulary” Concept

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Building on the paradigm shift from task-specific GNNs to universal Graph Foundation Models (GFMs) outlined in the previous section, realizing this vision requires a rigorous redefinition of what constitutes a foundational architecture in the graph domain. While GFMs are characterized by their capacity for universal generalization, scalability, and zero-shot adaptability, their development has historically been impeded by the inherent heterogeneity of graph data. Unlike text or images, graphs vary drastically in size, density, feature spaces, and structural patterns, making direct knowledge transfer exceptionally challenging. To overcome these barriers and enable the cross-domain versatility promised by GFMs, the research community has coalesced around a unifying theoretical framework: the concept of a “graph vocabulary.” This concept posits that beneath the surface diversity of graphical data lie invariant, transferable units that serve as the atomic building blocks for a universal graph representation, analogous to tokens in language or patches in vision.

The core proposition of the graph vocabulary is to facilitate positive knowledge transfer across graphs with disparate structural patterns. As highlighted in recent literature, the primary obstacle in developing GFMs is not merely the volume of data but the mechanism by which a model recognizes and utilizes commonalities between structurally dissimilar graphs [11]. Without such a mechanism, models risk negative transfer, where pre-training on one domain degrades performance on another due to conflicting structural biases. The graph vocabulary addresses this by decomposing complex graphs into fundamental substructures or semantic units that encode invariance. These units act as a shared lexicon, allowing a model trained on molecular structures, for instance, to recognize analogous relational patterns in social networks or knowledge graphs, provided those patterns map to the same elements within the vocabulary.

This theoretical construction rests on three essential pillars: network analysis, theoretical foundations, and stability. From the perspective of network analysis, the vocabulary consists of recurring motifs or subgraphs that capture local topological signatures. Theoretical foundations ensure that these units possess the necessary mathematical properties to support inductive reasoning and establish generalization bounds. Stability refers to the robustness of these vocabulary units against perturbations in the graph structure, ensuring that learned representations remain consistent even when the input graph undergoes minor modifications. By grounding the vocabulary in these pillars, researchers aim to create GFMs that adhere to neural scaling laws, where performance predictably improves with increased model size and data diversity [11].

The definition of these transferable units extends beyond mere topological motifs to include semantic and relational abstractions. In the context of Knowledge Graphs (KGs), for example, the challenge of non-overlapping entity and relation vocabularies has driven the development of models capable of learning universal relational representations. Approaches such as ULTRA demonstrate that by building relational representations as functions conditioned on their interactions, it is possible to achieve inductive generalization to unseen graphs with arbitrary relation vocabularies [12]. This aligns with the broader graph vocabulary concept, suggesting that the "words" of this vocabulary are not static identifiers but dynamic, context-aware functions that describe how entities interact regardless of their specific identity. Similarly, the integration of Large Language Models (LLMs) has introduced new paradigms for defining this vocabulary through natural language. Frameworks like MuseGraph and UniGraph leverage the rich semantic space of LLMs to create a unified description of graphs, effectively translating structural information into a textual vocabulary that LLMs can process and reason over [13; 8].

Furthermore, the concept of semantic units offers a granular approach to organizing knowledge graphs into identifiable, meaningful subgraphs that enhance cognitive interoperability and facilitate FAIR (Findable, Accessible, Interoperable, and Reusable) data principles [14; 15]. These semantic units can be viewed as higher-order elements of the graph vocabulary, encapsulating complex propositions rather than simple edges or nodes. By structuring graphs into these units, GFMs can operate at varying levels of granularity, distinguishing between assertional statements and compound collections of knowledge, thereby supporting more sophisticated reasoning capabilities. This organization also aids in mitigating "epistemic monoculture" by allowing models to learn from diverse, semantically rich representations rather than structural encodings.

The necessity of such a vocabulary is further underscored by the limitations of traditional transfer learning methods, which often assume stationary distributions or require significant fine-tuning. In contrast, a GFM equipped with a robust graph vocabulary can perform cross-dataset zero-shot transfer by mapping unseen graph structures onto its learned set of invariant units [16]. This capability is critical for scientific discovery, where data is often scarce, and the ability to generalize from related domains is paramount. The graph vocabulary thus serves as the bridge between the shallow, task-specific methods of the past and the deep, adaptable architectures of the future, providing the linguistic structure necessary for graphs to speak a universal language of relations and structures. As the field advances, the refinement of this vocabulary—through better tokenization strategies, more expressive subgraph encoders, and tighter integration with symbolic reasoning—will likely determine the ultimate success of Graph Foundation Models in achieving human-level generalization across the sciences [6]. Crucially, while this vocabulary provides the structural foundation for universality, its application in scientific domains demands that these abstract units be grounded in the specific physical laws and symmetries that govern natural phenomena, a transition explored in the following section.

1.3 Motivations for Scientific Discovery in Physics-Governed Domains

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Building upon the theoretical framework of a "graph vocabulary" established in the previous section, the transition from task-specific Graph Neural Networks (GNNs) to Graph Foundation Models (GFMs) in scientific discovery is not merely a trend driven by successes in natural language processing and computer vision; it is a necessity dictated by the unique structural and epistemological constraints of physics-governed domains. While the graph vocabulary provides the linguistic structure for universal representation, scientific systems—ranging from quantum mechanical interactions in materials to large-scale climate dynamics—demand that this vocabulary be grounded in universal physical laws. These systems are characterized by severe data scarcity in high-fidelity regimes and the critical need for robust generalization across varying spatial and temporal scales. Consequently, the motivation for developing GFMs in these fields centers on three critical drivers: the imperative to respect physical symmetries and conservation laws within the vocabulary itself, the capacity to overcome data limitations through few-shot and zero-shot learning paradigms, and the ability to generalize across multiscale systems governed by fundamental differential equations.

First and foremost, scientific domains demand models that inherently respect the symmetries and invariances of the physical universe, requiring a graph vocabulary that encodes these principles as atomic units. Unlike social or recommendation graphs where structural patterns may be stochastic or context-dependent, physical systems are bound by strict mathematical principles such as rotation, translation, and permutation equivariance. Standard black-box machine learning approaches often fail to generalize outside their training distribution because they do not encode these inductive biases, leading to predictions that violate fundamental conservation laws. The motivation for GFMs arises from the need to embed these symmetries directly into the architecture's fabric, ensuring that the "words" of the graph vocabulary are physically meaningful. As demonstrated in recent research, imposing symmetry group equivariance on machine learning architectures yields models that are more economical, interpretable, and trainable, effectively reducing the risk of erroneous extrapolation that deviates from data symmetries and physical laws [17]. Furthermore, specific applications in many-body systems have shown that models accounting for inherent symmetries and non-identical particles can accurately infer effective force laws and extract physical properties like mass and charge with high precision, pointing to new physics beyond current theoretical resolutions [18]. By moving towards foundation models that universalize these equivariant capabilities, the scientific community aims to create architectures where the learned vocabulary does not just fit data but adheres to the geometric and topological constraints of the physical world, ensuring that generated hypotheses and simulations remain physically plausible.

Secondly, the scarcity of high-quality, labeled data in scientific domains serves as a potent driver for the adoption of GFMs capable of few-shot and zero-shot learning, leveraging the transferability inherent in a robust graph vocabulary. In fields such as materials science, drug discovery, and high-energy physics, obtaining ground-truth data often requires expensive quantum mechanical calculations, sophisticated laboratory experiments, or massive supercomputing resources. For instance, generating training data for lattice quantum chromodynamics consumes a significant portion of open-science supercomputing power worldwide, making the creation of massive datasets for every new task prohibitive [19]. Traditional supervised learning methods struggle in these "small data" regimes, often overfitting to limited samples and failing to capture the underlying dynamics. GFMs address this by leveraging self-supervised pre-training on vast amounts of unlabeled or low-fidelity data, learning a universal "graph vocabulary" that can be adapted to specific tasks with minimal supervision [11]. This is particularly crucial for generalized few-shot learning scenarios, where models must recognize novel classes or predict properties of unseen materials based on limited examples. Approaches that explicitly model relationships between seen and novel classes or utilize semantic knowledge transfer have shown promise in mitigating class imbalance and improving adaptation [20]. Moreover, the ability to learn from invalid or constraint-violating data further enhances sample efficiency, allowing generative models to improve precision and constraint satisfaction even when valid high-fidelity samples are scarce [21].

Finally, the motivation for GFMs is rooted in the challenge of generalizing across varying scales and complex dynamical systems, a feat achievable only when the graph vocabulary supports multiscale abstraction. Scientific phenomena often exhibit multiscale behavior, where macroscopic properties emerge from microscopic interactions, requiring models that can bridge these scales without losing physical consistency. Conventional models trained on specific scales often fail to transfer knowledge to systems with different parameters or boundary conditions due to a lack of shared representational units. The vision for GFMs includes the development of context-informed dynamics models that can adapt to new physical contexts by conditioning on environmental parameters, enabling fast adaptation to unseen systems that share general dynamics but differ in specific contexts [22]. This capability is essential for solving partial differential equations (PDEs) in diverse settings, from fluid dynamics to reaction-diffusion processes, where the governing equations may be partially unknown or too complex to solve analytically. By encoding physics structure forcibly into deep learning frameworks, researchers aim to achieve high accuracy and robustness even in sparse data regimes, facilitating the discovery of governing laws and the simulation of complex spatiotemporal dynamics [23]. Additionally, the integration of theory-guided data science paradigms allows for the combination of scientific consistency with data-driven flexibility, producing models that are not only predictive but also scientifically interpretable, thereby advancing our understanding of novel domain insights [24].

In summary, the shift towards Graph Foundation Models in scientific discovery is motivated by the critical need to reconcile the data-hungry nature of deep learning with the data-scarce, law-governed reality of physical sciences. By prioritizing architectures that enforce physical symmetries within their vocabulary, enable learning from limited data through advanced transfer learning mechanisms, and generalize across multiscale dynamical systems, GFMs promise to transform scientific modeling from a process of curve-fitting to one of genuine discovery and reliable simulation. This paradigm shift sets the stage for the practical realization of cross-domain knowledge transfer, where the theoretical robustness of physics-informed GFMs converges with the adaptive strategies of Large Language Models and prompt-based mechanisms to accelerate discovery in physics-governed domains.

1.4 The Landscape of Cross-Domain Knowledge Transfer and Adaptation

1.4 Cross-Domain Adaptation Strategies: From Feature Alignment to Language-Guided Foundation Models

Building on the motivations for physics-informed generalization and data efficiency outlined in the previous section, the practical realization of Graph Foundation Models (GFMs) hinges on advanced cross-domain adaptation strategies. The landscape of knowledge transfer in graph machine learning has undergone a radical transformation, evolving from traditional domain adaptation techniques—often limited by rigid assumptions of feature space alignment—to sophisticated paradigms powered by Large Language Models (LLMs) and foundation architectures. Historically, transfer learning on graphs assumed that source and target domains shared similar feature distributions. However, the inherent heterogeneity of scientific data, where graph structures and node attributes vary significantly across domains (e.g., from molecular bonds to climate networks), frequently led to "negative transfer," where knowledge from a source domain inadvertently degraded performance on the target task [25]. To overcome these barriers and fulfill the promise of the universal "graph vocabulary" discussed earlier, recent advancements have introduced unified frameworks that leverage natural language as a semantic bridge, instruction tuning for task generalization, and prompt-based mechanisms for zero-shot adaptation.

A primary strategy emerging in this landscape is the unification of diverse graph domains through Text-Attributed Graphs (TAGs). By representing node features and structural contexts as natural language text, researchers can map heterogeneous graph data into a shared semantic space, effectively bypassing the feature misalignment that plagues traditional methods. The [8] framework exemplifies this approach, utilizing a cascaded architecture of language models and graph neural networks to train a foundation model capable of generalizing across unseen graphs and tasks. Similarly, the [7] framework proposes using text-attributed graphs to unify data from citation networks, molecular graphs, and knowledge graphs, enabling a single model to address node, link, and graph-level tasks simultaneously. This unification is critical for scientific discovery, where data scarcity in specific domains, such as rare disease modeling or novel material synthesis, necessitates the robust transfer of knowledge from data-rich domains, directly addressing the scarcity drivers identified in Section 1.3.

Complementing structural unification is the advent of instruction tuning tailored for graph structures. Inspired by the success of LLMs in following natural language commands, new frameworks guide models to understand complex topological features and perform diverse tasks without task-specific retraining. The [9] framework introduces a dual-stage instruction tuning paradigm that aligns LLMs with graph structural knowledge, enabling high generalization even when no information from the downstream graph data is available. Furthermore, [13] enhances this capability by distilling reasoning abilities from powerful LLMs to create task-specific Chain-of-Thought instruction packages, facilitating generic graph mining across different datasets. In the realm of molecular science, [26] demonstrates the efficacy of using natural language instructions to accomplish molecule-related tasks in a zero-shot setting, significantly outperforming baselines that lack adequate treatment of instructions. These instruction-based approaches allow scientific models to adapt to new hypotheses or experimental conditions simply by modifying the textual prompt, rather than retraining the entire network, thereby operationalizing the few-shot learning capabilities required in physics-governed domains.

To further facilitate zero-shot and few-shot adaptation, prompt-based mechanisms have been developed to bridge the gap between pre-training objectives and downstream tasks. Traditional fine-tuning often suffers from an objective mismatch, whereas prompt tuning reformulates downstream tasks to align with pre-training contexts. The [27] framework addresses this by designing a verbalizer-free prompting function that unifies pre-training and fine-tuning objectives, proving particularly effective in biological datasets. For scenarios involving heterogeneous graphs common in web and social sciences, [28] introduces a prompting function that integrates virtual class prompts and heterogeneous feature prompts to reformulate tasks, thereby mitigating negative transfer. Moreover, [29] proposes a multi-task prompting method that unifies the format of graph and language prompts, utilizing meta-learning to learn a robust initialization for varied graph applications. These mechanisms provide the flexible adaptation layer necessary to transition from the theoretical potential of GFMs to their practical deployment.

The challenge of cross-dataset generalization is also being tackled by frameworks specifically designed for zero-shot transferability. [16] introduces a framework that leverages language models to encode both node attributes and class semantics, ensuring consistent feature dimensions across datasets while employing a prompt-based subgraph sampling module to enrich structural information. Addressing the issue of structural differences that amplify distribution shifts, [25] reveals that transferring subgraph-level knowledge can mitigate negative transfer even when global structures differ, proposing Subgraph Pooling methods to effectively transfer knowledge across semantically similar but structurally distinct graphs. Additionally, [10] develops a unified graph tokenizer and a scalable graph transformer to generalize well on unseen graph data, utilizing LLM-enhanced data augmentation to alleviate data scarcity limitations. These approaches collectively ensure that the "vocabulary" learned by GFMs remains robust even when applied to entirely new scientific systems.

Finally, the integration of LLMs as teachers or collaborators has opened new avenues for efficient adaptation. [30] proposes a translator to bridge pretrained graph models and LLMs, empowering the LLM to make predictions based on language instructions for both pre-defined and open-ended tasks. For resource-constrained environments, [31] employs a knowledge distillation framework where an LLM teacher guides a student GNN, aligning hierarchically learned node features to boost accuracy while maintaining inference efficiency. These diverse strategies collectively define the current landscape, moving the field toward versatile, adaptable, and data-efficient graph foundation models. By establishing these cross-domain adaptation protocols, the field is now poised to bridge the historical gap between shallow, task-specific methods and the deep, unified architectures that characterize the next generation of scientific discovery tools, a transition explored in detail in the following section.

1.5 Bridging the Gap Between Shallow Methods and Deep Foundation Architectures

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Building upon the cross-domain adaptation strategies discussed previously, the evolution of graph machine learning for scientific discovery is further characterized by a profound architectural paradigm shift: moving from isolated, task-specific methodologies toward unified, scalable foundations capable of generalizing across diverse domains. Historically, the landscape was bifurcated between traditional shallow methods and early deep learning approaches, both facing significant limitations in the complex, data-scarce environments typical of scientific inquiry. Traditional shallow methods, such as label propagation and spectral clustering, rely on strong assumptions regarding graph homophily and smoothness. While computationally efficient, these approaches struggle to capture the intricate, non-linear dependencies inherent in physical systems. Recent investigations have surprisingly demonstrated that for certain transductive node classification tasks, combining shallow models with simple post-processing steps like "Correct and Smooth" can match or even exceed the performance of sophisticated Graph Neural Networks (GNNs) while utilizing a fraction of the parameters and training time [32]. However, the applicability of such shallow techniques is fundamentally constrained by their inability to learn rich, transferable feature representations from raw data, rendering them inadequate for tasks requiring deep semantic understanding or adaptation to unseen graph structures.

The advent of GNNs marked a transition to end-to-end supervised learning, enabling the direct encoding of structural and feature information through message-passing mechanisms. Despite their success in specific domains, standard supervised GNNs are often hindered by their reliance on abundant task-specific labeled data, a resource notoriously scarce in fields such as drug discovery, materials science, and genomics. When trained naively on limited datasets, these models frequently suffer from poor generalization, overfitting, and an inability to transfer knowledge across different graph distributions. This limitation spurred the development of self-supervised learning (SSL) strategies designed to leverage vast amounts of unlabeled data. Early SSL approaches primarily focused on contrastive learning or predictive tasks within a single domain. For instance, frameworks like Graph Contrastive Coding (GCC) were developed to capture universal topological properties by performing subgraph instance discrimination across multiple networks, demonstrating that pre-training could yield transferable representations superior to training from scratch [33]. Similarly, generative pre-training methods such as GPT-GNN factorize graph generation into attribute and edge components to learn inherent dependencies, significantly outperforming non-pre-trained models on downstream tasks [34].

However, the true bridge to the era of Graph Foundation Models (GFMs) lies in overcoming the fragmentation of these earlier pre-training efforts. While methods like GCC and GPT-GNN established the viability of pre-training, they often remained tied to specific graph types or required fine-tuning strategies that were not universally optimal. The transition to GFMs represents a move toward "foundation" capabilities: models pre-trained on massive, heterogeneous corpora of graph data that can be adapted to novel scientific tasks with minimal supervision. This shift addresses the critical gap where traditional shallow methods lack expressivity and standard supervised GNNs lack data efficiency. The emergence of GFMs is driven by the recognition that scientific graphs, despite their domain-specific differences, share underlying structural and physical principles that can be learned jointly. Frameworks such as UniGraph exemplify this transition by unifying node representations through Text-Attributed Graphs (TAGs), allowing a single model to generalize across diverse domains and achieve zero-shot or few-shot performance that rivals supervised baselines [8]. By leveraging Large Language Models (LLMs) for instruction tuning, these foundation models can interpret complex topological features through natural language, effectively bridging the semantic gap between graph structure and scientific context [9].

Furthermore, the architecture of GFMs is designed to mitigate the inherent limitations of deep GNNs, such as over-smoothing and the difficulty of training very deep networks. While earlier works introduced techniques like residual connections and dilated convolutions to enable deeper GCNs [35; 36], foundation models integrate these architectural innovations with scalable pre-training objectives. The "pre-train and prompt" paradigm, distinct from the rigid fine-tuning of the past, allows for flexible adaptation. Instead of retraining the entire network, GFMs utilize prompting mechanisms to align pre-trained knowledge with downstream objectives, effectively narrowing the gap between pre-training and task-specific goals [37; 38]. This approach is particularly vital for scientific applications where data distributions shift significantly between training and deployment, such as in modeling dynamic functional connectivity in the brain or predicting properties of novel molecular conformers.

The distinction between shallow methods, supervised GNNs, and GFMs is also evident in their handling of data scale and diversity. Shallow methods often fail to scale to billion-edge graphs or capture long-range dependencies, while supervised GNNs are bottlenecked by label scarcity. In contrast, GFMs are built upon the premise that scaling model size and data diversity yields emergent capabilities. Recent studies on molecular graphs have confirmed that GNNs benefit tremendously from increasing scales of depth, width, and dataset diversity, observing substantial performance improvements when scaling to billions of parameters and massive unlabeled datasets [39]. Moreover, advanced pre-training strategies now incorporate curriculum learning and data-active selection to optimize the quality of learned representations, ensuring that the model focuses on the most informative substructures rather than being diluted by noise [40; 41].

Ultimately, the transition to Graph Foundation Models represents a maturation of the field, moving from crafting specialized tools for narrow problems to building universal engines for scientific discovery. By synthesizing the structural awareness of shallow methods, the representational power of deep GNNs, and the generalization capabilities of foundation models trained on extensive unlabeled data, GFMs offer a robust framework for tackling the grand challenges of science. They enable the extraction of universal physical laws and structural motifs from heterogeneous data sources, facilitating zero-shot adaptation to new materials, drugs, and physical systems. This paradigm shift not only enhances predictive accuracy but also fosters a more data-efficient, interpretable, and scientifically grounded approach to machine learning, paving the way for autonomous discovery systems that can reason across the vast, interconnected landscape of scientific knowledge [6; 10].

2 Theoretical Foundations, Expressivity, and Advanced Architectures

2.1 Expressivity Limits and the Weisfeiler-Lehman Hierarchy

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The theoretical landscape of Graph Neural Networks (GNNs) is fundamentally anchored in the correspondence between message-passing mechanisms and classical graph isomorphism tests. At the heart of this analysis lies the seminal finding that standard Message Passing Neural Networks (MPNNs)—the most ubiquitous family of GNN architectures—are bounded in expressive power by the 1-dimensional Weisfeiler-Lehman (1-WL) test, also known as color refinement. This equivalence implies that if two non-isomorphic graphs cannot be distinguished by the 1-WL algorithm, no MPNN, regardless of its depth or width, can differentiate them either [42]. While MPNNs have achieved remarkable empirical success across diverse domains, this theoretical ceiling reveals critical limitations for scientific discovery, where distinguishing subtle structural nuances or counting specific substructures is often paramount.

The primary deficiency of the 1-WL test, and by extension standard MPNNs, is their inability to distinguish certain classes of non-isomorphic graphs. A canonical example involves regular graphs with identical degree sequences but distinct topological structures; the 1-WL test fails here because it relies solely on aggregating neighbor colors, which remain uniform in regular graphs [43]. Furthermore, the 1-WL framework struggles with counting specific substructures, such as triangles or cycles of length greater than three, which are decisive features in molecular property prediction and social network analysis. It has been rigorously proven that MPNNs cannot perform induced subgraph counting for substructures consisting of three or more nodes, nor can they distinguish graphs that differ only in the count of such motifs [44]. This deficiency is particularly problematic in chemistry, where the presence of specific ring systems (e.g., benzene rings) dictates molecular stability and reactivity. Consequently, relying solely on standard message-passing schemes risks overlooking crucial topological invariants inherent in physical systems.

To overcome these expressivity bottlenecks, researchers have explored the hierarchy of higher-order Weisfeiler-Lehman tests, specifically the k-WL and Folklore WL (k-FWL) tests. These extensions operate on k-tuples of nodes rather than individual nodes, thereby capturing higher-order interactions and complex structural patterns that 1-WL misses. Theoretical analysis demonstrates that the expressive power of k-WL strictly increases with k, offering a systematic pathway to more powerful graph learners [45]. For instance, the 2-WL test (and its equivalent, the Folklore 2-WL) can distinguish almost all random graphs and is capable of counting triangles and other small substructures, surpassing the capabilities of 1-WL [46]. Architectures inspired by these tests, often referred to as k-GNNs or Invariant Graph Networks (k-IGNs), have been shown to match the distinguishing power of their corresponding k-WL tests, providing a provably stronger alternative to standard MPNNs [47].

However, pursuing higher expressivity via the k-WL hierarchy incurs a steep computational cost. The space and time complexity of simulating k-WL scales as $O(n^k)$, where n is the number of nodes. This combinatorial explosion renders direct implementation impractical for even moderate values of k (e.g., $k = 3$) on large-scale scientific graphs [48]. While theoretical results confirm that k-IGNs are bounded in expressive power by k-WL and are equally powerful in distinguishing graphs, these resource requirements limit their applicability to small molecules or local subgraphs [47]. Recent efforts have sought to optimize the simulation of WL tests, showing that with specific activation functions and randomized architectures, the feature dimension required to simulate WL can be reduced to $O(\log n)$ or even constant dimensions; yet, the fundamental tensor operations for higher-order tuples remain a bottleneck [49; 50].

Within this hierarchy, the Folklore WL (k-FWL) variant offers a distinct perspective, often providing greater distinguishing power than standard k-WL for the same order k. Research indicates that k-FWL can be simulated by GNNs operating on k-tuples, and extensions like (k, t)-FWL have been proposed to create a more flexible design space that balances expressiveness and complexity [48]. Despite these advancements, a critical debate persists regarding the practical necessity of chasing the highest rung on the WL ladder. Some empirical studies suggest that for many standard graph datasets, the expressiveness of 1-WL is nearly sufficient, as classification accuracy upper bounds approach 100% even for simple WL-based models [51]. This finding implies that the gap between theoretical expressivity and practical generalization is not always linear; enhancing a model's ability to distinguish pathological non-isomorphic graphs does not automatically translate to better performance on real-world scientific tasks where data distributions and noise play significant roles [52].

Nevertheless, for scientific discovery applications involving complex physical laws, symmetry constraints, and rare substructures, the limitations of 1-WL are often unacceptable. The inability to count cycles or distinguish regular graphs can lead to physically inconsistent predictions. Therefore, the field has pivoted towards developing "practically expressive" architectures that approximate the power of higher-order WL tests without incurring the full $O(n^k)$ cost. Strategies include injecting local graph parameters, utilizing subgraph sampling policies, and employing equivariant subgraph aggregation networks to capture higher-order features efficiently [53; 54]. These approaches aim to bridge the gap between the theoretical elegance of the k-WL hierarchy and the computational realities of modeling large-scale scientific systems. The following section details how these strategies—specifically through subgraph decomposition and local parameter encoding—have been operationalized into advanced architectures that overcome expressive bottlenecks while maintaining scalability.

2.2 Advanced Architectures via Subgraph and Local Parameter Encoding

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Building on the theoretical limitations of the 1-WL test and the prohibitive $O(n^k)$ complexity of direct higher-order k-WL simulations discussed in Section 2.1, the field has pivoted toward a pragmatic design space: architectures that approximate higher-order expressivity through subgraph decomposition and local parameter encoding. These strategies aim to inject critical structural information—such as motif counts and complex connectivity patterns—into the learning process without incurring the exponential memory and time costs associated with processing full k-tuples. By strategically decomposing graphs or enriching node features with precomputed local invariants, these methods bridge the gap between the scalability of standard MPNNs and the expressive power required for rigorous scientific discovery.

A primary paradigm in this domain involves decomposing the input graph into a collection of subgraphs and processing them using equivariant set-based architectures. The Equivariant Subgraph Aggregation Networks (ESAN) framework exemplifies this approach by representing each graph as a set of subgraphs derived from predefined policies, such as node deletion or edge removal [54]. By applying an equivariant neural architecture to this set, ESAN can distinguish graphs that standard MPNNs cannot, effectively surpassing the 1-WL bound. Theoretical analysis within this work establishes novel variants of the WL test to provide lower bounds on the expressiveness of such subgraph-based methods, proving that the choice of subgraph selection policy and the equivariant architecture directly influences the model's distinguishing power [54]. To mitigate the increased computational burden of processing multiple subgraphs, stochastic sampling schemes have been proposed, allowing these powerful architectures to remain scalable for larger datasets [54]. This line of research has further matured with the establishment of a complete expressiveness hierarchy for subgraph GNNs via Subgraph Weisfeiler-Leman (SWL) tests, which categorizes node-based subgraph GNNs into equivalence classes with strictly growing expressivity [55]. This analysis reveals that while subgraph GNNs are potent, an inherent gap remains compared to the 2-FWL test, guiding future architectural designs to close this discrepancy [55].

Parallel to set-based aggregation, another significant line of inquiry focuses on explicitly encoding local graph parameters or counting specific substructures to enrich node representations directly within the message-passing scheme. Graph Substructure Networks (GSN) introduce a topologically aware mechanism that incorporates substructure counts into the feature space, enabling the detection of motifs that standard MPNNs and even some higher-order variants like 2-WL miss for induced subgraphs of size three or more [56; 44]. Theoretical investigations reveal nuanced relationships between architecture and substructure type; for instance, while MPNNs and 2-Invariant Graph Networks (2-IGNs) fail at induced-subgraph counting for certain motifs, they can successfully count star-shaped subgraphs [44]. By integrating these counts, GSNs achieve universality under sufficient conditions and demonstrate state-of-the-art performance on molecular and social network tasks without adhering to the rigid WL hierarchy [56]. Complementing this, the concept of Local Graph Parameters enables GNNs to incorporate structural information through a one-time preprocessing step, offering a computationally efficient middle ground [57]. This framework precisely characterizes the distinguishing power of such models in terms of finite variable logics, demonstrating that adding cheap-to-compute local parameters can significantly boost performance across diverse tasks [57]. Furthermore, recursive pooling techniques have been proposed to count subgraphs of size k by exploiting graph sparsity, offering a flexible trade-off between computational cost and expressive power that surpasses low-order GNN limitations [58].

To further refine the efficiency of substructure utilization, recent innovations focus on injecting local connectivity information without running full GNNs on every decomposed subgraph. The Union Subgraph Neural Network (UnionSNN) identifies a specific structure called the "union subgraph," which captures the complete connectivity picture of a 1-hop neighborhood around an edge [59]. By designing a shortest-path-based descriptor to encode high-order connectivities within these union subgraphs, UnionSNN creates a plugin module that can be infused into existing MPNNs or Transformers, strictly increasing their power beyond 1-WL while maintaining competitive efficiency [59]. Similarly, the Nested Graph Neural Networks (NGNNs) framework represents a graph via rooted subgraphs rather than rooted subtrees, applying a base GNN to each local subgraph to learn representations that encode more complex structural contexts [60]. This approach is proven to discriminate almost all r-regular graphs where 1-WL fails, yet it only introduces a constant-factor increase in time complexity [60]. For scenarios requiring explicit modeling of connections among motifs, Subgraph Networks (SGN) have been optimized via sampling strategies to create scale-controllable models with higher diversity [61]. These sampling-based SGNs reduce time complexity by two orders of magnitude compared to full SGN models while doubling performance enhancements in graph classification tasks [61]. Moreover, frameworks like SUREL+ have transitioned from walk-based to set-based representations to eliminate node duplication, achieving substantial speedups in subgraph-based representation learning for link prediction and higher-order pattern detection [62].

Collectively, these advanced architectures demonstrate that by strategically injecting local structural knowledge and leveraging subgraph decompositions, it is possible to overcome the expressive bottlenecks of standard GNNs while maintaining the scalability required for large-scale scientific discovery. However, while these methods successfully enhance topological expressivity, they primarily operate on abstract graph structures, often treating nodes and edges as discrete entities. This abstraction overlooks the continuous geometric embeddings inherent in many scientific domains, where atomic systems are governed by fundamental physical symmetries in Euclidean space. Consequently, the next evolution in architectural design must address not only topological complexity but also geometric consistency, leading to the development of Geometric Deep Learning and Equivariant Models for 3D systems, as detailed in the following section.

2.3 Geometric Deep Learning and Equivariant Models for 3D Systems

2.3 Geometric Deep Learning and Equivariant Models for 3D Systems

While the subgraph and local parameter encoding strategies discussed in Section 2.2 successfully enhance the topological expressivity of GNNs beyond the 1-WL limit, they primarily operate on abstract graph structures, often neglecting the continuous geometric embeddings inherent in scientific data. In domains such as computational chemistry, material science, and physics, atomic systems are not merely collections of nodes and edges but are embedded in Euclidean space, governed by fundamental physical symmetries. Consequently, standard permutation-invariant GNNs are insufficient; models must adhere to the rigorous constraints of rotations, translations, and reflections to ensure physical consistency. This necessity has catalyzed the development of Geometric Graph Neural Networks, a specialized class of architectures that encode these symmetries as inductive biases. A comprehensive taxonomy distinguishes between invariant networks, which output scalar quantities independent of the coordinate frame (sufficient for properties like formation energy), and equivariant networks, which preserve the transformation properties of vector and tensor features throughout the message-passing process [63]. The latter is indispensable for predicting vector-valued outputs such as forces, dipoles, and dynamics, ensuring predictions remain valid regardless of the system's spatial orientation.

A foundational challenge in this domain is balancing geometric expressivity with computational efficiency. Early approaches often circumvented the complexity of equivariant operations by projecting geometric features into invariant scalars via distance matrices or local frames, allowing standard GNNs to process geometric information. For instance, research has demonstrated that projecting equivariant features onto a local frame can decouple symmetry requirements from the network architecture, enabling ordinary GNNs to achieve state-of-the-art accuracy with significantly reduced memory and inference time compared to complex equivariant baselines [64]. Similarly, methods utilizing complete local frames constructed via cross-product operations have been proposed to efficiently approximate geometric quantities while maintaining SE(3) equivariance, effectively trading high-order tensor complexity for computationally efficient frame constructions [65]. However, reliance solely on pairwise distances has proven theoretically insufficient; it has been shown that "Vanilla" distance-based GNNs are geometrically incomplete and cannot distinguish certain symmetric graph structures, prompting the development of higher-order models like k-DisGNNs that exploit the rich geometry contained within the full distance matrix to achieve universality [66].

To overcome the limitations of scalar-only representations and fully capture the nuances of 3D interactions, modern architectures increasingly employ equivariant convolutions that operate directly on geometric tensors. The Neural Equivariant Interatomic Potentials (NequIP) model exemplifies this shift, utilizing E(3)-equivariant convolutions to interact geometric tensors, thereby capturing a more faithful representation of atomic environments and achieving remarkable data efficiency—often outperforming existing models with orders of magnitude fewer training samples [67]. Further advancements have sought to streamline these computationally intensive tensor operations; for example, the QHNet architecture introduces an innovative design that reduces the number of required tensor products by 92% while preventing the exponential growth of channel dimensions, thus reconciling the conflicting demands of efficiency and strict equivariance [68]. Alternative approaches leverage scalar-based parameterizations, proving that universally approximating polynomial functions equivariant under Euclidean and Lorentz groups can be constructed using lightweight collections of scalar products and contractions, offering a simple yet scalable pathway for modeling classical physics [69].

Beyond static property prediction, modeling the temporal evolution of physical systems requires incorporating continuity and dynamical laws into the network architecture. Traditional discrete-time GNNs often fail to capture the smooth trajectories inherent in physical dynamics. To address this, the Second-order Equivariant Graph Neural Ordinary Differential Equation (SEGNO) was introduced to incorporate second-order motion laws and continuity directly into the message-passing framework. By learning a unique, bounded trajectory between states rather than discrete mappings, SEGNO significantly improves generalization in complex dynamical systems [70]. Extending this concept to the continuous time domain, the Equivariant Graph Neural Operator (EGNO) framework models dynamics as functions over time using equivariant temporal convolutions in the Fourier space, marking a paradigm shift from next-step prediction to learning solution operators for 3D dynamics [71].

The integration of geometric algebra and attention mechanisms has also yielded powerful architectures for small point clouds and molecular systems. Geometric Algebra Attention Networks utilize products of terms from geometric algebra combined with attention mechanisms to systematically account for rotation invariance and permutation equivariance, demonstrating utility across physics and biology [72]. Similarly, Spatial Attention Kinetic Networks (SAKE) propose a functional form using neurally parametrized linear combinations of edge vectors to achieve E(n)-equivariance, offering a computationally lighter alternative to spherical harmonics-based methods while maintaining competitiveness in many-body system modeling [73]. Furthermore, the incorporation of directional message passing, as seen in DimeNet, explicitly embeds spatial directionality into messages using spherical Bessel functions and harmonics, outperforming distance-only models by capturing angular potentials crucial for empirical force fields [74].

Finally, the scope of equivariant learning extends to constrained systems and generative modeling, bridging the gap towards more holistic scientific simulations. Graph Mechanics Networks (GMN) address the challenge of geometrically constrained systems by representing forward kinematics through generalized coordinates, ensuring that constraints are implicitly encoded while maintaining equivariance [75]. In the realm of generation, E(n) Equivariant Normalizing Flows (E-NFs) provide a continuous-time invertible framework for jointly generating molecular features and 3D positions, setting new benchmarks in log-likelihood for particle systems [76]. Collectively, these advancements illustrate a maturing field where architectural innovations are increasingly driven by the need to strictly respect the geometric and dynamical principles of the physical world. While these models excel at encoding Euclidean symmetries and continuous dynamics, they typically operate on pairwise or local geometric relations. This sets the stage for the next evolution in structural representation: Topological Deep Learning, which generalizes these concepts to simulate multi-way interactions on simplicial and cell complexes, as discussed in the following section.

2.4 Topological Deep Learning on Simplicial and Cell Complexes

2.4 Topological Deep Learning on Simplicial and Cell Complexes

While the geometric models discussed in Section 2.3 successfully encode Euclidean symmetries and continuous dynamics, they typically operate on pairwise or local geometric relations, limiting their ability to capture the rich, multi-way dependencies prevalent in complex scientific systems ranging from protein folding to brain connectomes. To overcome this structural bottleneck, Topological Deep Learning (TDL) has emerged as a transformative paradigm, generalizing neural architectures to operate on higher-order topological domains such as simplicial complexes, cell complexes, and hypergraphs. These structures provide a rigorous mathematical framework for encoding polyadic relationships, allowing models to reason about groups of entities interacting simultaneously rather than merely as an aggregation of dyadic links. The foundational shift involves moving beyond the node-centric view to define signals and convolutions on edges, triangles, and higher-dimensional cells, thereby unlocking the ability to detect intrinsic topological features that are invisible to traditional graph-based and purely geometric methods.

Simplicial complexes represent one of the most prominent domains in this expansion, serving as high-dimensional generalizations of graphs that explicitly encode ordered relations between vertices at multiple resolutions. Unlike standard graphs, simplicial complexes can model higher-order interactions where a group of nodes forms a unified structural unit, such as a triangle or a tetrahedron. The development of Simplicial Neural Networks (SNNs) marks a critical advancement in this field, defining appropriate convolution operations that leverage the combinatorial structure of these complexes to process data living on simplices of arbitrary dimension [77]. These networks facilitate the analysis of complex data types, including vector fields and n-fold collaboration networks, by performing message passing not just between nodes, but also between higher-order simplices via their boundary and coboundary operators. However, early implementations of SNNs were often rigidly tied to the static combinatorial structure of the underlying complex, limiting their adaptability. To address this, Simplicial Attention Networks (SAT) were introduced, employing dynamic attention mechanisms to weight interactions between neighboring simplices. This approach allows the model to adapt to novel structures and, through signed attention mechanisms, ensures orientation equivariance, a crucial property for maintaining physical consistency in models operating on chain complexes [78].

A central mathematical tool enabling these advancements is the Hodge Laplacian, a multi-relational operator that generalizes the graph Laplacian to higher dimensions. The Hodge Laplacian is instrumental in signal processing on simplicial complexes, facilitating operations such as Fourier analysis, signal denoising, and interpolation by leveraging the unique spectral properties of the domain [79]. By spectrally manipulating the combinatorial k-dimensional Hodge Laplacians, researchers have developed models capable of learning functions parametrized by homological features. For instance, the Dist2Cycle model utilizes these spectral properties to perform homology localization, effectively measuring the distance of each simplex from the nearest optimal homology generator, a task impossible for pairwise GNNs [80]. Furthermore, the spectral decomposition of the Hodge Laplacian allows for the identification of simplicial communities, providing a powerful method for extracting community structures of arbitrary dimension from social and biological networks [81].

While simplicial complexes offer a robust framework, they impose strict constraints on the types of higher-order relations allowed (e.g., if a triangle exists, all its edges must exist). Cell complexes relax these constraints, offering a more flexible topology that can model ring-like structures and other complex shapes without requiring the filling of all sub-simplices. The generalization to regular cell complexes has led to the development of Cellular Isomorphism Networks (CINs) and their enhanced variant, CIN++, which incorporate direct interactions between cells within the same layer. This enhancement allows for a more comprehensive representation of higher-order and long-range interactions, achieving state-of-the-art results in chemistry benchmarks where ring structures are prevalent [82]. Theoretical analysis demonstrates that architectures based on cell complexes, collectively known as CW Networks, possess expressive power strictly greater than the 1-Weisfeiler-Lehman test and comparable to the 3-WL test, effectively distinguishing non-isomorphic graphs that stump standard GNNs [83].

The versatility of TDL extends to hypergraphs, another powerful abstraction for higher-order interactions. Although hypergraphs and simplicial complexes both capture multi-way relations, they induce different collective dynamics; for example, higher-order interactions may enhance synchronization in hypergraphs while suppressing it in simplicial complexes, highlighting the importance of choosing the correct topological representation for the specific physical system being modeled [84]. Recent surveys provide comprehensive guides on Hypergraph Neural Networks (HNNs), categorizing them by their message-passing schemes and training strategies, and demonstrating their efficacy in domains ranging from recommendation systems to genomics [85]. Innovations such as Sheaf Hypergraph Networks further enrich this space by introducing cellular sheaves to hypergraphs, creating linear and non-linear Laplacians that provide more expressive inductive biases than standard hypergraph diffusion [86].

Moreover, the integration of these topological concepts has led to novel approaches for mitigating common GNN failures such as over-squashing and limited expressivity. Topological Neural Networks utilize higher-dimensional structures to create shortcuts for information flow, decoupling the computational graph from the input graph structure and thus alleviating bottlenecks associated with long-range dependencies [87]. Additionally, frameworks like the Differentiable Cell Complex Module enable latent topology inference, allowing models to learn the optimal higher-order cell complex structure directly from data in an end-to-end fashion, rather than relying on a predefined topology [88]. These developments collectively signify a profound shift in scientific machine learning, where the explicit modeling of topology via Hodge Laplacians and combinatorial complexes enables the discovery of latent physical laws and structural patterns that remain obscured in pairwise representations. However, while expanding the structural domain addresses expressivity and long-range connectivity, the deep architectures required to process these complex topologies introduce new challenges regarding information propagation stability. Specifically, the interplay between depth and topology can exacerbate phenomena like over-smoothing and over-squashing, necessitating the geometry and dynamics-driven mitigation strategies explored in the following section.

2.5 Mitigating Over-Smoothing and Over-Squashing through Geometry and Dynamics

2.5 Mitigating Over-Smoothing and Over-Squashing through Geometry and Dynamics

While the expansion of GNNs into higher-order topological domains, as discussed in the preceding section, addresses the limitation of pairwise interactions, the effective deployment of deep architectures within these complex structures remains constrained by two interrelated phenomena: over-smoothing and over-squashing. Over-smoothing refers to the convergence of node representations to indistinguishable vectors as network depth increases, effectively erasing the discriminative features necessary for tasks like node classification [89]. Conversely, over-squashing describes a bottleneck where information from distant nodes is compressed into fixed-size vectors, rendering the model insensitive to long-range dependencies critical for understanding complex physical systems [90]. Recent theoretical advancements reveal that these issues are not merely architectural flaws but are intrinsically linked to the spectral properties of the graph Laplacian and the underlying geometric curvature of the data manifold. Specifically, research indicates an inevitable trade-off between these two phenomena, as strategies to alleviate one often exacerbate the other without careful geometric intervention [91].

A primary avenue for addressing these limitations involves leveraging discrete differential geometry, particularly through the lens of graph curvature. Theoretical analyses have established a profound connection between local graph geometry and the onset of these failure modes. For instance, over-smoothing is strongly associated with positive graph curvature, where rapid mixing causes features to homogenize, while over-squashing is linked to negative curvature, which creates bottlenecks that impede information flow [92]. This geometric insight has spurred the development of curvature-aware graph rewiring techniques. Methods utilizing Ollivier-Ricci curvature, such as the Batch Ollivier-Ricci Flow, dynamically modify the graph topology during training to simultaneously address both issues by smoothing out negative curvature bottlenecks and reducing excessive positive curvature [92]. Similarly, approaches based on Augmented Forman-Ricci curvature offer a computationally efficient alternative, providing linear-time scalability for large-scale scientific graphs while effectively characterizing and mitigating these effects through targeted edge augmentation and removal [93]. Furthermore, the Stochastic Jost and Liu Curvature Rewiring (SJLR) algorithm demonstrates that performing edge additions and removals stochastically during training can preserve fundamental graph properties while achieving a suitable compromise between the spectral gap requirements for smoothing and squashing [91].

Beyond static rewiring, dynamic formulations based on Partial Differential Equations (PDEs) offer a robust framework for controlling information propagation, bridging the gap between discrete message passing and continuous physical processes. By modeling GNN layers as discretizations of continuous diffusion processes, researchers can explicitly manage the Dirichlet energy of node embeddings, which serves as a quantitative measure of feature smoothness. Standard message-passing acts as a low-pass filter, monotonically decreasing Dirichlet energy and leading to over-smoothing [94]. To counteract this, novel architectures introduce high-pass filtering components or reverse the time direction of the graph heat equation to enhance feature sharpness [95]. For example, the Multi-Scaled Heat Kernel based GNN (MHKG) amalgamates diverse filtering functions to control the trade-off between smoothing and squashing, demonstrating that enhancing node feature sharpness can inadvertently increase susceptibility to over-squashing unless carefully balanced [95]. Additionally, Framelet Augmentation strategies have been developed to strictly enhance Dirichlet energy through multi-scale representations, allowing deep GNNs to maintain discriminative power even on heterophilous graphs where traditional models fail [96].

The interplay between network depth, width, and topology further complicates these dynamics, necessitating a holistic view of architectural design. Theoretical work proves that while increasing network width can mitigate over-squashing, it comes at the cost of heightened sensitivity, whereas increasing depth often exacerbates the issue due to vanishing gradients and topological bottlenecks [97]. Consequently, some approaches focus on optimizing the embedding space geometry itself rather than the input graph. Riemannian GNNs, which embed nodes into manifolds of variable curvature, have shown promise in alleviating over-squashing by aligning the geometry of the embedding space with the intrinsic topology of the graph, particularly for structures exhibiting negative curvature [98]. Moreover, implicit graph neural diffusion networks formulated as fixed-point equations of Dirichlet energy minimization problems provide conditions under which over-smoothing can be avoided during both training and inference, ensuring unique equilibria and strong generalization bounds [99].

Alternative strategies involve modifying the aggregation mechanics to prevent the exponential convergence of features. Techniques such as DropEdge randomly remove edges during training to reduce the convergence speed of over-smoothing and act as a data augmenter, effectively relieving information loss [100]. Similarly, the concept of “Anti-Symmetric Deep Graph Networks” utilizes ordinary differential equations to design stable, non-dissipative architectures that preserve long-range information and prevent gradient vanishing, enabling the training of networks with dozens of layers [101]. Other methods, such as Graph-Coupled Oscillator Networks (GraphCON), model the system as nonlinear controlled oscillators, proving that zero-Dirichlet energy steady states are unstable in their formulation, thereby naturally mitigating over-smoothing [102]. Collectively, these geometry and dynamics-driven approaches represent a paradigm shift from treating GNNs as static aggregators to viewing them as controllable dynamical systems. By resolving the conflicts between depth, expressivity, and stability, these methods lay the necessary groundwork for the rigorous theoretical guarantees regarding universality, stability, and generalization explored in the following section.

2.6 Theoretical Foundations of Universality, Stability, and Generalization

2.6 Theoretical Foundations of Universality, Stability, and Generalization

Building upon the geometry and dynamics-driven strategies detailed in the previous section—which resolve conflicts between depth, expressivity, and stability to enable deep architectural deployment—it is essential to establish the rigorous theoretical guarantees that underpin these models. The reliable application of Graph Neural Networks (GNNs) in scientific discovery rests on three foundational pillars: universality, stability, and generalization. These properties collectively determine whether a model can approximate complex physical laws, remain robust against noisy experimental data or uncertain graph topologies, and perform reliably on unseen systems. Recent theoretical advancements have moved beyond empirical observations to provide the mathematical formalism necessary to validate the efficacy of GNNs in physics-governed domains.

The concept of universality establishes the conditions under which GNN architectures can approximate arbitrary functions defined on graph-structured data, effectively defining the ceiling of their expressive power. While early results linked this capability to the Weisfeiler-Lehman (WL) test for graph isomorphism, recent work has expanded this understanding to node-level tasks and varying graph sizes. It has been proven that GNNs act as universal approximators in probability for node classification and regression tasks, capable of approximating any measurable function that satisfies the 1-WL equivalence, even when target functions are discontinuous [103]. Furthermore, the role of input features and positional encodings is critical; on large random graphs, the inclusion of appropriate positional encodings allows equivariant GNNs to converge to a broader function space, effectively extending universality results to more general stochastic graph models [104]. For scientific applications involving symmetry, such as particle physics or molecular dynamics, the universality of invariant and equivariant maps is paramount. Theoretical frameworks utilizing intermediate polynomial layers and generalized Stone-Weierstrass theorems have demonstrated that specific GNN architectures can uniformly approximate continuous maps that are invariant or equivariant with respect to linear representations of compact groups, including permutations and rigid Euclidean motions [105]. This ensures that GNNs can theoretically learn any physical law that adheres to specific symmetry groups, provided the architecture is sufficiently expressive. Additionally, it has been shown that a single set of parameters can define a GNN that approximates functions uniformly well on graphs of varying sizes, further supporting their applicability to diverse scientific systems [106].

Complementing expressivity, stability addresses the sensitivity of GNN outputs to perturbations in the underlying graph topology, a crucial feature for scientific data where connectivity may be estimated with error or subject to dynamic fluctuations. A core finding is that GNNs composed of integral Lipschitz filters combined with pointwise nonlinearities achieve a unique balance: they are stable to small topological changes while remaining discriminative of high-frequency information, a dual property unattainable by linear graph filters alone [107]. This stability extends to relative perturbations of the graph shift operator, where the output difference is bounded by the magnitude of the relative change in topology, explaining the success of GNNs in scenarios with uncertain graph structures [108]. In the context of large-scale systems, stability analysis via graphons—limit objects of convergent graph sequences—reveals that the transference error decreases asymptotically with graph size, justifying the application of models trained on moderate-sized graphs to massive scientific networks [109]. Moreover, rigorous bounds have been established for stochastic perturbations, such as random link losses, showing that the expected output difference scales linearly with the link loss probability, thereby providing a handle to improve robustness through architectural choices like filter width and depth [110]. Importantly, there exists a quantifiable trade-off between stability and representational capacity; architectures with fewer degrees of freedom, such as those with perfect alignment between parameter matrices and the graph shift operator, tend to be more stable but less expressive, necessitating careful design for specific scientific tasks [111]. Furthermore, stability guarantees have been extended to directed graphs and graph-coarse-graining procedures, demonstrating that stability holds under broader conditions than previously assumed [112].

Generalization in GNNs, particularly in out-of-distribution (OOD) settings common in scientific discovery, is governed by the interplay of architectural bias, optimization dynamics, and data distribution. Theoretical analyses using PAC-Bayesian frameworks indicate that generalization bounds for GNNs are tightly linked to the spectral norm of weights and the maximum node degree, suggesting that controlling these factors is essential for robust performance [113]. In transductive learning settings, classical measures like VC dimension may yield trivial bounds, whereas transductive Rademacher complexity offers a more precise explanation of generalization behavior, highlighting the specific role of graph convolutions in reducing error on stochastic block models [114]. However, GNNs often fail in OOD scenarios by exploiting spurious correlations rather than causal mechanisms. To address this, causal representation frameworks have been proposed to distinguish stable correlations from environment-specific biases, enabling models to generalize across distribution shifts by focusing on invariant subgraph structures [115]. Additionally, the implicit bias of training algorithms plays a significant role; for instance, linear equivariant networks trained via gradient descent converge to solutions with low-rank Fourier coefficients, effectively regularizing the model in a way that promotes generalization on symmetric tasks [116]. The phenomenon of "double descent" has also been characterized in GNNs, where the risk is shaped by the complex interplay between graph noise, feature noise, and the number of training labels, challenging traditional complexity measures [117]. Recent work also suggests that generalization bounds can be improved by considering the largest singular value of the feature diffusion matrix rather than just the maximum degree, offering tighter predictive metrics for model performance [118].

Synthesizing these theoretical strands reveals that the success of GNNs in science is not merely empirical but grounded in their ability to balance expressivity with stability and to leverage inductive biases for generalization. The geometric interventions discussed in the preceding section provide the structural mechanisms to achieve the stability and depth required for these theoretical guarantees to hold in practice. Future advancements in scientific AI will depend on refining these foundations to handle dynamic, heterogeneous, and multi-modal systems where the assumptions of static graphs and i.i.d. data no longer hold, ensuring that models remain trustworthy and physically consistent in high-stakes discovery pipelines [119].

3 Physics-Informed, Geometry-Aware, and Causal Learning Frameworks

3.1 Physics-Informed Graph Networks and Discrete PDE Solvers

3.1 Physics-Informed Graph Networks and Discrete PDE Solvers

The integration of physical laws into machine learning frameworks has revolutionized the simulation of complex scientific systems, particularly in data-scarce or noisy regimes. While Physics-Informed Neural Networks (PINNs) based on feed-forward architectures established a foundational paradigm for embedding partial differential equations (PDEs) into loss functions via automatic differentiation, they often encounter significant bottlenecks when applied to systems with unstructured data or complex geometries. In such scenarios, Graph Neural Networks (GNNs) offer distinct advantages due to their inherent ability to operate on irregular domains and mesh-like structures. This subsection reviews the evolution from standard PINNs to Physics-Informed Graph Networks (PIGNs) and discrete solvers that leverage graph exterior calculus, Galerkin methods, and variational principles to address forward and inverse problems on unstructured meshes [120].

A primary limitation of conventional PINNs is their reliance on point-wise formulations and fully connected networks, which struggle with scalability and the rigorous enforcement of boundary conditions on irregular domains. Furthermore, the infinite search space in continuous function approximation often leads to non-convex optimization landscapes that are difficult to navigate. To overcome these challenges, researchers have developed discrete frameworks that combine the flexibility of GNNs with the mathematical rigor of traditional numerical methods. For instance, the framework known as Physics-Informed Graph Neural Galerkin Networks utilizes graph convolutional networks (GCNs) alongside a variational structure of PDEs. By employing piecewise polynomial basis functions, this approach reduces the dimensionality of the search space, facilitates training convergence, and allows for the strict imposition of boundary conditions without the need for tuning penalty parameters, a common pain point in classic PINNs [121]. This discrete formulation effectively bridges the gap between data-driven learning and classical finite element methods, enabling the solution of both linear and nonlinear PDEs on complex geometries.

Beyond discrete variational formulations, a more profound integration of physics involves the use of graph exterior calculus to construct differential operators that inherently respect conservation laws and topological structures. Traditional machine learning approaches often enforce physics weakly through penalization terms in the loss function, which may not guarantee physical realizability, especially in small data regimes. In contrast, the Data-Driven Exterior Calculus (DDEC) framework utilizes graphs as coarse-grained mesh surrogates that inherit desirable conservation properties from combinatorial Hodge theory. In this approach, the missing metric information required for PDE discretization is learned directly from data, allowing the resulting models to enforce physics to machine precision even when the network is poorly trained or data is sparse [122]. This method provides mathematical guarantees of well-posedness and has been successfully applied to learn $H(div)$ and $H(curl)$ systems representative of subsurface flows and electromagnetics, demonstrating a robust pathway for constructing structure-preserving surrogate models.

The application of these principles extends to various specific architectures designed to handle the unique demands of scientific computing. For example, the Physics-Driven GraphSAGE method incorporates the Galerkin method and piecewise polynomial nodal basis functions to solve computational problems governed by irregular PDEs. This approach employs graph representations of physical domains to reduce the demand for evaluated points through local refinement and introduces distance-related edge features to improve convergence in the presence of singularities and oscillations [123]. Similarly, the RBF-MGN framework integrates Radial Basis Function Finite Difference (RBF-FD) schemes with GNNs to construct high-precision difference formats that guide model training, effectively addressing the difficulties CNNs face with unstructured meshes [124]. These hybrid approaches highlight the versatility of graph-based methods in adapting classical numerical discretizations to a learnable context.

Furthermore, the capability of GNNs to solve inverse problems on unstructured meshes has been significantly advanced. Differentiable graph network simulators, such as Graph Network-based Simulators (GNS), represent physical domains as graphs where particles are nodes and interactions are edges. These models not only accelerate forward simulations by orders of magnitude compared to traditional solvers but also enable the solution of inverse problems through automatic differentiation. By interleaving learned surrogates with Material Point Methods (MPM), these frameworks can minimize errors while satisfying conservation laws, achieving substantial speedups in tasks like granular flow prediction and material parameter identification [125]. The ability to iteratively update material properties, such as friction angles, by computing gradients of loss functions based on target runouts exemplifies the power of differentiable physics-embedded graph networks in design and optimization contexts.

The scalability of these methods to large-scale engineering problems remains a critical focus. Recent advancements have demonstrated the feasibility of training GNN surrogates on 3D meshes with millions of nodes through domain decomposition strategies that are mathematically equivalent to training on the whole domain under specific conditions [126]. This progress, combined with higher-order numerical integration techniques, paves the way for applying PIGNs to real-world computational fluid dynamics simulations that were previously intractable for standard deep learning models. Additionally, frameworks like Physics-Informed Boundary Integral Networks (PIBI-Nets) reduce the dimensionality of the problem by requiring collocation points only at the boundary, offering a data-driven approach that outperforms standard PINNs in high-dimensional settings [127].

In summary, the transition from continuous, point-wise PINNs to discrete, graph-based solvers represents a paradigm shift in scientific machine learning. By leveraging graph exterior calculus, Galerkin methods, and hybrid numerical-learning architectures, Physics-Informed Graph Networks provide a robust, scalable, and physically consistent framework for solving forward and inverse problems on unstructured meshes. These developments not only enhance the accuracy and efficiency of simulations but also expand the applicability of deep learning to complex, multi-scale physical systems governed by fundamental laws [120]. Having established how physical laws can be encoded via discrete operators and variational principles on graphs, the discussion now turns to how these concepts are further refined by explicitly enforcing geometric symmetries and conservation laws directly within the neural architecture itself. As the field progresses, the synthesis of topological deep learning with rigorous physical constraints promises to unlock new capabilities in autonomous scientific discovery and high-fidelity simulation.

3.2 Enforcing Physical Symmetries and Conservation Laws in Architectures

3.2 Enforcing Physical Symmetries and Conservation Laws in Architectures

Building upon the discrete operators and variational principles introduced in the previous subsection, which embed physical laws primarily through loss functions or specific numerical discretizations, this section explores a more intrinsic approach: hard-coding physical symmetries and conservation laws directly into the neural architecture itself. While Physics-Informed Graph Networks (PIGNs) leverage graph exterior calculus and Galerkin methods to respect physics in the optimization objective, architectural enforcement ensures that the model's hypothesis space is restricted to functions that inherently satisfy geometric transformation rules—such as translation, rotation, and permutation equivariance—regardless of the training data distribution. This paradigmatic shift from soft constraints in the loss landscape to hard inductive biases in the network structure significantly enhances data efficiency, guarantees robust generalization under distributional shifts, and ensures that predictions remain physically consistent across arbitrary coordinate systems, a critical requirement for modeling atomic systems, fluid dynamics, and rigid body mechanics.

A cornerstone of this architectural evolution is the development of E(n)-equivariant graph neural networks, which strictly preserve equivariance with respect to the Euclidean group E(n). These architectures ensure that if the input geometry is rotated or translated, the output features transform correspondingly, a property indispensable for modeling 3D atomic systems where the absolute orientation of a molecule is arbitrary. The paper [128] introduces a foundational model that achieves this equivariance without relying on computationally expensive higher-order representations in intermediate layers, thereby maintaining scalability while delivering competitive performance in dynamical systems modeling and molecular property prediction. Building upon this, [73] proposes an alternative functional form utilizing neurally parametrized linear combinations of edge vectors. This approach, known as SAKE, avoids the heavy computational burden of spherical harmonics while still universally approximating node environments, demonstrating that rigorous equivariance can be achieved with significantly improved computational efficiency for many-body system modeling. Furthermore, the need for lightweight yet accurate models in large-scale simulations is addressed by [129], which introduces LEIGNN. This model employs a scalar-vector dual representation to encode equivariant features, successfully leveraging geometric symmetry information to achieve ab initio accuracy with the computational efficiency of classical molecular dynamics across diverse states of matter.

However, strict equivariance is not always sufficient or appropriate for all physical scenarios, particularly when external fields break spatial symmetries. In many real-world systems, symmetries are partially broken by forces such as gravity, which introduces a preferred direction in space. Enforcing full rotational equivariance in such contexts can inadvertently hinder the model's ability to learn the correct physical dynamics. To address this, recent research has moved towards subequivariant models that relax strict symmetry constraints to accommodate broken symmetries. The work [130] explicitly tackles this challenge by proposing a backbone that transitions from full equivariance to subequivariance. This architecture incorporates external fields like gravity into the message-passing mechanism, allowing the model to distinguish between directions parallel and perpendicular to the gravity vector. By introducing a subequivariant object-aware message passing scheme, this approach effectively models interactions between objects of varying shapes and sizes, achieving superior generalization and data efficiency compared to models that either ignore symmetry or enforce it excessively. Similarly, [131] provides a critical theoretical perspective, demonstrating that while exact equivariance is ideal for unseen symmetries, approximately equivariant networks can outperform their exact counterparts when the training data symmetries do not perfectly match the network's built-in constraints, highlighting the nuance required in designing symmetry-aware architectures for complex, heterogeneous systems.

Beyond particle systems, the enforcement of geometric constraints is paramount in modeling rigid body dynamics and mechanical systems where objects are connected by joints or hinges. Standard GNNs often struggle to maintain these hard constraints over long simulation rollouts, leading to physically implausible configurations that violate conservation of momentum or energy. The paper [75] addresses this limitation by proposing the Graph Mechanics Network (GMN). This architecture uniquely represents forward kinematics information using generalized coordinates, which implicitly and naturally encodes geometrical constraints such as fixed distances between connected nodes. By developing a general form of orthogonality-equivariant functions, GMN ensures that the dynamics of these constrained systems are learned in a manner that respects both the combinatorial complexity of the interactions and the underlying geometric restrictions. This results in significantly improved prediction accuracy and constraint satisfaction for systems involving particles, sticks, and hinges, as well as real-world human motion capture data.

The theoretical underpinnings of these architectures are further solidified by comprehensive surveys and taxonomies that categorize geometric GNNs based on their handling of symmetries. [63] provides a structured overview, distinguishing between invariant networks, equivariant networks in Cartesian and spherical bases, and unconstrained networks, aiding practitioners in selecting the appropriate inductive bias for their specific scientific problem. Additionally, [132] delves into the mathematical foundations, connecting group equivariant networks to gauge equivariant networks on manifolds, thereby offering a unified formalism for understanding how symmetries can be encoded in deep learning models operating on non-Euclidean domains. The importance of these symmetries is also evident in fluid flow modeling, where [133] demonstrates that while equivariant architectures are powerful, modeling invariant quantities can sometimes yield more accurate long-term predictions for fluid evolution, suggesting a delicate balance between enforcing equivariance and learning invariant representations depending on the specific physical quantity being predicted.

In summary, the enforcement of physical symmetries and conservation laws in GNN architectures represents a crucial refinement of the physics-informed paradigm, evolving from soft penalization to structural guarantees. By explicitly embedding the principles of E(n) symmetry, accommodating broken symmetries like gravity, and rigorously enforcing geometric constraints in mechanical systems, these advanced architectures provide a robust foundation for scientific discovery. They ensure that learned models are not only accurate within the training distribution but also generalize reliably to unseen physical configurations, adhering to the fundamental laws of nature. Having established how geometric symmetries can be hard-coded to ensure transformational consistency, the discussion now turns to a deeper layer of physical understanding: causal reasoning. While the architectural enforcement of symmetries addresses *how* physical quantities transform, it does not inherently explain *why* they interact or distinguish genuine causal mechanisms from spurious correlations, a gap that the following subsection aims to bridge through latent structure inference and causal discovery.

3.3 Causal Reasoning and Latent Structure Inference in Physical Systems

3.3 Causal Reasoning and Latent Structure Inference in Physical Systems

While the architectural enforcement of symmetries discussed in the previous subsection provides a robust inductive bias for geometric consistency, it primarily addresses *how* physical quantities transform rather than *why* they interact. The integration of causal reasoning into graph-based scientific discovery represents a critical evolution beyond these correlation-based or symmetry-constrained approaches. In complex physical systems, observational data often contains spurious correlations induced by hidden confounders, environmental shifts, or measurement noise, which can lead to models that fail when deployed in out-of-distribution scenarios even if they respect geometric symmetries. To address this, recent advancements focus on distinguishing genuine causal physical mechanisms from statistical artifacts, leveraging differentiable frameworks that encode implicit domain knowledge within latent spaces and techniques capable of discovering governing differential equations directly from data. This shift from enforcing known symmetries to inferring unknown causal structures serves as a vital bridge toward the energy-based variational principles discussed in the following section, where the goal is not merely to identify relationships but to formulate them within globally consistent physical functionals.

A primary challenge in causal discovery is the presence of latent variables that obscure the true generative process. In many scientific domains, such as climate modeling or biological pathway analysis, not all relevant factors are observable, rendering standard graph learning insufficient. To mitigate this, researchers have developed architectures that explicitly model latent confounders to recover invariant causal structures. For instance, the method proposed in [134] iteratively derives proxies for the latent space from the residuals of an inferred model, thereby improving the structural inference of Gaussian graphical models and enhancing the identifiability of causal effects even when the causal sufficiency assumption is violated. Similarly, [135] addresses the ill-posed nature of causal discovery with hidden variables by assuming a linear structural equation model with independent latent factors, utilizing the method of moments and coupled tensor decomposition to provide guaranteed solutions for joint causal discovery and latent variable inference.

Beyond static structural learning, dynamic physical systems require the identification of causal relationships within time-series data, often characterized by non-linearity and non-stationarity. The

[136] framework introduces a deep learning architecture designed to simultaneously discover contemporaneous and lagged causal relations in such complex environments. By incorporating custom causal layers and acyclicity constraints, this approach overcomes the limitations of traditional methods that assume stationarity, offering a robust tool for analyzing intricate dependencies in fields ranging from epidemiology to environmental science. Furthermore, the challenge of learning causal structures from interventional data where the specific interventions are unknown is addressed in [137]. This work frames the problem as learning a shared causal graph among an infinite mixture of intervention structural causal models, enabling the discovery of causal relations even when the correspondence between samples and interventions is entirely latent.

The fusion of causal reasoning with physics-informed machine learning has yielded powerful frameworks for discovering the underlying laws of nature, moving closer to the variational objectives seen in energy-based methods. A notable approach is the [138], which incorporates implicit physics knowledge provided by domain experts into the latent space of a graph network. This architecture, validated on climate prediction tasks, demonstrates that encoding physical behaviors implicitly can significantly improve inductive learning and multi-step predictions, effectively bridging the gap between data-driven patterns and theoretical physical constraints. In a similar vein, [139] proposes a novel architecture that forcibly encodes known physics knowledge through a coercive mechanism rather than penalty-based regularization. This ensures that the learned model rigorously obeys given physical laws, exhibiting remarkable robustness against data noise and scarcity while generalizing better than state-of-the-art models.

Discovering the explicit mathematical form of governing equations from sparse and noisy observations is another frontier where causal and physics-informed methods converge, laying the groundwork for defining the energy functionals central to the next subsection. The [140] framework utilizes two deep neural networks: one to approximate the unknown solution and another to represent the nonlinear dynamics, thereby avoiding unstable numerical differentiation and accurately learning underlying dynamics like the Navier-Stokes equations. Building on this, [141] integrates deep neural networks with sparse regression to identify key derivative terms and parameters, successfully discovering PDE systems even under conditions of extreme data scarcity. To further enhance robustness against noise, [142] introduces a deep convolutional-recurrent network that encodes prior physics knowledge to reconstruct high-fidelity data before performing sparse regression, ensuring the discovery of explicit governing equations is not misled by measurement errors.

Uncertainty quantification is intrinsic to reliable causal discovery in physical systems, as it allows scientists to assess the credibility of discovered laws before embedding them into conservative energy frameworks. [143] presents a Bayesian approach comprising "leaf" modules for spatiotemporal data and a "root" module expressing a nonparametric distribution over governing operators. This framework quantifies the reliability of learned physics as a posterior distribution over operators, propagating uncertainty to novel initial-boundary value problems. Complementing this, [144] employs latent variable models and adversarial inference to train probabilistic representations constrained by partial differential equations, effectively characterizing uncertainty in system outputs due to input randomness or observation noise without requiring expensive repeated sampling.

Finally, the interpretability of these causal models is paramount for scientific acceptance and the formulation of human-understandable physical principles. [145] demonstrates how graph networks with physically motivated inductive biases can learn message representations equivalent to true force vectors, allowing symbolic regression to recover explicit algebraic equations such as Newton's law of gravitation from implicit knowledge. This capability to translate latent neural representations into human-interpretable symbolic laws marks a significant step toward autonomous scientific discovery, where models not only predict but also explain the causal mechanisms driving physical phenomena. Collectively, these advances illustrate a paradigm shift towards graph learning frameworks that are not only predictive but also causally sound, physically consistent, and interpretable. By identifying the correct causal operators and governing equations, these methods provide the essential building blocks for the energy-based formulations and variational principles explored in the subsequent section, ensuring that the global optimization objectives are grounded in discovered, rather than assumed, physical truths.

3.4 Energy-Based Formulations and Variational Principles on Graphs

3.4 Energy-Based Formulations and Variational Principles on Graphs

Building upon the causal structures and explicit governing equations identified in the previous subsection, the next critical step is to embed these discovered physical truths into a globally consistent optimization framework. While causal reasoning distinguishes genuine mechanisms from spurious correlations, and equation discovery provides the mathematical form of interactions, the actual learning process requires an objective function that ensures stability and physical fidelity. Traditional approaches, epitomized by standard Physics-Informed Neural Networks (PINNs), often rely on minimizing the residuals of governing partial differential equations (PDEs) at collocation points. However, this residual-minimization paradigm frequently encounters significant hurdles, including the numerical instability of computing high-order derivatives, sensitivity to noise, and difficulties in strictly enforcing complex boundary conditions. To address these limitations, a robust alternative has emerged: energy-based formulations and variational principles. These approaches shift the optimization objective from satisfying differential equations point-wise (the strong form) to minimizing global variational energy functionals derived from the underlying physics (the weak form). This subsection explores how implementing these energy-based methods through graph neural architectures offers superior stability, data efficiency, and strict adherence to conservation laws, serving as the theoretical bedrock for the robust optimization strategies discussed in the following section.

A primary advantage of transitioning to energy-based formulations is the reduction in the order of derivatives required during training. In many physical systems, the strong form of the governing equations involves second or higher-order spatial derivatives. Computing these via automatic differentiation in deep networks can lead to vanishing or exploding gradients and increased computational cost. By contrast, variational formulations, such as those based on the principle of minimum potential energy, typically involve only first-order derivatives. For instance, in the context of coupled field problems like thermo-mechanical systems or induction heating, energy-based PINNs (ePINNs) have demonstrated that minimizing the variational energy functional is not only more computationally efficient but also requires fewer hyperparameters compared to residual-based counterparts [146]. This efficiency is critical when dealing with multi-physics interactions where the coupling between different fields (e.g., magnetic and thermal) introduces additional complexity into the loss landscape, a challenge that simpler residual methods often fail to navigate smoothly.

Furthermore, energy-based methods provide a natural and rigorous framework for enforcing boundary conditions and ensuring physical consistency, addressing a key weakness of standard PINNs. Traditional approaches often treat boundary conditions as soft constraints added to the loss function, which can lead to violations if the weighting coefficients are not tuned perfectly. Energy-based approaches, however, can construct admissible functions that inherently satisfy essential boundary conditions without the need for penalty terms. The Conservative Energy Method based on Neural Networks (CENN) exemplifies this by utilizing radial basis functions and neural subdomains to construct solutions that strictly adhere to boundary constraints while minimizing potential energy [147]. This method has shown superior accuracy in handling heterogeneous materials and complex geometries, outperforming strong-form PINNs in scenarios involving singularities and discontinuities, thereby ensuring that the global functional remains well-posed.

In the realm of dynamical systems and time-dependent phenomena, preserving geometric structures such as symplecticity and energy conservation is paramount for long-term prediction accuracy. Standard neural networks often fail to conserve these quantities, leading to drift and unphysical behavior over extended simulation times. Variational Integrator Graph Networks address this by unifying energy constraints with high-order symplectic variational integrators within a graph neural network architecture [148]. By embedding the discrete Lagrangian mechanics directly into the message-passing mechanism, these networks ensure that the learned dynamics respect the conservation of momentum and energy. Empirical studies reveal that this unifying framework significantly outperforms existing methods in data-efficient learning scenarios, as the inductive bias of energy conservation guides the model to correct physical trajectories even from noisy observations, effectively leveraging the causal insights from the previous section to maintain trajectory integrity.

The benefits of variational principles extend further to fracture mechanics and topology optimization, where the evolution of discontinuities poses severe challenges for residual-based methods. In brittle fracture modeling, minimizing the variational energy of the system allows for a more robust imposition of boundary conditions and avoids the high-order derivative issues associated with phase-field equations [149]. Similarly, in topology optimization, dynamically configured PINNs that leverage energy-based objectives can efficiently navigate the design space, reducing the computational burden of data acquisition while maintaining high fidelity in displacement predictions [150]. These applications highlight how variational formulations provide a stable foundation for handling non-smooth and evolving domains, which are common in real-world scientific discovery.

Moreover, the stability of energy-based models is enhanced by their ability to handle noisy data more gracefully than residual-based models. Since the energy functional represents a global property of the system, it acts as a natural regularizer against local measurement errors. Research has shown that physical regularizations based on conservation laws, when properly integrated into the variational framework, can recover robust performance even when training data contains significant noise, whereas standard residual minimization might overfit to these errors [151]. Additionally, the use of mixed formulations, which combine strong and weak forms within an energy minimization context, has proven effective for thermo-mechanically coupled systems, offering a balanced approach that captures both local fluxes and global energy states [152]. This inherent robustness to noise complements the uncertainty quantification techniques discussed earlier, providing a dual layer of reliability for scientific models.

Finally, the flexibility of graph-based representations allows these energy-based principles to be applied to irregular domains and unstructured meshes, bridging the gap between traditional finite element methods and deep learning. By defining the variational problem on a graph where nodes represent mesh points and edges encode spatial relationships, methods like Physics-Informed Graph Neural Galerkin Networks can solve forward and inverse problems on complex geometries with greater scalability than fully connected networks [121]. The combination of variational principles with graph architectures thus represents a frontier in scientific machine learning, offering a pathway to models that are not only accurate but also physically grounded, stable, and capable of generalizing across diverse and complex scientific domains. While these energy-based formulations provide a theoretically stable foundation, their practical deployment still faces challenges related to optimization dynamics and data sampling, which are the focus of the subsequent subsection.

3.5 Robustness, Sampling Strategies, and Optimization Dynamics for Physics Models

3.5 Robustness, Sampling Strategies, and Optimization Dynamics for Physics Models

While the energy-based formulations discussed in the previous subsection provide a theoretically stable foundation by minimizing global variational functionals, the practical deployment of physics-informed machine learning models in scientific discovery remains contingent upon resolving critical challenges in optimization dynamics, sampling strategies, and data robustness. Even with superior loss landscapes derived from variational principles, the trajectory from formulation to robust solution is frequently hampered by the sensitivity to collocation point distribution, the stiffness of multi-scale systems, and the ubiquity of noisy observational data. A primary bottleneck in training Physics-Informed Neural Networks (PINNs) and related surrogate models lies in the selection and distribution of collocation points, which serve as the anchors for enforcing differential equation constraints. Traditional fixed or density-based sampling strategies often fail to capture the multiscale nature of physical phenomena, particularly in systems exhibiting sharp boundary layers or chaotic dynamics. To mitigate these issues, recent advancements have shifted towards adaptive and causality-aware sampling mechanisms. For instance, standard adaptive sampling that relies solely on residual loss magnitude can be insufficient, as it may neglect the temporal ordering inherent in dynamical systems. Addressing this, researchers have introduced novel adaptive causal sampling methods that explicitly incorporate temporal causality into the point selection process. This approach has been shown to significantly improve prediction performance, achieving accuracy gains of up to two orders of magnitude with fewer collocation points for high-order nonlinear partial differential equations such as the Cahn-Hilliard and KdV equations [153].

Complementing these sampling innovations, the optimization landscape of physics-informed models often presents rugged terrains riddled with local minima and stiffness, especially when dealing with multi-scale solutions where different spatial or temporal regions evolve at vastly different rates. In such scenarios, attempting to learn high-loss regions—such as sharp interior layers in convection-diffusion-reaction problems—early in the training process can lead to convergence failures, even when utilizing robust energy-based objectives. Counter-intuitively, strategies that prioritize easier, non-layer regions while downplaying difficult layer regions during initial training phases have proven more effective. This curriculum learning paradigm allows the network to establish a robust global approximation before refining its focus on complex local features, thereby preventing the model from getting stuck in poor local minima [154]. Similarly, lightweight curriculum strategies for distributing collocation points have demonstrated success in recovering full two-dimensional magnetohydrodynamic solutions from partial samples, significantly reducing training time while enhancing solution quality [155]. These findings underscore the necessity of dynamic, stage-wise optimization protocols over static training regimes to fully leverage the stability offered by advanced architectural designs.

Beyond optimization and sampling, the robustness of these frameworks against noisy sensor data and unknown measurement errors represents another critical frontier. Real-world scientific data is rarely clean; sensors inherently introduce noise that can distort the learned dynamics if not properly accounted for, potentially undermining the physical consistency ensured by variational principles. Classic filtering approaches often rely on strong assumptions about frequency characteristics that may not hold for complex physical processes. In response, physics-informed denoising models have been developed that leverage the algebraic relationships governed by underlying physical laws, such as second-order differential equations between sensor measurements. By integrating these physics constraints directly into the denoising objective, it is possible to recover clean signals from noisy inputs without requiring ground-truth clean data, a capability demonstrated effectively in domains ranging from inertial navigation to HVAC control [156]. Moreover, the presence of noise in both inputs and outputs necessitates rigorous uncertainty quantification (UQ). Bayesian approaches have been extended to PINNs and neural operators to quantify uncertainty arising from noisy spatial-temporal coordinates and input functions, ensuring that models remain reliable even when faced with significant measurement perturbations [157].

In scenarios where data is sparse or the system involves unresolved subgrid processes, stochastic parameterization and history-based closures offer a path forward. Traditional machine learning parameterizations often fail to capture the stochasticity inherent in chaotic systems like weather patterns. To address this, Bayesian history-based parameterizations utilizing Hamiltonian Monte Carlo Markov Chain sampling have been proposed. These methods incorporate memory effects to account for non-instantaneous responses and provide trustworthy uncertainty quantifications, successfully predicting skillful forecasts in models like Lorenz '96 even under noisy and sparse observation conditions [158]. Additionally, active learning strategies play a pivotal role in enhancing model robustness when data collection is expensive. Active operator inference techniques have been developed to judiciously sample high-dimensional trajectories, aiming to minimize mean-squared errors by actively reducing the impact of noise on the inferred low-dimensional dynamical models [159].

Finally, the integration of causal reasoning into the sampling and learning process helps distinguish true physical mechanisms from spurious correlations induced by noise. Causality-based learning approaches that alternate between identifying model structures and recovering unobserved variables have shown unique advantages in coping with indirect couplings and stochastic noise in complex turbulent systems [160]. By combining adaptive causal sampling, curriculum learning for multi-scale fields, and robust Bayesian or physics-constrained denoising techniques, the field is moving towards physics-informed models that are not only theoretically sound but also practically resilient in the face of the messy, noisy, and multiscale reality of scientific data. These advancements collectively form a robust toolkit for overcoming the optimization and data quality hurdles that have historically limited the application of deep learning in high-stakes scientific domains, setting the stage for the next generation of graph foundation models capable of generalizing across diverse scientific disciplines.

4 The Era of Graph Foundation Models: Pre-training, Scaling, and Prompting

4.1 Unified Self-Supervised Pre-training Strategies for Graphs

4.1 Unified Self-Supervised Pre-training Strategies for Graphs

The evolution of Graph Neural Networks (GNNs) has been defined by a critical paradigm shift from task-specific, supervised learning to versatile, self-supervised pre-training frameworks capable of capturing universal graph semantics. In the early stages of graph representation learning, models were predominantly trained on isolated datasets with abundant labeled data, severely limiting their transferability to out-of-domain scenarios or tasks with scarce annotations—a common constraint in scientific discovery. To overcome these limitations, the community has increasingly embraced self-supervised learning (SSL) as a foundational strategy, transitioning from single-task objectives to unified frameworks that synergize generative and contrastive learning philosophies. This evolution is the prerequisite for developing Graph Foundation Models (GFMs) that can generalize across heterogeneous domains, from molecular chemistry to social network analysis, without relying on extensive manual labeling, thereby setting the stage for the scalable architectures discussed in the subsequent section.

Initially, pre-training strategies were often fragmented, focusing either on node-level or graph-level tasks in isolation. Research demonstrated that naive strategies, which pre-train GNNs at the level of either entire graphs or individual nodes independently, frequently yield limited improvements and can even lead to negative transfer on downstream tasks [161]. Recognizing this disconnect, early unified approaches sought to capture both local and global representations simultaneously. For instance, Graph Contrastive Coding (GCC) emerged as a pioneering framework designed to capture universal network topological properties across multiple networks by framing the pre-training task as subgraph instance discrimination [33]. By leveraging contrastive learning to empower GNNs to learn intrinsic structural representations, GCC laid the groundwork for transferring knowledge across diverse graph domains, proving that pre-training on a collection of diverse datasets could achieve competitive performance compared to task-specific models trained from scratch.

Parallel to contrastive methods, generative approaches have gained significant traction, drawing inspiration from the success of masked modeling in natural language processing. A landmark development in this direction is the introduction of Masked Graph Modeling, exemplified by GraphMAE. Unlike traditional Graph Autoencoders that focus on reconstructing graph structures, GraphMAE shifts the objective to feature reconstruction with a high masking ratio, utilizing a scaled cosine error to ensure robust training [162]. This approach addresses critical issues in generative SSL, such as reconstruction objectives and training robustness, consistently outperforming contrastive baselines across numerous benchmarks. Theoretical analyses have further illuminated the efficacy of masked graph modeling, establishing close connections between generative reconstruction and contrastive learning, suggesting that node-level reconstruction implicitly performs context-level contrastive learning [163]. Building upon this, advanced variants like GraphMAE2 introduced decoding-enhanced strategies, employing multi-view random re-mask decoding and latent representation prediction to regularize feature reconstruction and improve discriminability [164].

To further unify the learning process, researchers have explored frameworks that integrate both generative and contrastive paradigms, recognizing their complementary nature. While generative models excel at capturing local details through reconstruction, contrastive methods are superior at extracting global structural information. The Graph Contrastive Masked Autoencoder (GCMAE) framework explicitly unifies these branches, allowing the model to exploit global information extracted by the contrastive branch to aid the generative reconstruction task [165]. Similarly, the PARETOGNN framework advances this unification by employing multiple gradient descent algorithms to reconcile different learning philosophies, actively learning from manifold pretext tasks to minimize conflicts and enhance task generalization [4]. These unified approaches demonstrate that combining disjoint yet complementary knowledge sources leads to more robust representations capable of handling diverse downstream applications.

Beyond simple binary combinations, recent innovations have focused on attribute-edge factorization and multi-pretext attention mechanisms to capture deeper semantic dependencies. The GPT-GNN framework introduces a generative pre-training strategy that factorizes the likelihood of graph generation into attribute generation and edge generation components [34]. By modeling the inherent dependency between node attributes and graph structure during the generative process, GPT-GNN effectively initializes GNNs with rich structural and semantic properties, significantly outperforming models without such pre-training on billion-scale graphs. Furthermore, to address the limitations of relying on single pretext tasks, the Multi-pretext Attention Network (MAN) was proposed to adaptively combine traditional augmentation-relied methods with graph-driven clustering, exploiting endogenous correlation information among samples to obtain more universal representations [166].

Addressing the complexity of real-world data, unified strategies have also expanded to handle heterogeneous graphs and dynamic masking schedules. The HGMAE model tackles the challenges of heterogeneous graphs by introducing metapath masking and adaptive attribute masking, alongside training strategies that incorporate positional feature prediction and target attribute restoration [167]. Additionally, the issue of uneven node significance in masking has been addressed by structure-guided masking strategies, such as StructMAE, which employs an easy-to-hard masking schedule based on structural scores to gradually guide the model in learning complex graph topology [168]. Other frameworks like UGMAE propose a unified approach incorporating adaptivity, integrity, complementarity, and consistency to resolve issues related to uneven node significance and unstable reconstructions [169]. Moreover, curriculum learning paradigms like MentorGNN have been introduced to automatically re-weigh graph signals across diverse structures, deriving an optimal curriculum for pre-training that minimizes human bias and maximizes generalization [40].

Collectively, these unified self-supervised pre-training strategies represent a fundamental shift towards building foundational graph models. By moving beyond single-task constraints and integrating generative reconstruction, contrastive discrimination, and multi-task attention mechanisms, these frameworks enable GNNs to learn transferable, high-quality representations from vast amounts of unlabeled data. This evolution provides the essential architectural and methodological backbone required for few-shot learning and zero-shot transfer in data-scarce scientific domains. Crucially, by establishing robust pre-training protocols, these strategies create the necessary conditions for the emergence of true Graph Foundation Models, where the focus shifts from designing specific pretext tasks to understanding how performance scales with model size, data volume, and task diversity—the central theme of the following section on neural scaling laws.

4.2 Neural Scaling Laws and Task Scaling in Graph Foundation Models

4.2 Neural Scaling Laws and Task Scaling in Graph Foundation Models

Building upon the unified self-supervised pre-training strategies discussed in Section 4.1, the emergence of Graph Foundation Models (GFMs) represents a paradigmatic shift from specialized, small-scale Graph Neural Networks (GNNs) to versatile architectures capable of generalizing across diverse domains. While Section 4.1 established the methodological backbone for learning transferable representations through generative and contrastive objectives, this section addresses the critical question of scalability: how do these models behave as we systematically increase model size, data volume, and task diversity? Central to this inquiry is the empirical validation of neural scaling laws within the graph domain, which describe predictable power-law relationships between model performance and key resources. Although scaling laws have been extensively characterized in natural language processing and computer vision, their manifestation in graph learning presents unique challenges due to the irregular, non-Euclidean nature of graph data. Recent empirical studies have begun to dismantle skepticism regarding GNN scalability, demonstrating that deep graph models benefit tremendously from increased scale, provided that scaling metrics are appropriately redefined for graph structures.

A foundational insight into graph scaling is the necessity of redefining data volume metrics. Unlike text or images, where token or pixel counts provide linear measures of data size, graphs exhibit highly irregular sizes and topological complexities. Research indicates that simply counting the number of graphs in a dataset is an ineffective metric for establishing scaling laws because it fails to capture the structural richness of individual instances. Instead, the total number of edges has been proposed as a more robust metric for data scaling, offering a unified view of scaling behaviors across fundamental tasks such as node classification, link prediction, and graph classification [170]. When utilizing edge count as the primary scaling variable, clear power-law relationships emerge, confirming that larger datasets yield diminishing but consistent returns in performance, mirroring trends observed in other modalities. Furthermore, investigations into molecular graphs have revealed that scaling model depth, width, and the diversity of pre-training datasets leads to substantial performance gains. For instance, scaling GNNs to one billion parameters and increasing dataset sizes eightfold has resulted in accuracy improvements exceeding 30%, challenging the notion that sparse operations in GNNs inherently limit their scalability [39].

Beyond the traditional axes of model and data size, the concept of "task scaling" has emerged as a critical frontier for Graph Foundation Models, serving as a conceptual bridge to the multimodal integration discussed in Section 4.3. Task scaling posits that increasing the diversity and number of pre-training tasks can serve as a potent alternative to merely increasing parameter counts, potentially enabling robust generalization with smaller, more efficient models. This hypothesis draws inspiration from findings in language modeling, where scaling the number of tasks to over a thousand significantly improved zero-shot generalization, suggesting that task diversity can substitute for massive model scales [171]. In the context of graphs, this implies that a GFM trained on a heterogeneous mixture of tasks—ranging from molecular property prediction to social network analysis and traffic forecasting—can learn transferable structural representations that generalize to unseen domains. The efficacy of task scaling relies on the model's ability to distill invariant graph properties across disparate domains, effectively creating a "graph vocabulary" of reusable patterns that facilitates the cross-domain reasoning required for the instruction-tuning paradigms explored next.

However, the realization of these scaling laws is not without constraints. The relationship between model size and performance is not always monotonic; phenomena such as "scaling law collapse" can occur, often attributed to overfitting when the model capacity outpaces the effective information content of the training data [170]. Moreover, the depth of graph models plays a distinct role compared to other domains; while deeper networks generally improve performance, the optimal depth is heavily influenced by the specific graph topology and the presence of long-range dependencies, requiring careful architectural design to avoid issues like over-smoothing. The interplay between model size and data quality is also paramount. Just as in language models where data curation strategies must account for compute budgets to avoid the diminishing utility of repeated low-quality data [172], graph pre-training requires principled data subsampling and curation. Model-agnostic subsampling methods that leverage graph topology, such as estimating edge importance via graph conductance, have shown promise in improving training efficiency without relying on pilot models [173].

Furthermore, the theoretical underpinnings of scaling suggest that the intrinsic dimensionality of the learned representations plays a crucial role in generalization, particularly in low-data regimes typical of scientific discovery. Pre-training has been shown to implicitly minimize the intrinsic dimension of the solution space, allowing large models to be fine-tuned effectively even with limited labeled data [174]. This suggests that task scaling in GFMs may work by guiding the model toward a low-dimensional manifold of universal graph features, thereby enhancing sample efficiency for downstream scientific tasks. Additionally, the trade-off between parameter scaling and data scaling must be optimized; recent resource models indicate that for composite tasks, resources allocated to subtasks grow uniformly as models scale, suggesting that multi-task graph learning can benefit from balanced scaling strategies [175].

In conclusion, the era of Graph Foundation Models is being defined by a nuanced understanding of neural scaling laws that extend beyond simple parameter counting. By redefining data metrics to account for graph irregularity, leveraging task diversity as a scaling axis, and addressing the complexities of overfitting and data curation, researchers are paving the way for GFMs that achieve human-level or superhuman performance in scientific discovery. The synergy between massive scale and diverse task exposure promises to unlock new capabilities in predicting molecular dynamics, designing novel materials, and simulating complex physical systems. Yet, achieving this potential requires more than just scaling; it necessitates efficient mechanisms to adapt these massive models to specific scientific queries and to integrate them with the rich textual knowledge inherent in scientific literature. This necessity drives the evolution toward the "Pre-train, Prompt, and Predict" paradigm and the integration of Large Language Models, where scaling laws meet multimodal alignment, as detailed in the following section on bridging modalities through graph-text alignment and instruction tuning.

4.3 Bridging Modalities: Graph-Text Alignment and Instruction Tuning

4.3 Bridging Modalities: Graph-Text Alignment and Instruction Tuning

While Section 4.2 established that scaling model parameters, data volume (via edge counts), and task diversity is fundamental to unlocking the potential of Graph Foundation Models (GFMs), the sheer scale of these models necessitates efficient and versatile mechanisms for downstream adaptation. Simply increasing scale does not automatically guarantee seamless interoperability with the rich textual knowledge inherent in scientific domains. Consequently, the convergence of Large Language Models (LLMs) and Graph Neural Networks (GNNs) represents a pivotal evolution, moving beyond isolated scaling efforts toward integrated, general-purpose systems capable of multimodal reasoning. A central challenge in this integration lies in the fundamental modality gap: LLMs excel at processing sequential, unstructured text, while graphs embody complex, non-Euclidean relational structures. Bridging this divide requires sophisticated alignment mechanisms that translate topological information into formats understandable by language models, alongside instruction-tuning paradigms that empower these models to reason effectively over graph structures. Recent advancements have categorized these integration strategies into distinct roles for LLMs, including acting as enhancers, predictors, or alignment components, each offering unique pathways for multimodal synthesis [176].

A primary approach to bridging modalities involves the utilization of Text-Attributed Graphs (TAGs), where node features are inherently textual, thereby providing a natural anchor for alignment. In this context, the UniGraph framework exemplifies a robust strategy for creating a cross-domain graph foundation model. By leveraging a cascaded architecture of Language Models and GNNs, UniGraph unifies node representations through text, allowing the model to generalize across diverse domains and unseen graph structures. This approach employs self-supervised training objectives, such as Masked Graph Modeling, to capture both local and global graph properties without relying on labeled data. Furthermore, UniGraph introduces graph instruction tuning using LLMs, which empowers the model with zero-shot prediction capabilities, effectively transferring knowledge from pre-training tasks to downstream scientific applications [8]. This methodology underscores the potential of using natural language as a universal interface to harmonize heterogeneous graph data, directly addressing the data scarcity issues highlighted in the previous discussion on scaling laws.

Beyond simple feature augmentation or unified text interfaces, more nuanced alignment techniques employ dedicated projectors to map graph embeddings directly into the token embedding space of LLMs. The GraphGPT framework illustrates this paradigm by introducing a dual-stage instruction tuning process accompanied by a lightweight graph-text alignment projector. This projector serves as a critical bridge, grounding textual instructions with structural graph knowledge. GraphGPT operates by exploring self-supervised graph structural signals and task-specific instructions, guiding the LLM to comprehend complex topological features. This alignment allows the model to adapt to various downstream tasks, demonstrating superior generalization in both supervised and zero-shot scenarios compared to traditional baselines [9]. Similarly, the LLaGA (Large Language and Graph Assistant) model reorganizes graph nodes into structure-aware sequences, which are then mapped to the token space via a versatile projector. This design enables LLaGA to retain the general-purpose nature of LLMs while maintaining high fidelity to graph structures, achieving state-of-the-art performance across multiple datasets and tasks [177].

Instruction tuning has emerged as a dominant mechanism for enabling LLMs to perform complex graph reasoning, effectively operationalizing the "task scaling" concepts discussed earlier. The MuseGraph framework advances this by proposing a diverse instruction generation mechanism that distills reasoning capabilities from powerful LLMs to create task-specific Chain-of-Thought instruction packages. By employing an adaptive input generation strategy to encapsulate key graph information within token limits, MuseGraph facilitates generic graph mining across different tasks and datasets. This approach ensures that the training process is both effective and generalizable, significantly enhancing the accuracy of downstream tasks while preserving the generative powers of LLMs [13]. Complementing this, the GraphTranslator framework addresses the dilemma of handling both pre-defined and open-ended tasks by training a translator to align pretrained graph models with LLMs. This translator converts node representations into tokens, allowing the LLM to make predictions based on natural language instructions, thereby providing a unified perspective for diverse graph analytics [30].

However, the efficacy of these alignment strategies depends heavily on how graph structures are encoded into natural language, a factor that can influence the scaling behaviors observed in Section 4.2. Research indicates that the choice of encoding method significantly impacts performance, with studies showing that optimizing graph-to-text encoders can boost reasoning accuracy by substantial margins [178]. For instance, some approaches convert graphs into graph-syntax trees or linearized sequences to facilitate processing by LLMs, enabling training-free graph reasoning through in-context learning [179]. Conversely, other studies suggest that LLMs may not always interpret structural prompts as intended, often treating them as contextual paragraphs rather than explicit relational maps. This observation highlights the necessity for careful prompt design and the development of specialized alignment modules that explicitly enforce structural understanding [180].

Furthermore, the integration of LLMs and graphs is not unidirectional; it also involves leveraging graph structures to enhance LLM reasoning capabilities, creating a synergistic loop. Frameworks like Graph Chain-of-Thought (Graph-CoT) augment LLMs by encouraging iterative reasoning on graphs, combining LLM inference with graph execution steps to answer complex queries grounded in relational knowledge [181]. Similarly, the GraphAdapter approach utilizes a GNN as an efficient adapter for LLMs, introducing minimal trainable parameters to model TAGs effectively while preserving the zero-shot inference capabilities of the base language model [182]. These hybrid architectures demonstrate that aligning modalities is not merely about translating data formats but about creating synergistic systems where the semantic depth of LLMs and the structural rigor of GNNs complement each other.

Despite these advances, challenges remain in ensuring that LLMs truly comprehend graph topology rather than relying on spurious correlations or superficial textual cues. Evaluations have shown that while LLMs can comprehend graph data described in natural language, they often struggle with complex structural reasoning tasks, particularly as graph size and complexity increase [183]. Consequently, future research in graph-text alignment must focus on developing more robust grounding mechanisms, such as the graph-text grounding projectors seen in GraphGPT, and refining instruction-tuning datasets to cover a broader spectrum of topological reasoning patterns. Addressing these alignment limitations is a prerequisite for the next evolutionary step: moving from static fine-tuning or basic prompting toward the dynamic "Pre-train, Prompt, and Predict" paradigm, which seeks to minimize objective gaps and maximize adaptability in scientific discovery, as detailed in the following section.

4.4 The "Pre-train, Prompt, and Predict" Paradigm for Graphs

4.4 The "Pre-train, Prompt, and Predict" Paradigm for Graphs

Building upon the graph-text alignment and instruction-tuning mechanisms detailed in Section 4.3, the field is undergoing a profound paradigmatic shift from the traditional "pre-train and fine-tune" workflow to the emerging "pre-train, prompt, and predict" framework. While fine-tuning has historically been the dominant strategy for adapting pre-trained models, it often suffers from a fundamental misalignment between the objectives of the pre-training phase and those of specific downstream applications. This objective gap can lead to negative transfer, where the rich structural knowledge acquired during pre-training is partially forgotten or distorted during the aggressive parameter updates of fine-tuning. To address these limitations—and to operationalize the efficient adaptation strategies necessitated by the scale discussed in Section 4.2—prompt tuning has emerged as a parameter-efficient alternative. By freezing the backbone pre-trained model and instead optimizing a small set of learnable vectors, or "soft prompts," this approach guides the model toward the target task without eroding its foundational knowledge. In the context of graph foundation models, this paradigm is particularly critical for scientific discovery, where data scarcity and the need for robust generalization across heterogeneous graph structures are paramount.

A central challenge in adapting prompt tuning from natural language processing to graph domains is the lack of a unified framework that can accommodate diverse graph pre-training methods and downstream tasks. Unlike text, where prompts can be naturally inserted as token sequences, graphs require sophisticated reformulations to bridge the disparity between pretext tasks (such as node classification or link prediction) and downstream applications. Recent advancements have proposed reformulating both pre-training and downstream tasks into a common format to facilitate this alignment. For instance, the SGL-PT framework introduces a strong and universal pre-training task that synthesizes the complementary merits of generative and contrastive self-supervised graph learning. By designing a novel verbalizer-free prompting function, SGL-PT successfully unifies the training objectives, allowing the model to reformulate downstream graph classification tasks in a format similar to the pretext task. This approach effectively mitigates the objective gap and has demonstrated superior performance over traditional fine-tuning methods, particularly on biological datasets where labeled data is extremely sparse [27].

To further enhance the adaptability of graph foundation models, researchers have explored multi-view contrastive prompting strategies. These methods leverage the inherent multi-view nature of graph data, constructing contrasts that capture relevant structural and attribute information from different perspectives. By integrating prompt tuning with multi-view graph contrastive learning, models can better capture global and local graph properties without relying on extensive labeled data. This strategy not only bridges the gap between pretexts and downstream tasks but also ensures that the learned prompts encode invariant features that are robust to structural perturbations [184]. The efficacy of such contrastive approaches is further supported by findings in vision-language models, where contrastive prompt learning has been shown to prevent the learning of spurious representations and improve generalization to unseen concepts by simultaneously employing counterfactual generation and contrastive optimization [185]. Although originally developed for vision-language tasks, the underlying principle of using contrastive signals to refine prompt representations is directly transferable to the graph domain, offering a pathway to more robust scientific models.

Another significant innovation within this paradigm is the development of dual-prompt mechanisms designed to align disparate modalities or task representations, extending the alignment concepts introduced in Section 4.3. In scenarios where graph data is accompanied by textual attributes or when integrating graph structures with large language models, maintaining alignment between different modalities is crucial. Dual-modality prompt tuning approaches, which learn text and visual (or structural) prompts simultaneously, have proven effective in preventing the misalignment that often occurs when tuning only a single branch. By generating class-aware visual prompts dynamically through cross-attention between text prompts and graph node embeddings, these mechanisms ensure that the final representations concentrate on target concepts while preserving the pre-trained model's general knowledge [186]. Furthermore, methods like Multi-modal Deep-symphysis Prompt Tuning (MuDPT) extend this concept by learning a transformative network that allows for deep hierarchical bi-directional prompt fusion, thereby achieving a synergistic alignment of representations that outperforms uni-modal tuning strategies [187].

The shift to prompting also necessitates careful consideration of initialization and optimization dynamics, especially in few-shot scientific settings where data is limited. Poor initialization can lead to convergence issues or overfitting, undermining the benefits of the foundation model. Strategies such as gradient-regulated meta-prompt learning have been proposed to jointly learn an efficient soft prompt initialization and a lightweight gradient regulating function. This ensures that the prompts adapt quickly to new tasks while maintaining strong cross-domain generalizability, a feature essential for applying graph foundation models to novel physical systems [188]. Additionally, the use of model ensembles and sample-specific knowledge transfer has shown promise in improving few-shot performance by leveraging the collective competence of source prompts rather than relying on a single fused representation [189].

Ultimately, the "pre-train, prompt, and predict" paradigm represents a maturation of graph machine learning, moving towards models that are not only powerful but also adaptable and efficient. By minimizing the objective gap through verbalizer-free functions, leveraging multi-view contrasts, and employing dual-prompt mechanisms, this paradigm enables graph foundation models to retain their pre-trained knowledge while seamlessly adapting to the diverse and complex demands of scientific discovery. While this framework provides a robust mechanism for parameter-efficient adaptation, it serves as a critical precursor to the even more data-frugal regimes discussed next. As the field progresses toward scenarios where even prompt optimization may be too costly or data-intensive, the focus shifts to zero-shot and few-shot learning via in-context reasoning, where models must adapt solely through contextual cues without any parameter updates, as detailed in the following section.

4.5 Zero-Shot and Few-Shot Learning via In-Context Reasoning

4.5 Zero-Shot and Few-Shot Learning via In-Context Reasoning

While the "pre-train, prompt, and predict" paradigm detailed in the previous subsection establishes a parameter-efficient framework for adapting Graph Foundation Models (GFMs), the extreme data scarcity characteristic of scientific discovery often necessitates capabilities beyond standard prompt tuning. In domains where labeled data is prohibitively expensive or entirely absent, the field is increasingly relying on zero-shot and few-shot learning driven by in-context reasoning. Unlike traditional supervised learning or even fine-tuned prompting, which require parameter updates or extensive labeled support sets, in-context reasoning enables models to adapt to unseen graph structures and novel scientific tasks solely by leveraging contextual information provided at inference time. This capability is increasingly interpreted through the lens of implicit Bayesian inference, where the model infers a latent task distribution from a limited set of demonstrations without altering its weights. Recent theoretical advancements suggest that large language models (LLMs), when adapted for graph tasks, operate as latent variable models that implicitly perform this Bayesian inference to select optimal strategies based on provided demonstrations [190]. This perspective provides a rigorous mathematical grounding for why in-context learning succeeds in graph domains, suggesting that the "context" serves as a proxy for updating the posterior distribution over task-specific parameters, thereby bridging the gap left by static pre-training.

In the specific realm of graph-structured data, meta-learning has emerged as a critical mechanism to facilitate this rapid adaptation, addressing the heterogeneity of label spaces and the scarcity of examples for rare phenomena such as novel protein functions or rare disease subtypes. To bridge the gap between auxiliary graph datasets and target tasks with limited samples, frameworks like GSHOT have been developed. GSHOT employs a meta-learning framework specifically designed for few-shot labeled graph generative modeling, allowing the model to transfer meta-knowledge from similar auxiliary domains and quickly adapt to unseen graph datasets through self-paced fine-tuning [191]. This approach is particularly vital for scientific discovery, where the ability to generate high-fidelity graphs from limited examples can accelerate hypothesis generation in drug discovery and materials science. Furthermore, the integration of graph structures into meta-learning extends beyond generation to classification and link prediction. For instance, the Gated Propagation Network (GPN) explicitly models the relationships between output dimensions (classes) as a graph, allowing prototypes to propagate messages between related classes to improve few-shot classification accuracy [192]. Similarly, the FAITH framework constructs hierarchical task graphs to capture correlations among meta-training tasks, thereby enhancing generalization to target tasks with disjoint label sets [193]. These methods collectively demonstrate that explicitly modeling task and class relationships via graph structures significantly outperforms independent task assumptions in few-shot settings, offering a structural complement to the semantic alignment sought in prompt tuning.

Beyond direct meta-learning on graph structures, the synergy between LLMs and graph learning has opened new avenues for data augmentation and synthetic data generation, effectively acting as a force multiplier for few-shot learning in data-scarce regimes. In many scientific applications, the bottleneck is not the model architecture but the absence of sufficient training instances. LLMs can be leveraged to generate synthetic training data that adheres to specific domain schemas, thereby enabling few-shot prompting strategies that rival fully supervised approaches. For example, in dialog state tracking, frameworks utilizing LLMs to synthesize natural, coherent dialogues with annotations have been shown to recover nearly 98% of the performance achieved with humanannotated data [194]. While this example originates from NLP, the principle is directly transferable to scientific graph tasks, where LLMs can generate synthetic molecular graphs or protein interaction networks conditioned on textual descriptions of physical properties. This capability allows researchers to overcome the “cold start” problem in new scientific domains by creating high-quality synthetic datasets that guide the foundation model’s adaptation, laying the groundwork for the principled data curation strategies discussed in the subsequent section.

Moreover, the concept of in-context learning has been extended to the “many-shot” regime, where the availability of larger context windows allows models to learn from hundreds or thousands of examples provided directly in the prompt. This evolution challenges the traditional boundaries of few-shot learning, demonstrating that models can effectively learn high-dimensional functions and override pretraining biases when provided with sufficient in-context examples, even without weight updates [195]. In the context of graphs, this suggests that future GFMs could ingest large subgraphs or entire small-scale scientific networks as context to perform zero-shot inference on larger, related systems. Additionally, specialized techniques such as Transductive Linear Probing have shown that transferring pretrained node embeddings via contrastive learning can outperform meta-learning methods in few-shot node classification, even in self-supervised settings where ground-truth labels are scarce [196]. This highlights a divergent yet complementary path to in-context reasoning: leveraging robust pretrained representations that require minimal adaptation, thus serving as a bridge between pure in-context methods and the adaptive prompting techniques required for ultra-low-resource regimes.

The robustness of these in-context and few-shot methods is further enhanced by strategies that address distributional shifts and data imbalance, ensuring reliability in volatile scientific environments. Bayesian approaches, such as Bayesian Task-Adaptive Meta-Learning, introduce balancing variables to adaptively weigh meta-knowledge against task-specific learning, ensuring robustness when unseen tasks differ significantly from the training distribution [197]. Similarly, methods like Instance Credibility Inference exploit the distribution support of unlabeled instances to iteratively refine classifiers in few-shot settings, providing a statistical backbone to the more complex neural approaches [198]. As the field moves toward autonomous scientific discovery, the integration of these in-context reasoning mechanisms—ranging from implicit Bayesian inference and hierarchical meta-learning to LLM-driven synthetic data generation—will be indispensable. They provide the flexibility required to navigate the vast, sparse, and dynamically evolving landscape of scientific data, enabling models to make reliable predictions and generate novel hypotheses even when faced with entirely unseen graph structures and minimal observational evidence, setting the stage for the advanced adaptive prompting and data curation techniques explored next.

4.6 Adaptive Prompting and Data Curation for Low-Resource Regimes

4.6 Adaptive Prompting and Data Curation for Low-Resource Regimes

Building upon the in-context reasoning and meta-learning mechanisms discussed in Section 4.5, which enable GFMs to adapt to novel tasks without weight updates, this subsection addresses the complementary challenge of optimizing the adaptation interface itself when data is not just scarce, but critically sparse. While in-context learning leverages demonstrations within the prompt to infer task distributions, ultra-low-data regimes in scientific discovery—such as rare disease modeling or novel material characterization—often demand more than just contextual examples; they require robust, parameter-efficient adaptation strategies that prevent overfitting while maximizing the utility of every available sample. In these scenarios, standard fine-tuning fails due to instability, and zero-shot inference lacks domain specificity. Consequently, the field is converging on synergistic strategies that combine adaptive prompting mechanisms with principled data curation to bridge the gap between static pre-training and dynamic scientific application.

A critical evolution from the implicit Bayesian inference described previously is the development of explicit adaptive prompting frameworks designed to generalize from limited examples without succumbing to the volatility of few-shot learning. Traditional prompt tuning, which learns soft prompts to condition frozen models, remains highly sensitive to initialization and prone to memorizing sparse label distributions. To mitigate this, recent advances introduce gradient-regulated meta-prompt learning, which jointly optimizes prompt initialization and a gradient regularization function to ensure cross-domain generalizability. For instance, the Gradient-RegulAted Meta-prompt learning (GRAM) framework demonstrates that by meta-learning an efficient soft prompt initialization using only unlabeled data, models can achieve robust adaptation across diverse tasks while preventing learnable tokens from overfitting to limited samples [188]. Similarly, the Self-sUpervised meta-Prompt learning framework with MEta-gradient Regularization (SUPMER) leverages self-supervised meta-learning to derive universal prompt initializations, effectively transforming raw gradients into domain-generalizable directions [199]. These approaches are particularly vital for scientific graphs, where the structural diversity of molecular or physical systems necessitates prompts that capture invariant topological features rather than transient label patterns.

Beyond gradient regulation, the architectural efficiency of the prompts themselves must be optimized to accommodate the computational constraints often present in scientific research. In scenarios where model scales are moderate or resources are limited, the inclusion of non-informative or negative prompt tokens can degrade performance. The XPrompt model addresses this by employing hierarchical structured pruning to eliminate detrimental tokens, thereby closing the performance gap between prompt tuning and full fine-tuning even at smaller scales [200]. Furthermore, decomposing prompts into low-rank matrices, as seen in Decomposed Prompt Tuning (DePT), allows for significant reductions in memory and time costs while maintaining competitive accuracy, a crucial factor for scaling these methods to large scientific graph datasets [201]. For black-box scenarios where backpropagation is infeasible, identifying optimal subspaces through meta-learning on similar source tasks can guarantee that prompt optimization yields high-performing results on target scientific tasks [202]. These techniques refine the "prompt" component of the adaptation loop, ensuring that the interface between the foundation model and the specific scientific task is both lightweight and highly expressive.

Parallel to refining the prompting mechanism, principled data curation stands as a cornerstone for enhancing model performance when the primary bottleneck is the absence of high-quality training signals. While Section 4.5 highlighted the potential of LLMs to generate synthetic data, the sheer volume of generated samples is insufficient; the quality and relevance of augmented data are paramount. Large Language Models have emerged as powerful engines for synthesis, yet their output requires rigorous filtering to ensure downstream utility. The Curated LLM (CLLM) approach illustrates a principled curation mechanism that leverages learning dynamics, confidence scores, and uncertainty metrics to filter LLM-generated tabular data, resulting in high-quality augmented datasets that outperform conventional generators in low-data regimes [203]. This concept of "superfiltering," where a smaller, weaker model is used to select high-quality instruction data for training a larger model, has also shown promise in accelerating the tuning process while improving final performance [204].

The interplay between data quantity, quality, and computational budget is governed by neural scaling laws, which dictate that data curation cannot be agnostic of available resources. As the utility of high-quality data diminishes upon repetition, strategies must dynamically balance the inclusion of unseen, lower-quality data to maximize model performance within specific compute constraints [172]. In scientific contexts, where data generation may involve expensive simulations or wet-lab experiments, iterative data enhancement strategies like LLM2LLM offer a viable path forward. This method iteratively fine-tunes a student model, identifies failure cases, and uses a teacher LLM to generate targeted synthetic data based on these errors, thereby amplifying the signal from challenging examples and significantly boosting performance in low-data settings [205]. This creates a feedback loop where the model actively guides the curation of its own training data, complementing the passive observation inherent in standard in-context learning.

Moreover, the integration of retrieval-based knowledge transfer further refines data utilization by optimizing how existing knowledge is accessed and applied. Techniques such as In-Context Sampling (ICS) optimize the construction of multiple prompt inputs to produce confident predictions, demonstrating that leveraging multiple prompts together can consistently enhance performance beyond single-prompt strategies [206]. Additionally, retrieval-based knowledge transfer paradigms enable extremely small-scale models to leverage knowledge stores constructed from large LLMs, effectively bridging the gap between massive pre-training and resource-constrained deployment [207]. By combining adaptive, gradient-regulated prompting with sophisticated, iterative data curation pipelines, researchers can overcome the severe barriers of data scarcity. These strategies not only enhance the sample efficiency of scientific AI but also ensure that the resulting models are robust, generalizable, and capable of driving breakthroughs in data-scarce domains, setting the stage for the next generation of autonomous scientific agents.

5 Trustworthiness, Interpretability, and Robustness in High-Stakes Science

5.1 Explainability and Counterfactual Reasoning in Graph-Based Scientific Models

5.1 Explainability and Counterfactual Reasoning in Graph-Based Scientific Models

The deployment of Graph Neural Networks (GNNs) in high-stakes scientific domains, ranging from molecular property prediction to climate modeling, necessitates a rigorous transition from opaque black-box predictors to transparent, interpretable systems. In these critical applications, understanding the "why" behind a model's decision is as paramount as the accuracy of the prediction itself. While predictive performance provides the foundation for scientific utility, true trustworthiness requires that models not only yield correct results but also offer intelligible rationales that align with domain knowledge. Consequently, the field has witnessed a surge in post-hoc and self-explainable frameworks designed to elucidate the decision boundaries of graph-based models. Among these, counterfactual reasoning has emerged as a particularly powerful paradigm, offering actionable recourse by identifying minimal perturbations to input graphs that would alter the model's output. This approach aligns closely with human cognitive processes, where understanding a phenomenon often involves exploring hypothetical "what-if" scenarios. However, generating effective counterfactuals for scientific graphs requires a delicate balance between plausibility, diversity, and adversarial robustness, ensuring that the generated explanations are not only mathematically valid but also physically meaningful and actionable. This focus on interpretability serves as a crucial precursor to the uncertainty quantification strategies discussed in the following section, as understanding the factors driving a prediction is often the first step in assessing its reliability.

Post-hoc explanation methods generally seek to identify subgraphs or features most correlated with a specific prediction. Yet, traditional attribution methods often fail to provide actionable insights or robust justifications in the presence of noise. Counterfactual explanations address these limitations by explicitly constructing alternative instances that flip the prediction, thereby defining the local decision boundary more precisely. For instance, in the context of GNNs, early methods focused on edge deletions to generate minimal counterfactuals, demonstrating that removing a small number of crucial edges could significantly alter predictions with high accuracy [208]. While effective, such approaches are often transductive and limited to specific perturbation types, failing to generalize to unseen data or incorporate edge additions which may be necessary for scientifically valid structures. To overcome these constraints, recent advancements have introduced inductive algorithms capable of predicting counterfactual perturbations directly, incorporating both edge deletions and additions to achieve better results and scalability [209].

A critical challenge in generating counterfactuals for scientific models is ensuring their plausibility within the underlying data manifold. In domains governed by physical laws, a counterfactual molecule or material structure that violates chemical valency or thermodynamic stability is useless, regardless of its ability to flip a classifier's decision. The generation of plausible counterfactuals requires models that respect the intrinsic distribution of the data. Techniques leveraging generative adversarial networks (GANs) and variational autoencoders have been adapted to ensure that generated counterfactuals remain within the realistic data distribution, avoiding the creation of artifacts that mislead interpretation [210]. Furthermore, the concept of "unjustified" counterfactuals—instances disconnected from ground-truth data—poses a significant risk in scientific validation. Research indicates that many state-of-the-art approaches fail to differentiate between justified and unjustified examples, potentially leading to erroneous scientific conclusions [211]. Therefore, robust frameworks must integrate mechanisms to verify that counterfactuals are continuously connected to valid scientific states.

Beyond plausibility, the diversity of counterfactual explanations is essential for providing a comprehensive view of model behavior. Relying on a single counterfactual instance may obscure alternative pathways to a different outcome, limiting the actionable recourse available to scientists. Frameworks based on determinantal point processes have been proposed to generate diverse sets of counterfactuals, ensuring that users are presented with multiple feasible actions rather than a singular, potentially suboptimal solution [212]. This diversity is particularly valuable in drug design or materials discovery, where multiple structural modifications might achieve a desired property, each with different synthesis costs or side effects. Moreover, the trade-offs between plausibility, the intensity of changes, and the adversarial power of the counterfactual must be carefully managed. Multi-objective optimization frameworks have been developed to balance these competing desiderata, ensuring that explanations are both realistic and sufficiently impactful to reveal model vulnerabilities [213].

Robustness against noise and adversarial attacks is another cornerstone of trustworthy explainability in science. Scientific data is often noisy, and explanations that are brittle to minor perturbations can lead to unstable interpretations. It has been shown that independently optimizing correlations for single inputs can lead to overfitting noise, resulting in explanations that do not align with human intuition or physical reality. To mitigate this, novel methods model the common decision logic of GNNs across similar input graphs, producing explanations that are naturally robust to noise and reflect the global decision boundaries rather than local artifacts [214]. Additionally, the stability of explainers under label noise is a critical concern; even minor levels of noise can substantially degrade the quality of generated explanations, necessitating robust training protocols and evaluation metrics [215]. Addressing these stability issues is intrinsically linked to the broader challenge of uncertainty quantification, as unstable explanations often correlate with high epistemic uncertainty in the model's predictions.

Finally, the integration of semantic knowledge and causal reasoning enhances the interpretability of counterfactuals. By representing data as semantic graphs and leveraging knowledge graphs to define conceptual distances, explanations can be aligned with high-level scientific concepts rather than low-level pixel or atom modifications [216]. Furthermore, combining sub-symbolic GNN approaches with symbolic reasoning allows for the incorporation of domain-specific constraints and causality, bridging the gap between statistical correlation and mechanistic understanding [217]. As the field evolves, the focus is shifting towards global counterfactual reasoning, which seeks representative counterfactual graphs that explain model behavior across entire datasets, offering high-level insights and reducing the cognitive load on human experts [218]. These advancements collectively pave the way for GNNs that are not only predictive but also trustworthy partners in scientific discovery, establishing a necessary foundation of transparency before proceeding to the rigorous quantification of uncertainty and out-of-distribution detection.

5.2 Uncertainty Quantification and Out-of-Distribution Detection

5.2 Uncertainty Quantification and Out-of-Distribution Detection

Building upon the interpretability frameworks discussed in the previous section, which elucidate the "why" behind model predictions, the next critical pillar of trustworthy scientific AI is quantifying the "how much" we can rely on those predictions. In high-stakes domains such as molecular dynamics, biological pathway modeling, and complex physical system simulation, a point estimate is insufficient; rigorous quantification of predictive uncertainty is essential to distinguish between reliable scientific insights and potential artifacts arising from model limitations or data noise. This necessity drives the adoption of sophisticated frameworks capable of disentangling aleatoric uncertainty, stemming from inherent data stochasticity and measurement noise, from epistemic uncertainty, which arises from model ignorance due to limited training data or distribution shifts. While the preceding section focused on generating intelligible rationales, this section addresses the statistical confidence required to act on those rationales, serving as a vital bridge to the privacy-preserving distributed learning paradigms discussed subsequently, where uncertainty estimation becomes even more complex due to data fragmentation.

A fundamental approach to addressing these challenges involves Bayesian frameworks that unify the treatment of both uncertainty sources. By incorporating probabilistic links and noise in node features to model aleatoric uncertainty, and utilizing probability distributions over model parameters to capture epistemic uncertainty, researchers can construct robust GNNs [219]. In specific scientific domains such as materials science, where interatomic potentials are learned to accelerate molecular dynamics, it is critical to detect unreliable predictions to prevent cascading errors in simulations. Advanced ensemble methods have been developed to produce calibrated uncertainty estimates for both energy and forces, ensuring that the predicted uncertainty correlates strongly with the expected error on diverse conformations far from equilibrium [220]. Furthermore, to enhance the efficiency of uncertainty estimation without sacrificing accuracy, novel single-model approaches like G-ΔUQ have been proposed. These methods extend stochastic centering frameworks to structured data, demonstrating superior performance in obtaining calibrated confidence indicators under various distribution shifts compared to traditional ensemble techniques [221].

Beyond parameter-based uncertainty, recent advancements have explored the information latent within the internal representations of neural networks. It has been demonstrated that the distribution of latent representations can serve as a proxy for epistemic uncertainty, effectively identifying out-of-distribution (OOD) inputs by measuring the "surprise" of observing specific hidden states [222]. This perspective is particularly valuable when computational constraints prevent the use of expensive sampling methods during inference. Complementing this, gradient-based approaches offer an orthogonal source of information for uncertainty quantification. By analyzing gradient metrics, researchers can derive confidence estimates that capture information distinct from output-based methods, thereby improving the probabilistic reliability of detectors in complex scenarios [223]. Extending this logic to graph-structured data, gradient-based uncertainty attribution methods have been introduced to not only quantify uncertainty but also identify the specific input regions contributing most to prediction instability, offering actionable insights for model refinement that synergize with the counterfactual explanations detailed previously [224].

For scientific applications requiring formal statistical guarantees, Conformal Prediction (CP) has emerged as a powerful, distribution-free framework. CP provides valid prediction sets that contain the true label with a desired probability, regardless of the underlying model architecture. However, the efficiency of these sets—measured by their size—is highly dependent on model calibration. Recent studies have shown that incorporating temperature scaling parameters into Bayesian GNNs within the CP framework can significantly reduce the inefficiency of prediction sets while maintaining validity [225]. This is particularly crucial in spatiotemporal domains, such as traffic forecasting or epidemic modeling, where distribution shifts are common. In these contexts, physics-informed structural causal models have been integrated with conformal prediction to upper bound coverage differences under distributional shifts, ensuring robust decision-making even when conditional distributions change [226]. The applicability of CP extends to Earth observation and environmental monitoring, where it offers a model-agnostic way to generate statistically valid prediction regions without requiring access to the underlying training data or model internals [227], [228].

The challenge of OOD detection is further complicated by the nuanced nature of uncertainty sources. While epistemic uncertainty is traditionally used to flag OOD samples, recent findings suggest that aleatoric uncertainty can also signal specific types of distribution shifts, particularly those involving ambiguous or ill-defined class labels. A principled Bayesian approach combining both uncertainty types with outlier exposure has been shown to outperform methods relying on a single source [229]. Moreover, the reliability of uncertainty estimates themselves must be scrutinized, as standard techniques using ReLU classifiers can fail to detect OOD data reliably due to the piece-wise affine nature of these networks and the saturation of activation functions [230]. To mitigate such failures, ensemble-based strategies and evidential deep learning have been employed to better disentangle uncertainties and provide more faithful representations of model ignorance, although care must be taken regarding the identifiability and convergence of second-order loss functions [231], [232].

In dynamic scientific systems, such as those encountered in reinforcement learning for robotic control or power grid optimization, disentangling uncertainties allows agents to accelerate exploration via epistemic uncertainty while learning risk-sensitive policies based on aleatoric uncertainty [233], [234]. Similarly, in node classification tasks on graphs, Bayesian uncertainty propagation mechanisms have been developed to embed uncertainty directly into the message-passing process, penalizing training examples with high predictive uncertainty and improving robustness against topological perturbations [235]. As the field moves towards autonomous scientific discovery, the integration of these diverse uncertainty quantification strategies—ranging from Bayesian ensembles and gradient analysis to conformal prediction and evidential learning—forms the bedrock of trustworthy AI. These methods ensure that models not only predict accurately and explainably but also possess the self-awareness to recognize their own limitations. This capability is a prerequisite for the secure, decentralized collaboration explored in the following section, where quantifying uncertainty across distributed, privacy-constrained data silos becomes paramount for maintaining system integrity.

5.3 Privacy-Preserving Federated Learning for Distributed Scientific Data

5.3 Privacy-Preserving Federated Learning for Distributed Scientific Data

While the preceding section established methods for quantifying model uncertainty and detecting out-of-distribution inputs to ensure predictive reliability, a parallel challenge in high-stakes scientific discovery lies in securing the data ecosystems from which these models learn. The accelerating pace of breakthroughs in biomedical research, genomics, and clinical diagnostics is increasingly driven by massive datasets, yet their utility is often severely constrained by stringent privacy regulations, ethical concerns, and the fragmentation of data across isolated institutional silos. In environments where the centralization of sensitive patient records or proprietary genomic sequences is infeasible due to legal frameworks like GDPR and HIPAA, Federated Learning (FL) has emerged as a transformative paradigm. FL enables collaborative model training where data remains localized, exchanging only model parameters or gradients. However, transitioning from centralized to distributed learning introduces new vulnerabilities regarding privacy leakage, security threats from malicious participants, and trust deficits in aggregation mechanisms. Consequently, robust scientific AI requires the integration of advanced cryptographic techniques and decentralized ledger technologies to complement the uncertainty-aware frameworks discussed previously.

A primary concern in collaborative scientific training is that the exchange of model updates, such as gradients, can inadvertently leak sensitive information about the underlying training data through inference or reconstruction attacks. To mitigate this, Differential Privacy (DP) has been widely adopted as a rigorous mathematical framework to quantify and limit privacy loss. By injecting calibrated noise into the learning process, DP ensures that the contribution of any single data point remains indistinguishable. In the context of medical imaging, for instance, researchers have demonstrated that applying differential privacy techniques to federated setups for brain tumor segmentation can effectively prevent model inversion attacks while maintaining high segmentation accuracy [236]. Furthermore, recent advancements have focused on optimizing the trade-off between privacy guarantees and model utility. Novel algorithms, such as differentially private similarity-weighted aggregation, have been developed specifically for multi-modal MRI analysis, offering an additional layer of protection during the global weight aggregation phase without compromising the robustness of the diagnostic models [237]. Similarly, concentrated differential privacy mechanisms have been proposed to provide tighter end-to-end privacy guarantees with minimal degradation to model convergence, addressing the high cost of privacy in utility-sensitive scientific applications [238].

While differential privacy protects against statistical inference, it does not inherently secure the communication channel against a curious or malicious aggregator who might tamper with model updates or attempt to reconstruct data from unencrypted gradients. Secure Multi-Party Computation (SMPC) and Homomorphic Encryption (HE) offer stronger cryptographic assurances by allowing computations to be performed on encrypted data. For example, in histopathology image analysis, a cluster-based SMPC framework enables hospitals to split and share model weights among peers such that no single entity can retrieve another's local update, thereby eliminating the risk of privacy leakage at the cost of increased communication overhead [239]. Moreover, fully homomorphic encryption has been utilized in neuroimaging studies to compute global models entirely within encrypted domains, preventing even the federation controller from accessing raw parameters and thwarting membership inference attacks [240]. Hybrid approaches combining SMPC with functional encryption have also shown significant efficiency gains, reducing training time and data transfer volume by substantial margins while maintaining strict privacy guarantees, making them viable for resource-constrained scientific environments [241].

Beyond cryptographic protections, the architectural reliance on a centralized server introduces a single point of failure and potential bottlenecks for trust and accountability, echoing the reliability concerns addressed in uncertainty quantification. Blockchain technology has been increasingly integrated to decentralize the aggregation process, ensuring immutability, transparency, and auditability of the training lifecycle. In scenarios involving multiple healthcare consortia, blockchain-enabled architectures facilitate secure model storage and parameter exchange without a trusted third party, effectively mitigating risks associated with single-point failures and malicious data poisoning [242]. Specific frameworks have been designed to detect and penalize dishonest behavior through smart contracts, utilizing reward-and-slash mechanisms to incentivize honest participation and deter poisoning attacks in distributed networks [243]. Furthermore, blockchain can serve as a provenance registry, logging data-model lineage to enhance accountability and fairness, which is critical when diverse institutions contribute heterogeneous data sources [244].

The synergy of these technologies is particularly evident in complex scientific applications requiring cross-silo collaboration. For instance, in genomic research involving high-dimensional Genome-Wide Association Studies (GWAS), federated singular value decomposition algorithms have been developed to handle privacy and computational constraints without exchanging sample-specific vectors [245]. Additionally, generative data exchange protocols utilizing blockchain and differential privacy allow institutions to share synthetic data generators instead of explicit gradients, enhancing resilience against inference attacks while improving task accuracy in cross-silo federations [246]. As scientific datasets continue to grow in volume and sensitivity, the convergence of federated learning with differential privacy, secure computation, and blockchain represents a foundational shift towards trustworthy, decentralized AI. These integrated frameworks not only safeguard patient confidentiality and proprietary scientific data but also foster a collaborative ecosystem where knowledge can be shared securely. This foundation of data privacy and system integrity is a prerequisite for the subsequent discussion on defending these systems against active adversarial attacks and data perturbations.

5.4 Robustness Against Adversarial Attacks and Data Perturbations

5.4 Robustness Against Adversarial Attacks and Data Perturbations

Building upon the foundations of privacy-preserving federated learning established in the previous section, which secures data ecosystems against passive inference and unauthorized access, this section addresses the more active and malignant threat of adversarial interventions. While federated learning safeguards the confidentiality of distributed scientific data, it does not inherently guarantee the integrity of the model against malicious actors seeking to manipulate outcomes. In high-stakes scientific domains such as molecular dynamics, climate modeling, and network intrusion detection, the deployment of Graph Neural Networks (GNNs) necessitates a rigorous examination of robustness against both intentional attacks and natural data perturbations. Unlike traditional Euclidean data, graph-structured data presents unique vulnerabilities stemming from the discrete nature of topological connections and the deep interdependence of node features. Adversarial threats in this context are generally categorized into poisoning attacks, which compromise the training phase, and evasion attacks, which target the model at inference time. Furthermore, the risk extends beyond mere performance degradation to include privacy breaches through model extraction and inversion, creating a complex trustworthiness challenge that bridges the concerns of the preceding privacy discussion and the forthcoming fairness analysis.

Poisoning attacks represent a critical threat where adversaries manipulate training data to embed backdoors or degrade generalization capabilities. In graph learning, this often involves modifying the graph structure or node attributes to mislead the learning process. Research has demonstrated that combining poisoning with evasion strategies can create symbiotic attack vectors that substantially improve the efficacy of adversarial interventions [247]. Specific studies have shown that attackers can poison training labels without altering the graph structure itself, injecting hidden backdoors that achieve near-perfect attack success rates while maintaining model functionality on benign samples [248]. Additionally, reinforcement learning agents have been employed to execute data poisoning attacks on graph classification tasks, exploiting the discrete nature of graph data where direct gradient optimization is challenging [249]. The vulnerability is not limited to static graphs; dynamic environments, such as those found in network intrusion detection, are susceptible to structural perturbations that can compromise system security even when feature-based defenses are robust [250].

Evasion attacks, conversely, aim to fool trained models by introducing subtle perturbations to input graphs, often exploiting the message-passing mechanism inherent in GNNs. Theoretical analysis reveals that attackers tend to add inter-class edges to corrupt node features, leveraging the smoothing effect of message passing to blur dissimilarities between nodes [251]. Gradient-based methods have been adapted to handle the discrete domain of graphs, with algorithms like Nettack demonstrating that even a few perturbations can significantly drop classification accuracy [252]. To improve efficiency, differentiable graph attack methods have been proposed, utilizing continuous relaxation to generate effective attacks with significantly reduced computational costs and memory footprints compared to meta-learning approaches [253]. Furthermore, black-box scenarios have been addressed through reinforcement learning frameworks that learn to inject malicious nodes or modify structures without access to model gradients, proving effective even with limited attack budgets [254] [255].

Beyond performance manipulation, the privacy of scientific data stored within graph models is under threat from model extraction and inversion attacks, creating a direct intersection with the privacy concerns discussed in Section 5.3. Adversaries can steal proprietary GNN models by querying them, effectively creating surrogate models that mimic the victim's behavior with high fidelity [256]. More insidiously, model inversion attacks aim to reconstruct sensitive training data, such as private edges representing personal associations or financial dealings, from model outputs or explanations [257] [258]. These privacy vulnerabilities are exacerbated by the release of post-hoc feature explanations, which can inadvertently reveal structural information about the training graph [259]. This overlap highlights that robustness and privacy are not isolated pillars; a model susceptible to inversion attacks undermines the guarantees provided by federated learning, just as a model vulnerable to poisoning may propagate biases that affect fairness.

To mitigate these diverse threats, a variety of defense mechanisms have been developed. Adversarial training remains a cornerstone strategy, where models are exposed to adversarial examples during training to enhance robustness. Advanced frameworks incorporate co-adversarial perturbations on both weights and features to flatten loss landscapes, thereby improving generalization and stability against sharp minima [260]. Decentralized adversarial training has also been proposed for multi-agent systems, leveraging the coordination power of linked agents to enhance robustness against heterogeneous attack models [261].

Randomized smoothing has emerged as a powerful tool for providing certified robustness guarantees. By treating the model as a black box and sampling noise, this method can certify robustness against structural perturbations. However, recent advancements propose gray-box certificates that exploit the specific message-passing architecture of GNNs to provide stronger guarantees against adversaries controlling entire nodes [262]. Extensions of this concept include node-aware bi-smoothing frameworks designed specifically to certify robustness against graph injection attacks, a previously unexplored vulnerability [263]. It is crucial, however, to ensure the integrity of the randomness sources themselves, as compromised random number generators can be exploited to falsify robustness certificates [264].

Other defensive strategies focus on graph structure learning and filtering. Methods like Pro-GNN jointly learn a clean structural graph and a robust model by enforcing properties such as low-rankness and sparsity, effectively filtering out adversarial perturbations [265]. Similarly, spectral methods like GARNET construct base graphs resistant to attacks by leveraging weighted spectral embedding and probabilistic pruning, offering scalability to large graphs [266]. Frequency-based approaches, such as Mid-pass filtering, have also shown efficacy by isolating robust mid-frequency signals from adversarial noise [267]. Furthermore, co-training frameworks that integrate structural and feature views can diversify sub-models, enhancing ensemble robustness without sacrificing performance on clean data [268].

Despite these advancements, the field faces an ongoing arms race between attackers and defenders. Comprehensive benchmarking efforts, such as the Graph Robustness Benchmark (GRB), are essential to standardize evaluation protocols and ensure that defense mechanisms are tested under realistic and diverse threat models [269]. As scientific discovery increasingly relies on autonomous graph-based systems, ensuring robustness against both malicious interventions and natural data perturbations remains a paramount challenge. Moreover, the measures taken to ensure robustness must be carefully balanced against other trustworthiness criteria; for instance, aggressive filtering of data to remove perturbations may inadvertently distort the representation of underrepresented groups, thereby introducing bias. This intricate interplay leads naturally to the next critical dimension of trustworthy scientific AI: fairness, bias mitigation, and the responsible curation of datasets.

5.5 Fairness, Bias Mitigation, and Responsible Dataset Curation

5.5 Fairness, Bias Mitigation, and Responsible Dataset Curation

While the previous section addressed the resilience of Graph Neural Networks (GNNs) against malicious adversarial interventions, this section turns to the subtler, yet equally critical, challenge of intrinsic bias and unfairness embedded within scientific data and models. The deployment of GNNs and Graph Foundation Models (GFMs) in high-stakes domains necessitates a rigorous examination of fairness, as scientific graph data inherently possesses complex topological dependencies that can propagate and exacerbate historical biases. Unlike standard machine learning tasks assuming independent and identically distributed (i.i.d.) data, the relational nature of graphs means that discrimination against a specific node can influence the representations and predictions of its neighbors, creating a cascading effect of unfairness throughout the network. Consequently, addressing fairness in scientific discovery requires moving beyond simple attribute removal to adopt sophisticated stochastic optimization frameworks, causal reasoning, and responsible dataset curation strategies.

A primary obstacle to achieving fair scientific models is the reliance on Empirical Risk Minimization (ERM), which optimizes for average performance and often fails to protect underrepresented subgroups or handle biased sampling distributions. In scenarios where training samples are drawn under biased conditions, such as Γ-biased sampling, standard ERM approaches may yield decision rules that perform poorly upon deployment [270]. To mitigate this, researchers have developed distributionally robust optimization (DRO) frameworks that minimize the worst-case risk across a family of test distributions, thereby ensuring robustness against sampling biases [270]. Furthermore, integrating fairness constraints into stochastic optimization has historically been challenging due to the incompatibility of complex fairness regularizers with mini-batch training. Recent advancements have introduced unified stochastic optimization frameworks, such as those based on f-divergence measures, which provide theoretical convergence guarantees while enabling fair empirical risk minimization even with small batch sizes [271]. These methods are particularly critical for scientific applications where data is often scarce and must be processed in stochastic settings to accommodate large-scale graph structures.

The impact of sensitive factors on model generalization extends beyond static attributes to include dynamic and latent factors of variation that may not be explicitly labeled. In many scientific contexts, sensitive factors causing model failure are unknown *a priori*, necessitating unsupervised discovery and intervention mechanisms. Approaches that discover these sensitive factors and apply interventions such as augmentation, coherence, and adversarial training have shown promise in improving robustness and reducing sensitivity to unobserved variations [272]. Moreover, the interplay between fairness and privacy is paramount, especially in biomedical and genomic graphs where sensitive information leakage poses severe ethical risks. Novel model-agnostic debiasing frameworks, such as MAPPING, utilize distance covariance-based constraints to simultaneously reduce feature and topology biases while confining the risks of attribute inference attacks, thus addressing the dual challenges of fairness and privacy in non-i.i.d. graph structures [273]. This intersection foreshadows the complex trade-offs discussed in the subsequent section, where optimizing for one pillar of trustworthiness may inadvertently compromise another.

Responsible dataset curation is equally critical, as algorithms inevitably learn and amplify the biases present in their training data. The debate between algorithmic bias and data bias highlights that neither DRO nor data curation alone offers a universal solution; rather, a synergistic approach is required [274]. One promising direction involves the generation of representative and fair synthetic data. By incorporating fairness constraints into self-supervised learning processes, generative models can produce unlimited synthetic samples that retain the structural relationships of the original data while controlling for historical biases regarding gender, race, or other protected attributes [275]. This approach allows scientists to train models on "worlds we strive to live in" rather than being constrained by biased historical records. Additionally, frameworks for generating synthetic data with specific bias combinations enable researchers to systematically analyze the impact of different bias types on model performance and fairness metrics, providing a controlled environment for stress-testing scientific models [276].

In the context of graph generation specifically, traditional unsupervised methods often suffer from representation disparity, where protected groups contribute less to the reconstruction loss and consequently receive higher errors. To address this, fairness-aware graph generative models like FairGen have been proposed, which jointly train label-informed generation modules with fair representation learning to mitigate disparity and boost downstream task performance through equitable data augmentation [277]. Furthermore, the curation of datasets must consider the "fair use" of group attributes; personalizing models with sensitive attributes can sometimes harm group-level performance if not managed with collective preference guarantees [278].

Finally, the construction of responsible datasets must account for the limitations of subgroup sample complexity. As the number of sensitive variables increases, the intersectional subgroups required for metric-fair learning may become too small to support accurate classification, necessitating human intervention in data collection and the adoption of individual fairness definitions where group fairness is statistically infeasible [279]. Guidelines for responsible curation must therefore include protocols for identifying data gaps, ensuring diverse representation across intersecting identities, and applying preprocessing techniques such as maximum entropy-based adjustments to controllably modify representation rates while maintaining proximity to the empirical distribution [280]. By integrating these stochastic optimization frameworks, causal interventions, and rigorous curation standards, the scientific community can develop Graph Foundation Models that are not only accurate but also equitable. However, as we will explore next, achieving this equilibrium requires navigating the intricate interdependencies and inevitable trade-offs among fairness, privacy, robustness, and explainability.

5.6 Interdependencies and Trade-offs Among Trustworthiness Pillars

5.6 Interdependencies and Trade-offs Among Trustworthiness Pillars

Building upon the specific strategies for fairness, bias mitigation, and responsible dataset curation discussed in the previous section, it becomes evident that optimizing for a single dimension of trustworthiness in isolation is insufficient and often counterproductive. The deployment of Graph Neural Networks (GNNs) and Graph Foundation Models (GFMs) in high-stakes scientific domains reveals a complex web of interdependencies among fairness, privacy, robustness, and explainability. Treating these dimensions as independent optimization objectives frequently leads to suboptimal or even detrimental outcomes, creating a multidimensional tension where enhancing one metric can inadvertently degrade others. This phenomenon, often described as a "three-way knot" of conflicting optimization vectors, implies that improvements in privacy, fairness, or predictive performance often come at the expense of the others [281]. In scientific discovery, characterized by scarce, noisy, and historically biased data, navigating these trade-offs is not merely a theoretical exercise but a critical prerequisite for the responsible governance frameworks detailed in the subsequent section.

A primary challenge lies in the antagonistic relationship between privacy preservation and fairness. Techniques such as differential privacy, which are essential for protecting sensitive biomedical or genomic data in federated learning settings, introduce noise that can disproportionately affect underrepresented groups. This noise injection may obscure the very features necessary for effective bias mitigation, leading to a scenario where rigorous privacy guarantees come at the expense of equitable outcomes [282]. Conversely, the pursuit of fairness through reweighting or adversarial debiasing can inadvertently increase the risk of membership inference attacks, thereby compromising privacy. Research indicates that while predictive performance and fairness might be jointly optimized in certain regimes, this is frequently achieved only by sacrificing data privacy, suggesting that one of the three vectors will inevitably be penalized regardless of the optimization strategy employed [281]. Consequently, compositional interventions that blindly stack privacy and fairness modules without accounting for their interaction can lead to unintended consequences, such as models that are formally private yet systematically discriminatory.

The interplay between explainability and fairness presents another layer of complexity, particularly regarding counterfactual reasoning. While explanations are often proposed as mechanisms to enhance trust and detect bias, they can themselves become sources of unfairness or privacy leakage. For instance, instance-based strategies for generating counterfactual explanations can expose training data to "explanation linkage attacks," where adversaries reconstruct sensitive information from the provided rationales [283]. Moreover, the transparency offered by explanations can be deceptive; different algorithms may produce widely divergent counterfactuals for the same input, allowing malicious agents to "fairwash" an unfair model by selectively presenting explanations that hide sensitive features [284]. This disagreement problem underscores that transparency does not automatically equate to accountability. In fact, providing explanations for biased models without addressing the underlying procedural biases can increase human agreement with unfair decisions, effectively amplifying the harm rather than mitigating it [285].

Robustness further complicates this landscape, introducing non-linear effects when composed with other trustworthiness functions. Efforts to enhance adversarial robustness, such as smoothing techniques or adversarial training, can alter decision boundaries in ways that exacerbate existing disparities across demographic groups. Conversely, fairness interventions that constrain the model to treat groups equally may reduce its capacity to learn robust features that generalize well under distributional shifts. For example, applying a bias mitigation algorithm followed by a post-hoc explanation method may result in explanations that are less faithful to the now-debiased model, confusing stakeholders about the true drivers of the prediction [286]. Empirical studies have confirmed that no single model type performs optimally across all dimensions of accuracy, fairness, explainability, and robustness simultaneously, highlighting the inevitability of trade-offs in model selection [287].

To navigate these intricate trade-offs, the scientific community must move beyond siloed metrics toward holistic evaluation frameworks. Traditional assessments focusing solely on accuracy-fairness or privacy-utility dyads fail to capture the systemic risks posed by such compositional failures. A more comprehensive approach involves the development of unified trust factors that quantify the collective responsibility of a model, integrating robustness, drift susceptibility, and fairness into a single actionable metric [288]. Additionally, causal inference offers a principled pathway to disentangle these dependencies, allowing practitioners to analyze how interventions on one pillar causally affect others within the ML pipeline [289]. By adopting a sociotechnical perspective that considers the broader deployment context—including the psychological impact of explanations on different user groups and the long-term societal feedback loops of biased predictions—researchers can design systems that balance these competing demands more effectively [290; 291].

Ultimately, achieving trustworthy AI in scientific discovery requires acknowledging that these pillars are not independent levers but interconnected components of a complex system. Future research must prioritize the development of auditing frameworks that explicitly test for these interdependencies, ensuring that improvements in one domain do not precipitate crises in another. Only through such holistic rigor can we ensure that Graph Foundation Models serve as reliable, equitable, and safe instruments for advancing human knowledge [292; 293]. Recognizing these inevitable trade-offs sets the stage for the next critical step: shifting from static optimization to dynamic, systemic governance. As explored in the following section, managing these tensions requires proactive vulnerability assessment, red teaming, and lifecycle management to actively steer AI trajectories toward societal good.

5.7 Red Teaming, Governance, and Lifecycle Management for Trustworthy AI

Building upon the complex interdependencies and inevitable trade-offs among trustworthiness pillars discussed in the previous section, the deployment of Graph Foundation Models (GFMs) in autonomous scientific discovery demands a shift from static optimization to dynamic, systemic governance. As these models transition from experimental tools to core components of high-stakes domains like drug design, materials science, and climate modeling, ensuring their long-term reliability requires a framework that extends beyond traditional performance metrics. This framework must integrate proactive vulnerability assessment through red teaming, rigorous threat modeling throughout the development lifecycle, and standardized auditing protocols to mitigate risks associated with opaque and uncontrollable AI systems. The concept of "violet teaming" offers a critical evolution in this domain, proposing an integrative approach that combines adversarial vulnerability probing (red teaming) with solutions for safety and security (blue teaming) while prioritizing ethics and social benefit [294]. Unlike conventional red teaming which focuses solely on breaking systems, violet teaming aims to steer AI trajectories toward societal good by managing risks proactively by design, a philosophy essential for GFMs that may influence fundamental scientific understanding and where the trade-offs identified earlier must be actively managed rather than merely accepted.

Effective governance requires embedding safety considerations into every stage of the model's existence, from data curation to deployment and monitoring, thereby addressing the systemic risks highlighted by compositional failures. The "hourglass model of organizational AI governance" provides a structured approach for translating ethical principles into practice, connecting governance requirements at environmental and organizational levels directly to the AI system lifecycle [295]. For scientific GFMs, this implies that governance cannot be an afterthought; it must be intrinsic to the architecture to prevent the degradation of one trust pillar while optimizing another. This aligns with the need for an end-to-end framework for internal algorithmic auditing, where each stage of development yields documentation that forms a comprehensive audit report, thereby closing the accountability gap often found in large-scale AI deployments [296]. Such documentation is crucial for tracing emergent issues back to their source, particularly when GFMs exhibit unexpected behaviors in novel scientific contexts that arise from the non-linear interactions of fairness, privacy, and robustness constraints.

Red teaming specifically for scientific graph models must address unique vectors of attack and failure that exploit these interdependencies, including data poisoning in collaborative sensor fusion and adversarial perturbations in molecular property prediction. Recent advancements in automated safety scenario red teaming, such as the ASSERT framework, demonstrate the efficacy of using semantically aligned augmentation and adversarial knowledge injection to generate diverse test suites that expose robustness gaps [297]. While originally developed for language models, these methodologies are adaptable to graph structures, where adversarial attacks might involve manipulating node attributes or edge connections to induce catastrophic failures in physical simulations or to exploit the tension between explainability and privacy. Furthermore, the application of red teaming frameworks to maritime autonomous systems highlights the necessity of both proactive (secure by design) and reactive (post-deployment) evaluations to uncover vulnerabilities ranging from poisoning to adversarial patch attacks [298]. In the context of scientific discovery, where models may control robotic laboratories or simulate nuclear interactions, such rigorous stress testing is non-negotiable to ensure that robustness enhancements do not inadvertently amplify biases or compromise data integrity.

Threat modeling must also evolve to handle the dynamic and distributed nature of scientific data, particularly in federated settings where privacy-preserving techniques might introduce new attack surfaces. The rise of decentralized physical infrastructure networks suggests a need for governance models that can operate across heterogeneous stakeholders, leveraging mechanisms like device onboarding and reward/penalty systems to ensure network-wide vigilance [299]. In collaborative scientific environments, where multiple institutions contribute data to train a global GFM, the risk of compromised agents or malicious data injection is significant, especially when differential privacy noise could be manipulated to obscure bias. Bayesian methods for trust in collaborative multi-agent autonomy offer a mathematical foundation for estimating the trustworthiness of data sources and tracks, allowing systems to dynamically adjust their reliance on specific inputs based on observed behavior [300]. Integrating such trust estimation into the governance framework allows for real-time mitigation of threats without halting the entire discovery process, effectively balancing the trade-off between open collaboration and secure operation.

Standardized auditing protocols are essential for comparing the safety and reliability of different GFMs and for verifying that holistic trust factors are maintained across updates. The Foundation Model Transparency Index proposes a set of fine-grained indicators to assess transparency across upstream resources, model details, and downstream use, revealing significant gaps in current disclosure practices regarding downstream impacts and redress mechanisms [301]. Adopting similar indexing for scientific GFMs would compel developers to disclose training data provenance, computational carbon footprints, and known limitations regarding physical law adherence, providing the necessary context to interpret the trade-offs discussed in Section 5.6. Moreover, quantitative AI risk assessments provide a methodology for evaluating existing models even without access to their original construction details, analogous to a home inspector assessing an already-built structure [302]. This is particularly relevant for the scientific community, which often adopts pre-trained models from external sources and must verify their suitability for sensitive applications where hidden biases or privacy leaks could have profound consequences.

Finally, the governance framework must support dynamic certification schemes capable of adapting to the evolving capabilities of autonomous systems and the shifting landscape of trust trade-offs. Traditional static certification fails to address the dynamism of software updates and cooperative behaviors in multi-agent scientific systems [303]. A multi-layer trust governance framework, incorporating continuous monitoring and adaptive certification, is required to ensure that GFMs remain safe and ethical as they learn from new data and interact with changing environments. By synthesizing red teaming, lifecycle governance, and standardized auditing, the scientific community can foster an ecosystem where Graph Foundation Models drive breakthroughs while maintaining the highest standards of trustworthiness and accountability. This holistic approach ensures that the extraordinary capabilities of AI enrich humanity without succumbing to the catastrophic risks posed by inadequate precaution, effectively operationalizing the balanced management of competing trust pillars [294].

6 Scalability, Efficiency, and Hardware-Accelerated Deployment

6.1 Distributed Training Paradigms and Communication Optimization

6.1 Distributed Training Paradigms and Communication Optimization

The application of Graph Neural Networks (GNNs) to scientific discovery increasingly demands the processing of billion-scale graphs, ranging from massive molecular interaction networks to complex climate simulation meshes. As the scale of these datasets outgrows the memory capacity of single accelerators, distributed training across multi-GPU clusters and supercomputers has become a necessity. However, unlike standard deep learning tasks involving dense tensor operations on regular grids, GNN training on scientific graphs is plagued by irregular memory access patterns, severe load imbalance due to power-law degree distributions, and prohibitive communication overheads. Consequently, the field has evolved from naive data parallelism toward sophisticated distributed paradigms that fundamentally rethink graph partitioning, parallelization strategies, and communication protocols to achieve scalability. While the following section addresses the challenges of heterogeneous CPU-GPU co-design within single nodes, this section focuses on the architectural advancements required to scale training across multiple interconnected devices.

A primary bottleneck in distributed GNN training is the massive feature communication required during the neighborhood aggregation phase. When a graph is partitioned across multiple devices, the dependencies of nodes near partition boundaries necessitate frequent data exchange, which often stalls computation. To mitigate this, advanced graph partitioning strategies have been developed to minimize edge cuts while balancing the computational load. Research indicates that standard partitioning algorithms often fail to account for the specific dynamics of GNN training, leading to heterogeneous data distributions and class imbalances that degrade convergence. Addressing this, [304] proposes an edge-weighted partitioning technique that minimizes total entropy to improve accuracy, coupled with asynchronous personalization phases to adapt models to local data distributions. Furthermore, [305] systematically evaluates how factors such as mini-batch size and graph type influence partitioning effectiveness, demonstrating that invested partitioning time is significantly amortized by reduced training times and memory footprints. For extremely large graphs where even loading the topology is infeasible, streaming partitioning approaches like those in [306] allow processing billion-scale graphs on commodity workstations by ingesting edge streams rather than loading full graphs into memory.

Beyond partitioning, the choice of parallelism paradigm—data versus model parallelism—plays a critical role in scaling. While data parallelism is the standard approach, it suffers from diminishing returns as the number of GPUs increases due to the synchronization overhead of aggregating gradients and the “neighborhood explosion” problem. To overcome these inherent limitations, novel paradigms such as split-parallelism have emerged. [307] introduces a paradigm that decouples the sampling and training processes, allowing for a more efficient distribution of work that outperforms traditional data parallel systems like DGL and Quiver. Similarly, [308] pioneers the use of layer-level model parallelism for full-graph training. By partitioning GNN layers across GPUs and utilizing historical vertex embeddings to handle dependencies, GNNPipe reduces communication volume by a factor proportional to the number of layers, achieving significant speedups over graph parallelism while maintaining convergence.

Communication-avoiding algorithms represent another frontier in optimizing distributed training. The core idea is to restructure the sampling and aggregation steps to minimize or eliminate cross-device data transfer. One effective strategy involves matrix-based bulk sampling, which expresses the sampling process as a sparse matrix multiplication (SpGEMM). [309] demonstrates that this approach allows for communication-free sampling when the graph topology fits in memory and enables efficient scaling via communication-avoiding SpGEMM algorithms when the graph is distributed. This method has shown to be significantly faster than existing distributed extensions on large benchmarks. Another innovative approach is the use of vertex-cut partitioning to enable communication-free training. [310] presents CoFree-GNN, which duplicates node information across partitions to preserve structure without requiring cross-GPU communication during the forward pass, employing a reweighting mechanism to maintain model accuracy despite the distorted graph distribution.

Furthermore, optimizing the interaction between sampling and communication is vital for both dynamic and static scientific graphs. [311] highlights that for dynamic graphs, a graph difference-based strategy can significantly reduce CPU-GPU transfer times and keep communication volume linear with input size, achieving substantial speedups on multi-node systems. In the context of static large-scale graphs, [312] introduces a family of parallel algorithms based on 1D to 3D sparse-dense matrix multiplications that asymptotically reduce communication costs compared to previous methods, successfully training on protein networks with over a billion edges. Additionally, [313] synergizes a new graph partitioning method with a highly optimized sampling kernel to eliminate most communication rounds in distributed sampling, doubling training speeds without accuracy loss.

Finally, the integration of caching and hierarchical memory structures is essential for managing the data movement bottleneck in large clusters. Systems like [314] utilize shared preloading and expansion-aware sampling to reduce reliance on low-bandwidth external server communications. Similarly, [315] employs a hierarchical graph partitioning mechanism and a unified multi-GPU cache to minimize PCIe traffic, enabling efficient training of billion-scale graphs on single machines. These collective advances in partitioning, parallelism, and communication avoidance lay the groundwork for scaling across multi-node environments. However, as the demand for computational density grows, simply adding more nodes is insufficient without addressing the internal inefficiencies of individual compute units. This necessitates a shift toward the heterogeneous computing architectures and CPU-GPU co-design strategies discussed in the subsequent section, which further unlock performance by leveraging the complementary strengths of diverse processing units within the cluster nodes.

6.2 Heterogeneous Computing and CPU-GPU Co-Design

6.2 Heterogeneous Computing and CPU-GPU Co-Design

While Section 6.1 established the architectural foundations for scaling GNN training across multi-node clusters via advanced partitioning and communication avoidance, scaling to billion-edge scientific graphs often encounters bottlenecks even within individual compute nodes. The limitations of pure GPU-centric approaches—specifically constrained device memory and high-latency PCIe data transfers—necessitate a shift toward heterogeneous computing architectures that leverage the complementary strengths of Central Processing Units (CPUs) and GPUs. Unlike the inter-node communication challenges discussed previously, this section focuses on intra-node co-design strategies that unify the abundant host memory and flexible control logic of CPUs with the massive parallel throughput of GPUs. This hybrid paradigm moves beyond simple task offloading, instead employing sophisticated unified memory protocols, dynamic load balancing, and architectural specialization to maximize resource utilization. These single-node optimizations serve as a critical bridge: they resolve the internal inefficiencies that limit the effectiveness of the distributed systems described above, while laying the groundwork for the out-of-core storage frameworks discussed in the subsequent section, which address scenarios where even combined CPU-GPU memory is insufficient.

A primary challenge in heterogeneous systems is the efficient management of data movement between CPU host memory and GPU device memory, where traditional protocols often induce significant GPU idle time. To mitigate this, recent advancements propose unified CPU-GPU protocols that instantiate multiple training processes in parallel across both architectures. By allocating specific training processes to the CPU to collaborate directly with GPU-based computations, these protocols significantly reduce data transfer overhead and improve overall platform efficiency. For instance, novel load balancing mechanisms have been developed to dynamically adjust workloads between CPUs and GPUs at runtime, accounting for the varying performance characteristics of each processor. Such approaches have demonstrated speedups of up to 1.41x on platforms where the GPU moderately outperforms the CPU, proving particularly beneficial for researchers with limited access to high-end GPU clusters [316]. Furthermore, the integration of asynchronous task scheduling allows for the overlapping of communication and computation, effectively hiding the latency of data transfers behind useful computational work. This is achieved by integrating GPU-aware communication into asynchronous tasks, which reduces synchronization time and increases concurrency, although care must be taken to mitigate fine-grained overheads in strong scaling scenarios [317].

Beyond simple task distribution, advanced co-design strategies explore the potential of heterogeneous Stochastic Gradient Descent (SGD) algorithms that explicitly exploit the differences in computational power and memory hierarchy between CPUs and GPUs. Standard practices often relegate CPUs to preprocessing roles, leaving vast computational resources underutilized. In contrast, heterogeneous asynchronous SGD algorithms, such as "CPU+GPU Hogbatch" and "Adaptive Hogbatch," combine small batch processing on CPUs with large batch processing on GPUs. These methods aim to maximize simultaneous resource utilization while balancing model update distributions to ensure statistical convergence. Experimental results indicate that such frameworks can achieve faster convergence rates and higher resource utilization compared to standard TensorFlow implementations on both on-premises servers and cloud instances [318]. The success of these algorithms relies heavily on the ability to perform principled exploration of the design space, often utilizing generic frameworks that exploit asynchronous message passing to bridge the computational disparity between the two processor types.

The complexity of heterogeneous systems also extends to the.network-on-chip (NoC) level, where traditional communication infrastructures often fail to handle the divergent requirements of CPU and GPU traffic efficiently. Specialized hybrid NoC architectures, incorporating both wireline and wireless links, have been designed to optimize on-chip traffic patterns arising from deep learning workloads. These designs have shown remarkable improvements, achieving up to a 2.2x increase in network throughput and a 1.8x reduction in latency compared to optimized wireline mesh NoCs, translating into significant energy-delay product savings for CNN training workloads [319]. Moreover, the concept of fine-grained coherence specialization offers a pathway to further efficiency. By enabling low-complexity independent specialization of individual coherence requests, modern heterogeneous systems can reduce execution time by up to 61% and network traffic by up to 99%, addressing the inefficiencies of explicit DMA transfers and rigid unified coherent memory interfaces [320].

In the context of graph-specific applications, systems like HyScale-GNN exemplify the potential of single-node heterogeneous architectures for scaling GNN training to billions of edges. By employing a two-stage data pre-fetching scheme and dynamic resource management, HyScale-GNN effectively utilizes both processors and accelerators to overcome memory limitations. Evaluations on CPU-GPU and CPU-FPGA architectures reveal speedups of up to 2.08x and 12.6x respectively, outperforming state-of-the-art multi-node systems using only a single node [321]. Additionally, the adoption of region template abstractions provides a unified interface for managing complex data structures across hybrid computing nodes, enabling performance-aware scheduling that minimizes the impact of data transfers while maintaining scalability in large-scale image and graph analysis tasks [322].

Finally, the co-design landscape is expanding to include specialized hardware accelerators and composable architectures that allow for domain-specific optimization. The Composable On-Package GPU (COPAGPU) architecture, for example, leverages multi-chip-module disaggregation to support maximal design reuse and memory system specialization, offering substantial gains in training and inference performance compared to converged GPU designs [323]. Similarly, the integration of FPGAs into heterogeneous pipelines offers unique advantages for embedded and edge scenarios, where energy efficiency is paramount. Hybrid FPGA-GPU acceleration methods have demonstrated superior energy efficiency and latency reduction over GPU-only solutions for various mobile-oriented neural networks, highlighting the versatility of heterogeneous co-design in diverse deployment environments [324]. As scientific discovery increasingly relies on massive, dynamic graph structures, the continued evolution of these heterogeneous co-design principles will be indispensable for unlocking the full potential of next-generation Graph Foundation Models. However, when graph sizes escalate into the terabyte range, surpassing even the combined memory limits of advanced heterogeneous nodes, the focus must shift from internal co-design to external storage integration. This transition motivates the discussion in the following section on out-of-core learning and storage-accelerated frameworks, which leverage non-volatile memory to extend the scalability horizon beyond physical RAM constraints.

6.3 Out-of-Core Learning and Storage-Accelerated Frameworks

6.3 Out-of-Core Learning and Storage-Accelerated Frameworks

While Section 6.2 detailed how heterogeneous CPU-GPU co-design alleviates memory capacity constraints through dynamic load balancing and unified protocols, the sheer scale of scientific graph data—ranging from massive biological interaction networks to high-resolution climate simulation meshes—often exceeds the combined memory limits of even the most advanced heterogeneous nodes. As graph sizes escalate into the terabyte range, surpassing both GPU memory and system DRAM, the field has pivoted toward out-of-core learning frameworks. These systems leverage high-performance non-volatile memory (NVM) and NVMe SSDs not merely as passive storage but as active, integrated components in the training pipeline. The primary challenge in this regime shifts from inter-processor coordination to overcoming the immense latency and bandwidth disparities between storage media and computational accelerators. Recent advancements demonstrate that with sophisticated system designs, single-node training on billion-scale graphs can achieve throughput comparable to in-memory systems, effectively democratizing access to large-scale graph learning without the prohibitive costs of distributed clusters.

A critical architectural shift in these storage-accelerated frameworks is the transition from CPU-centric to GPU-initiated data access. Traditional frameworks rely on the CPU to manage graph sampling, feature retrieval, and data transfer, creating a severe bottleneck where the GPU remains underutilized while waiting for data—a limitation that persists even in the heterogeneous setups discussed previously. To address this, novel systems have introduced GPU-initiated direct storage access mechanisms. For instance, the [325] system proposes a GPU-initiated asynchronous disk I/O stack that allows GPU threads to directly fetch graph data from SSDs. This design minimizes CPU intervention, utilizing only a fraction of GPU cores for I/O submission while leaving the majority available for computation, thereby achieving throughput that matches in-memory systems even for terabyte-scale graphs. Similarly, the [326] work introduces a GPU Initiated Direct Storage Access (GIDS) dataloader. By enabling GPU threads to bypass the CPU and fetch feature vectors directly from storage, GIDS eliminates expensive page-fault overheads and leverages GPU parallelism to tolerate storage latency, accelerating the training pipeline by up to 392 times compared to state-of-the-art CPU-managed dataloaders. The [327] further supports this paradigm by presenting a fine-grained software cache and high-throughput queues that enable massive numbers of concurrent GPU threads to issue I/O requests, significantly speeding up data-analytics workloads over CPU-initiated approaches.

Beyond direct access, the efficacy of out-of-core learning hinges on intelligent caching policies that mitigate the stochastic nature of graph neighborhood sampling. Naive caching strategies often fail to capture the access patterns inherent in multi-hop graph traversal, leading to excessive I/O churn that negates the benefits of fast storage. The [328] system addresses this by restructuring the training pipeline into separate sampling and gathering stages. This separation enables the implementation of Belady's algorithm, a provably optimal replacement policy for caching feature vectors, which drastically reduces data movement between main memory and disk. Ginex demonstrates that such theoretically grounded caching can yield a 2.11x higher training throughput on billion-scale datasets compared to extended SSD-based baselines. Complementing this, the [329] framework utilizes the entire storage hierarchy, including disk, to train graphs that fit within a single machine's SSD. By employing buffer-aware data orderings and partition caching, MariusGNN minimizes end-to-end training time and achieves accuracy comparable to in-memory training while operating up to 8 times faster than multi-GPU distributed solutions, highlighting the cost-efficiency of optimized single-node out-of-core systems. The foundational [330] work earlier established the viability of leveraging partition caching and interleaving data movement with computation to scale embedding training beyond single-machine memory limits.

To further enhance efficiency, modern frameworks employ deep pipelining and asynchronous I/O to overlap data loading with computation completely, ensuring that the high throughput of NVMe drives is fully exploited. The [325] system incorporates a deep GNN-aware pipeline that seamlessly integrates computation and communication phases, ensuring that GPU cycles are never idle waiting for data. This approach is echoed in the [331] work, which explores in-storage processing (ISP) architectures. SmartSAGE moves computation closer to the data by utilizing capacity-optimized NVM SSDs and an ISP-based co-design, effectively bypassing the memory bandwidth bottleneck and enabling large-scale training that was previously hampered by DRAM capacity constraints. Furthermore, the [332] system addresses full-graph training scalability by storing vertex data in CPU memory and employing a recomputation-caching-hybrid strategy. HongTu's deduplicated communication framework converts redundant host-GPU accesses into efficient inter-GPU transfers, significantly reducing communication overhead and enabling billion-scale full-graph training on multi-GPU servers.

The integration of these technologies—GPU-initiated access, optimal caching, and asynchronous pipelining—has fundamentally altered the landscape of scalable graph learning, bridging the gap between hardware limitations and algorithmic demands. Systems like [315] demonstrate that hierarchical graph partitioning and unified multi-GPU caching can automatically adapt to hardware specifications to maximize throughput on billion-scale graphs within a single machine. Additionally, the [333] system introduces a dynamic cache engine and improved graph partitioning to minimize feature retrieving traffic and cross-partition communication, outperforming existing systems by an average of 20.68 times. These advancements collectively prove that out-of-core learning is no longer a compromise for resource-constrained environments but a superior paradigm for handling the massive, sparse, and irregular data structures characteristic of modern scientific discovery. By effectively utilizing NVMe SSDs and computational storage, these frameworks enable researchers to train on datasets of unprecedented scale with single-node efficiency. This foundation of scalable data handling is essential before applying the algorithmic compression and condensation techniques discussed in the following section, which further optimize the computational burden of these massive models.

6.4 Algorithmic Efficiency, Graph Condensation, and Model Compression

6.4 Algorithmic Efficiency, Graph Condensation, and Model Compression

Building upon the storage-accelerated frameworks and out-of-core learning strategies detailed in Section 6.3, which enable the handling of terabyte-scale graphs on single nodes, the next critical frontier lies in reducing the intrinsic computational and memory complexity of the models themselves. While Section 6.3 addressed the "where" and "how" of data movement across storage hierarchies, this subsection focuses on the "what"—specifically, algorithmic transformations that shrink the problem size and streamline the computation prior to hardware deployment. As Graph Neural Networks (GNNs) and Graph Foundation Models (GFMs) scale to encompass billion-edge scientific networks and complex molecular dynamics, the computational burden of inference and training remains a primary bottleneck even with optimized I/O. Addressing these challenges requires a multi-faceted approach operating at the algorithmic level: graph condensation techniques that synthesize smaller, representative graphs; knowledge distillation frameworks that transfer graph intelligence to efficient Multi-Layer Perceptrons (MLPs); advanced quantization schemes tailored for irregular graph structures; and propagation mechanisms designed for sub-linear time complexity. These methods collectively prepare the ground for the specialized hardware accelerators discussed in Section 6.5, ensuring that the irregular and sparse nature of scientific graph data can be efficiently mapped onto next-generation computing substrates.

Graph condensation has emerged as a transformative paradigm for mitigating the scalability issues inherent in processing massive graph datasets, directly complementing the out-of-core strategies by reducing the data volume that must traverse the memory hierarchy. Traditional GNNs suffer from the "neighbor explosion" problem and high memory footprints when operating on large-scale adjacency matrices, challenges that persist even with efficient caching. Graph condensation addresses this by constructing a compact synthetic graph that preserves the essential structural and feature information of the original large graph, thereby enabling efficient training and inference without significant performance degradation. However, a significant limitation of early condensation methods was their transductive nature; they could only condense observed training nodes, failing to generalize to unseen inductive nodes, which necessitated retaining the original large graph during inference and undermining the benefits of condensation. To overcome this, recent advancements have introduced mapping-aware graph condensation, which explicitly learns a one-to-many node mapping from original nodes to synthetic nodes. This approach, exemplified by frameworks like

MCond, allows new nodes to be seamlessly integrated into the condensed graph structure, enabling direct information propagation on the synthetic graph. Such methods have demonstrated remarkable efficacy, achieving inference speedups of over 100x and reducing storage requirements by nearly 56x on large-scale datasets like Reddit, thus making inductive representation learning feasible on resource-constrained devices [334]. Furthermore, the efficiency of these condensed representations can be further enhanced by exploiting the sparse nature of graph operations. Algorithms that decompose matrices into products of sparse matrices with entries restricted to integer powers of two can drastically reduce computational effort while maintaining high approximation accuracy, a technique applicable to both neural networks and large-scale graph operations [335].

Complementing graph condensation is the strategy of knowledge distillation, which aims to transfer the representational power of complex GNNs into simpler, more efficient architectures such as MLPs. This process is crucial for deploying models on edge devices where the irregular memory access patterns of graph convolutions are prohibitive, even when supported by the high-throughput pipelines described previously. The success of distillation often hinges on the compatibility between the teacher model's precision and the student model's hardware constraints. Quantization plays a pivotal role here, yet standard linear quantization often leads to substantial accuracy drops in lightweight models. Research into quantization-friendly architectures reveals that specific design choices, such as modifying separable convolutions to be more robust to low-bit representation, can close the accuracy gap between floating-point and 8-bit integer models [336]. Moreover, advanced distillation frameworks like Quantization-aware Knowledge Distillation (QKD) coordinate the quantization and distillation processes through phased training—self-studying, co-studying, and tutoring—to recover full-precision accuracy even at extremely low bit-widths (e.g., 3-bit or 4-bit) [337]. For graph-specific applications, exploring reduced-precision operations, such as half-precision arithmetic, has shown promise in leveraging modern GPU Tensor Cores to reduce memory usage and runtime for Graph Convolutional Neural Networks without compromising task accuracy [338].

Beyond architectural simplification, algorithmic modifications to the propagation mechanism itself offer pathways to sub-linear complexity, addressing the irregular degree distributions found in real-world scientific graphs, such as protein interaction networks or citation graphs. These distributions often lead to severe workload imbalances during message passing, creating bottlenecks that pure hardware acceleration cannot fully resolve. Addressing this requires dynamic workload rebalancing and splitting strategies that polarize graph neighborhoods into denser or sparser regions, allowing for specialized processing engines to handle each regime efficiently. Such algorithm-hardware co-design approaches can significantly reduce off-chip memory accesses and improve utilization, yielding massive speedups compared to traditional CPU and GPU implementations [339]. Additionally, techniques that replace expensive matrix-vector multiplications with simpler addition and thresholding operations, such as margin-propagation networks, provide an alternative route to ultra-energy-efficient inference suitable for tiny ML applications on IoT devices [340].

Finally, the integration of stochastic and differentiable quantization methods allows for the automatic learning of optimal mixed-precision strategies across different layers and modules of a graph model. By employing differentiable bitwidth parameters as probability factors, models can automatically search for globally optimized precision configurations that minimize computational cost while maximizing accuracy, outperforming static quantization schemes on various hardware backends [341]. These algorithmic efficiencies, ranging from structural condensation to precision-aware distillation and adaptive propagation, form the bedrock of scalable graph learning. They ensure that the next generation of Graph Foundation Models can operate effectively within the strict latency and energy budgets of scientific discovery workflows. Crucially, by reducing the computational footprint and regularizing data access patterns, these techniques lay the necessary groundwork for the specialized hardware accelerators and photonic computing technologies explored in the following section, which are designed to exploit these optimized algorithmic properties to their fullest extent.

6.5 Specialized Hardware Accelerators and Photonic Computing

6.5 Specialized Hardware Accelerators and Photonic Computing

While algorithmic optimizations such as graph condensation, quantization, and adaptive propagation (discussed in Section 6.4) significantly reduce the computational footprint of Graph Neural Networks (GNNs) and Graph Foundation Models (GFMs), the inherent irregularity and sparsity of scientific graph data continue to challenge general-purpose hardware. Traditional CPU and GPU architectures, optimized for dense matrix operations, often suffer from low arithmetic intensity and unpredictable memory access patterns when executing sparse graph convolutions. To fully realize the efficiency gains promised by algorithmic advances, a paradigm shift toward specialized hardware accelerators and emerging photonic computing technologies is essential. These custom solutions are designed to explicitly exploit the structural properties of graph workloads, ranging from row-stationary sparse-dense matrix multiplication engines to silicon-photonic co-processors that leverage the speed of light for energy-efficient inference.

A primary avenue for hardware specialization involves redesigning accelerators for Sparse-Dense Matrix Multiplication (SpMM) and Sparse-Dense General Matrix-Matrix Multiplication (GEMM), the fundamental kernels of GNN aggregation and combination phases. Conventional systolic arrays, though effective for dense deep learning, experience low utilization during sparse graph processing due to stalled cycles caused by zero-valued inputs. To address this, researchers have developed row-stationary architectures that optimize data reuse and minimize memory movement. For instance, the GROW accelerator utilizes Gustavson's algorithm to architect a row-wise product based sparse-dense GEMM engine, effectively co-designing software and hardware to balance locality and parallelism for Graph Convolutional Networks (GCNs) [342]. Similarly, the RAMAN accelerator employs a novel dataflow inspired by Gustavson's algorithm to maximize input and output activation reuse, significantly reducing partial sum writeback traffic and enabling efficient inference on resource-constrained edge devices [343]. These designs demonstrate that aligning hardware dataflow with the specific algebraic properties of sparse graph operations yields substantial improvements in energy efficiency and throughput compared to generic dense accelerators.

Beyond electronic logic, silicon photonics has emerged as a transformative substrate for accelerating graph processing, offering superior bandwidth and energy efficiency by performing linear algebra operations in the optical domain. The unique ability of photonic circuits to execute matrix-vector multiplications at the speed of light with minimal heat dissipation makes them ideal for the massive scale of scientific graph models. GHOST represents a pioneering effort in this domain as the first silicon-photonic hardware accelerator specifically designed for GNNs. By separately implementing vertex-centric and edge-centric operations in the optical domain, GHOST achieves significant throughput and energy efficiency gains over state-of-the-art electronic accelerators, including GPUs and TPUs [344]. This approach is further validated by broader studies indicating that silicon-photonic accelerators can deliver at least a 10.2x improvement in throughput and a 3.8x enhancement in energy efficiency for both Large Language Models and graph processing tasks compared to conventional electronic platforms [345]. Furthermore, the integration of photonic components allows for novel architectures like SONIC, a sparse neural network inference accelerator that leverages silicon photonics to achieve up to 5.8x better performance-per-watt than state-of-the-art sparse electronic accelerators, highlighting the potential of optics to handle the sparsity inherent in scientific graph data [346]. Advanced concepts such as optical comb-based monolithic photonic-electronic linear-algebra accelerators are also being explored to support high-dimensional operations required by attention mechanisms in graph transformers, pushing the boundaries of computation density [347].

Complementing dedicated accelerators, heterogeneous Systems-on-Chips (SoCs) provide a flexible framework for the dynamic mapping of graph primitives to optimal compute units. These systems combine diverse processing elements—such as CPUs, GPUs, FPGAs, and AI Engines—to handle the alternating sparse aggregation and dense transformation steps characteristic of GNNs. The MaGNAS framework exemplifies this approach by employing a mapping-aware Graph Neural Architecture Search to identify model architectures and mapping strategies that maximize resource efficiency on heterogeneous MPSoCs, achieving significant latency speedups and energy savings compared to GPU-only deployments [348]. Similarly, heterogeneous architectures leveraging AMD Versal ACAP platforms have demonstrated the ability to accelerate GNN inference by dynamically assigning sparse primitives to Programmable Logic (PL) and dense primitives to AI Engines (AIE), resulting in superior performance over static CPU or GPU implementations [349]. The efficacy of combining FPGAs and GPUs in heterogeneous setups is further supported by evidence showing that hybrid acceleration methods can outperform standalone GPU acceleration in terms of both energy reduction and latency for deep learning tasks, suggesting a viable path for embedded scientific applications [324].

The landscape of specialized hardware also encompasses highly optimized sparse matrix kernels and flexible accelerator templates that adapt to varying sparsity patterns. Solutions like GE-SpMM introduce coalesced row caching and coarse-grained warp merging to optimize SpMM operations on GPUs, addressing the memory access bottlenecks that plague standard libraries [350]. Moreover, the concept of heterogeneous sparse accelerators, such as AESPA, proposes deploying multiple sub-accelerators within a single device to handle diverse workload characteristics, achieving better energy-delay products than homogeneous designs [351]. As scientific graphs grow in complexity and scale, the co-design of algorithms and these specialized hardware platforms—ranging from row-stationary electronic arrays to ultra-fast photonic co-processors and dynamic heterogeneous SoCs—forms a critical bridge to the next layer of scalability. This hardware foundation is particularly vital for the dynamic and mesh-based scientific simulations discussed in the following section, where the convergence of efficient acceleration and domain-specific decomposition will be central to deploying GNNs as viable, high-performance surrogates.

6.6 Scalability in Dynamic and Mesh-Based Scientific Simulations

6.6 Scalability in Dynamic and Mesh-Based Scientific Simulations

Building upon the specialized hardware accelerators and heterogeneous architectures discussed in Section 6.5, scaling Graph Neural Networks (GNNs) for scientific simulations requires addressing a distinct set of challenges rooted in the physical nature of the data. Unlike standard machine learning tasks, scientific domains such as computational fluid dynamics (CFD), structural mechanics, and subsurface flow modeling involve massive computational meshes with unstructured geometries that evolve dynamically over time. While Section 6.5 provided the hardware foundation to handle sparse and irregular operations, realizing the full potential of these accelerators for mesh-based physics demands a synergistic combination of algorithmic innovations, domain decomposition strategies, and adaptive resolution techniques. These software-level advances are essential to ensure that GNNs can operate efficiently within hardware constraints while maintaining the high fidelity and physical consistency required for scientific discovery.

A primary bottleneck in scaling mesh-based GNNs remains the memory limitation of modern accelerators, which often precludes training on full-scale industrial or scientific domains even with optimized hardware. To overcome this, domain decomposition has emerged as a critical strategy for distributing the computational load across the heterogeneous systems described previously. By partitioning large meshes into smaller subdomains, researchers can train models on manageable chunks while ensuring mathematical equivalence to training on the global domain under specific conditions. For instance, recent work has demonstrated the feasibility of training MeshGraphNets on 3D meshes containing millions of nodes by employing a domain decomposition approach that facilitates training on massive computational fluid dynamics problems, such as CO_2-capture simulations [126]. This approach not only enables the handling of larger graphs but also leverages parallelization across multiple GPUs or accelerator nodes, significantly reducing training time. Furthermore, distributed domain decomposition methods have been extended to physics-informed neural PDE solvers, enabling scalable inference on domains thousands of times larger than the training domain by leveraging pre-trained networks on smaller sub-regions [352]. These methods effectively decouple the problem size from hardware memory constraints, paving the way for solving large-scale partial differential equations (PDEs) in industrial settings.

Beyond spatial decomposition, handling the temporal evolution of dynamic systems requires mechanisms that can adapt to changing physical states without accumulating errors over long rollouts. A significant challenge in dynamic simulations is the multi-resolution nature of physical phenomena, where high-gradient regions require fine resolution while quiescent areas can be modeled coarsely. Uniform mesh approaches often waste computational resources by resolving the entire domain at the finest scale, creating unnecessary pressure on the memory bandwidth and compute units of the underlying hardware. To address this, adaptive strategies that jointly learn the physical evolution and the optimal spatial resolution have been developed. The Learning controllable Adaptive simulation for Multi-resolution Physics (LAMP) framework exemplifies this by utilizing a GNN-based actor-critic system to dynamically refine and coarsen the mesh during simulation, thereby devoting compute resources to highly dynamic regions and achieving substantial error reductions compared to uniform baselines [353]. Similarly, multiscale GNN frameworks that mimic conventional iterative multigrid solvers, coupled with adaptive mesh refinement (AMR), have shown promise in accelerating phase field fracture problems. These architectures employ downsampling and upsampling steps with skip connections to prevent over-smoothing, effectively managing the near-singular operators found in crack propagation scenarios [354].

The management of evolving structures and moving boundaries also benefits from difference-based update mechanisms and hybrid numerical-learning approaches. Traditional solvers often struggle with the computational cost of updating interactions in Lagrangian particle systems or deforming meshes, a task well-suited for the sparse acceleration capabilities highlighted in Section 6.5. Graph Network-based Simulators (GNS) have proven effective in learning local interaction laws through edge messages, allowing for generalization to new environments and particle counts significantly larger than those seen during training [355]. However, pure data-driven approaches can suffer from instability over long time horizons. To mitigate this, hybrid models that interleave learned surrogates with classical numerical methods have been proposed. For example, a hybrid GNS/Material Point Method (MPM) approach accelerates forward simulations by minimizing error on the surrogate while periodically invoking the MPM to satisfy conservation laws, achieving speedups of up to 24x compared to pure numerical simulations while maintaining physical validity [125]. Additionally, enhancing GNNs with components from standard Smoothed Particle Hydrodynamics (SPH) solvers, such as pressure and viscous force terms, has been shown to correct particle clustering instabilities and enable significantly longer and more stable rollouts in fluid dynamics [356].

Parallel surrogates for CFD and other continuum mechanics problems further illustrate the potential of scalable GNN architectures when aligned with efficient hardware execution. Multi-scale GNNs capable of encoding different spatial resolutions simultaneously have demonstrated the ability to infer long-term Navier-Stokes solutions across a range of Reynolds numbers, offering simulation speeds two to four orders of magnitude faster than traditional solvers [357]. These models are particularly effective in environmental fluid mechanics, where they can generalize to new domain geometries and complex boundary conditions inherent in oceanic and atmospheric processes [358]. Moreover, the integration of finite volume features and global geometry representations into graph convolutions has improved the accuracy of CFD surrogates by explicitly encoding cell characteristics like volume and face area, which are often ignored in standard message-passing schemes [359].

Finally, the scalability of these systems is increasingly supported by distributed training paradigms tailored for message-passing neural networks, which are essential for leveraging the heterogeneous SoCs and architectures discussed earlier. Techniques such as Nyström-approximation sampling combined with domain decomposition allow for the scaling of edge-based GNNs to graphs with hundreds of thousands of nodes, validating their efficacy on steady RANS simulations and Darcy flow datasets [360]. As scientific problems grow in complexity, the convergence of domain decomposition, adaptive resolution, and hybrid physics-learning frameworks will remain central to deploying GNNs as viable, high-performance surrogates for dynamic and mesh-based scientific simulations. However, demonstrating the true value of these scalable approaches requires more than just raw speed; it necessitates rigorous evaluation frameworks that account for scientific validity, reproducibility, and performance portability across diverse architectures, a topic we address in the following section.

6.7 Benchmarking Standards and Performance Profiling for Scientific Workloads

6.7 Benchmarking Standards and Performance Profiling for Scientific Workloads

While Section 6.6 detailed the algorithmic and architectural strategies required to scale Graph Neural Networks (GNNs) for dynamic and mesh-based simulations, the transition from theoretical scalability to practical deployment hinges on rigorous performance evaluation. As the field of GNNs and Graph Foundation Models (GFMs) matures towards high-stakes scientific discovery, a critical assessment reveals a significant disconnect between reported computational capacity and actual scientific utility. The primary challenge lies in the heterogeneity of modern computing infrastructures, where the adoption of diverse architectures to achieve exascale computing power has exacerbated the performance portability problem for scientific applications [361]. Although GNNs promise to accelerate simulations in fluid dynamics, materials science, and genomics, the lack of standardized metrics often leads to incoherent results that fail to provide clear guidance for designing the hardware and software architectures of tomorrow's supercomputing systems [361]. Consequently, there is an urgent need to establish an open repository of performance portability specifically tailored for graph-based scientific workloads, grounded in strict operating and reporting guidelines to ensure fair, comparable, and reproducible measures of efficiency.

A fundamental issue in current profiling practices is the over-reliance on theoretical peak performance or simplistic throughput metrics, such as Operations Per Second (OPS), which frequently fail to capture the complex reality of scientific graph workloads. It is trivial to construct benchmark scenarios that demonstrate massive speedups in specific metrics like sampling speed or energy efficiency while ignoring the metrics that truly matter for scientific validity, such as accuracy, convergence stability, and physical consistency [362]. For instance, an accelerator might achieve high throughput by reducing numerical precision or ignoring sparse data structures inherent in molecular graphs, thereby compromising the scientific integrity of the simulation. This pitfall is particularly acute in deep learning accelerators where optimizations targeting throughput can cause significant losses in model accuracy, rendering the acceleration useless for precision-dependent scientific tasks [363]. Therefore, operation-level performance profiling must evolve to include end-to-end evaluations that jointly consider accuracy, latency, and energy consumption, rather than isolating single dimensions of performance.

The gap between theoretical scalability and actual performance is further widened by the irregular control flow and complex memory access patterns characteristic of graph data, which pose distinct challenges compared to the regular grid-based simulations common in traditional High-Performance Computing (HPC). Scientific graphs often exhibit dynamic topologies and heterogeneous node attributes, leading to severe bottlenecks in memory bandwidth and interconnect latency. Studies on heterogeneous systems indicate that applications with random memory access patterns, typical of large-scale graph traversals, may suffer from performance degradation when relying solely on high-bandwidth memory without appropriate latency-hiding mechanisms [364]. Furthermore, the design of efficient code that scales well on supercomputers is a non-trivial task; processing speed alone is no longer sufficient, as the true metric of success is the "time-to-output" for a scientifically valid result [365]. This necessitates benchmarking suites that can simulate the full stack of scientific discovery, from data ingestion and graph construction to iterative message passing and final property prediction.

To address these challenges, the community must move towards unified benchmark suites that mirror the complexity of real-world scientific applications. Initiatives like MLPerf HPC have begun to address this by introducing benchmark suites for large-scale scientific machine learning that analyze data staging, algorithmic convergence, and compute performance jointly [366]. Similarly, the proposal for scalable evaluation methodologies, such as SAIH, emphasizes the need to augment problem sizes dynamically to understand performance trends as both data and models scale on HPC systems [367]. These approaches highlight that static benchmarks with fixed problem sizes are inadequate for evaluating the evolving landscape of GFMs, which must handle billion-edge graphs and dynamic temporal interactions. A robust benchmarking framework must therefore support scalability testing across diverse system sizes, ranging from single nodes to massive clusters, while accounting for the specific constraints of scientific domains, such as conservation laws and symmetry requirements.

Moreover, the evaluation of efficiency must extend beyond raw computational speed to include cost-effectiveness and environmental sustainability. The rising computational demands of modern AI systems pose serious environmental concerns, and progress in model efficiency has been impeded by practical challenges in evaluation and comparison across disparate hardware platforms [368]. A standardized arena for efficiency evaluation, similar to the "Pentathlon" proposed for natural language processing, is essential for scientific graphs. Such a platform would offer strictly controlled hardware environments and incorporate a suite of metrics targeting latency, throughput, memory overhead, and energy consumption, allowing researchers to make fair and reproducible efficiency comparisons [368]. This is crucial because the most efficient machine in terms of calculation time is not necessarily the most energy-efficient, and the optimal architecture often depends on the specific number of cores and processes employed [369].

Finally, the future of benchmarking in scientific graph learning requires a shift towards “benchmark science and engineering,” a discipline dedicated to establishing consistent metrics across multi-disciplines and developing meta-benchmarks to measure the benchmarks themselves [370]. This involves creating traceable and supervised learning-based benchmarking methodologies that can adapt to emerging computing paradigms, including quantum computers and neuromorphic accelerators [371]. By leveraging standardized execution traces, such as the Chakra schema, the community can facilitate agile benchmark creation and co-design of future AI systems, enabling the synthesis of synthetic workloads that protect proprietary information while accurately modeling future scientific scenarios [372]. Only through such rigorous, holistic, and standardized approaches can the field ensure that the deployment of Graph Foundation Models in scientific discovery is both computationally efficient and scientifically trustworthy.

7 Domain-Specific Applications Driving Scientific Breakthroughs

7.1 Accelerating Molecular Dynamics and Force Field Construction

7.1 Accelerating Molecular Dynamics and Force Field Construction

While generative models offer powerful mechanisms for designing novel molecular structures, as explored in the following section, their practical utility in scientific discovery is fundamentally anchored in the ability to accurately simulate and validate the physical behavior of these entities. Molecular Dynamics (MD) simulation stands as the cornerstone technique for elucidating the structural dynamics and thermodynamic properties of matter at the atomic scale. However, the widespread application of traditional MD is often constrained by the prohibitive computational cost associated with achieving long time-scale simulations using femtosecond integration steps, particularly when high accuracy is required. In conventional workflows, each simulation step necessitates numerous iterative computations to derive energy and forces from complex interaction potentials. To overcome these bottlenecks, the scientific community has increasingly turned to Graph Neural Networks (GNNs) to surrogate these expensive calculations, leading to the emergence of GNN-Accelerated Molecular Dynamics (GAMD). This paradigm shift enables the direct prediction of atomic forces given the system state—comprising atom positions and types—thereby bypassing the explicit evaluation of potential energy surfaces while maintaining rigorous physical fidelity [373].

The development of Neural Network Potentials (NNPs) has revolutionized this domain by offering quantum-mechanical accuracy at a fraction of the computational cost of *ab initio* methods. A prime example of this advancement is the Deep Potential Molecular Dynamics (DeePMD) framework, which utilizes a many-body potential generated by deep neural networks trained on first-principles data. By preserving natural symmetries such as rotation, translation, and permutation invariance, DeePMD achieves results indistinguishable from original quantum data while scaling linearly with system size [374]. The scalability of such models has been pushed to unprecedented limits; recent optimizations have enabled simulations of systems containing up to 100 million atoms with *ab initio* accuracy, reaching peak performances of 86 PFLOPS on supercomputers like Summit [375]. Further algorithmic and system innovations, including model tabulation and kernel fusion, have extended these capabilities to simulate systems with up to 17 billion atoms, opening new frontiers for studying mechanical properties in metals and semiconductor devices [376].

Despite these successes, ensuring the robustness and stability of NNPs over long simulation times remains a critical challenge. State-of-the-art models, while fast, can suffer from fidelity-scaling issues where unphysical force predictions accumulate, leading to simulation failure. To address this, recent works have integrated sharpness-aware minimization (SAM) into equivariant architectures like Allegro, resulting in the Allegro-Legato model. This approach enhances the smoothness of the loss landscape, significantly delaying the time-to-failure and allowing for stable simulations of larger atomic ensembles over extended durations [377]. Furthermore, stability-aware training frameworks, such as the Stability-Aware Boltzmann Estimator (StABlE), combine supervised learning with reference system observables to iteratively correct instabilities, ensuring that the learned potentials reproduce accurate structural and dynamic properties even with limited training data [378].

Beyond pure neural network potentials, hybrid approaches that combine the accuracy of NNPs with the efficiency of classical Molecular Mechanics (MM) have gained traction to address specific biomolecular contexts. The NNP/MM method, for instance, models specific regions of interest, such as a ligand in a protein complex, using high-fidelity neural potentials while treating the surrounding environment with efficient MM force fields. This strategy has demonstrated a five-fold increase in simulation speed, enabling microsecond-scale sampling for protein-ligand complexes that were previously inaccessible [379]. Similarly, the Grappa force field leverages graph attentional neural networks to predict MM parameters directly from molecular graphs, achieving chemical accuracy with the computational efficiency of traditional force fields, thus facilitating stable folding simulations of small proteins and stable dynamics of large biomolecules [380].

The construction of these force fields has also evolved towards end-to-end differentiable frameworks that eliminate the need for manual parameterization, bridging the gap between static structure prediction and dynamic simulation. Models like Espaloma utilize GNNs to perceive chemical environments and predict valence and nonbonded parameters continuously, allowing for the automatic generation of force fields that are self-consistently applicable to both biopolymers and small molecules [381]. This approach has been further refined in Espaloma-0.3, which reproduces quantum chemical properties across diverse chemical domains and supports accurate binding free energy predictions [382]. Additionally, the integration of electronic degrees of freedom and nonlocal effects, as seen in SpookyNet, allows these models to generalize across varying electronic states and chemical spaces, addressing limitations in earlier local models [383].

To further enhance data efficiency and transferability, novel pre-training strategies and featurization schemes have been introduced, laying the groundwork for more versatile foundation models. The Gaussian Multipole (GMP) featurization scheme provides a universal representation that interpolates between element types, enabling models to extrapolate to new elements with reasonable accuracy [384]. Moreover, denoise pre-training on nonequilibrium molecular conformations has proven effective in improving the transferability of equivariant GNNs, allowing models trained on small molecules to perform robustly on larger, more complex systems [385]. Collectively, these advancements demonstrate that GNN-accelerated MD is not merely a speedup tool but a transformative methodology that bridges the gap between quantum accuracy and macroscopic scales. By providing the high-fidelity dynamic trajectories and conformational ensembles necessary for validation, these methods create the essential physical substrate upon which the generative design strategies discussed in the next section rely.

7.2 Generative Models for De Novo Drug Design and Protein Engineering

7.2 Generative Models for De Novo Drug Design and Protein Engineering

While the preceding section established GNN-accelerated molecular dynamics as the essential physical substrate for validating atomic behavior, this section explores the complementary paradigm of *de novo* design, where generative artificial intelligence shifts the scientific workflow from the high-throughput screening of existing libraries to the creation of novel biological entities. This transformation is driven by sophisticated frameworks—including diffusion models, Generative Flow Networks (GFlowNets), and variational autoencoders (VAEs)—that navigate the vast, high-dimensional chemical and structural spaces inherent to biology. Unlike the simulation-focused approaches of Section 7.1, these models actively optimize for specific functional capabilities, target binding affinities, and physicochemical properties, thereby compressing the timeline from hypothesis to therapeutic candidate. However, the utility of these generated structures ultimately relies on the rigorous physical validation mechanisms discussed previously, creating a synergistic loop between design and simulation.

In the realm of small molecule drug design, the ability to generate molecules conditioned on specific protein targets represents a critical frontier. Traditional methods often struggle with the combinatorial explosion of chemical space, but modern graph-based generative models have introduced mechanisms to incorporate 3D structural information directly into the generation process. For instance, the development of target-aware architectures allows for the generation of ligands that exhibit high binding affinities to arbitrary protein targets by unifying sequential, topological, and geometrical protein representations [386]. Furthermore, the incorporation of explicit 3D protein-ligand contacts within relational graph architectures has been shown to significantly enhance ligand-target compatibility, producing molecules that are more stereochemically viable and recoverable in commercial databases compared to 2D-only methods [387]. To address the challenge of optimizing multiple competing objectives simultaneously—such as binding affinity, synthetic accessibility, and drug-likeness—researchers have deployed reinforcement learning frameworks that build molecules atom-by-atom within the constraint of a protein pocket [388]. Additionally, GFlowNets have emerged as a powerful tool for sampling diverse candidates proportional to their reward, with variants like TacoGFN demonstrating superior performance in generating novel hits with improved docking scores and synthesizability [389]. Recent advancements have further refined these approaches by introducing dynamic backtracking mechanisms to correct disadvantageous decisions during the generation process, thereby enhancing exploration in sparse reward domains [390].

Parallel to small molecule design, the field of protein engineering has witnessed a revolution through the application of equivariant diffusion models and flow matching techniques. The primary challenge in protein design lies in creating backbones that are not only novel and diverse but also "designable," meaning they can be realized by a valid amino acid sequence. Equivariant neural networks operating in 3D space have enabled the generation of oriented residue clouds that capture the complex geometric constraints of protein folding. Notably, models like Genie utilize discrete-time diffusion to generate protein backbones that surpass existing methods in terms of novelty, diversity, and designability [391]. The adoption of SE(3)-equivariant diffusion models has further solidified this progress, allowing for the principled generation of rigid body motions and frames that maintain physical symmetries essential for structural stability [392]. Beyond pure diffusion, flow matching paradigms such as FoldFlow offer stable and efficient training regimes for generating protein backbones up to 300 amino acids, leveraging Riemannian optimal transport to construct simple and stable flows over the SE(3) group [393].

A significant advancement in this domain is the shift from generating static structures to modeling dynamic conformational ensembles, which are crucial for understanding protein function and bridge the gap to the simulation techniques discussed in Section 7.1. Traditional molecular dynamics simulations are often computationally prohibitive for exhaustive sampling, but generative models now serve as efficient surrogates to populate these ensembles. For example, force-guided SE(3) diffusion models like ConfDiff incorporate physical priors to generate protein conformations that preserve high fidelity to equilibrium distributions while ensuring rich diversity [394]. Similarly, frameworks that repurpose accurate single-state predictors like AlphaFold into flow-based generative models, such as AlphaFlow, have demonstrated the ability to capture conformational flexibility and higher-order ensemble observables with faster convergence than replicate MD trajectories [395]. To manage the computational complexity of these large-scale systems, latent diffusion models have been proposed to embed proteins into condensed latent spaces before applying equivariant diffusion, effectively reducing modeling complexity while maintaining the capacity to generate novel, designable structures [396]. Furthermore, the acceleration of inference in these complex models has been addressed by learning latent representations of protein structure, which can reduce inference time by three-fold without sacrificing the quality of generated ligands [397].

The optimization of drug properties remains a central theme, with multi-objective strategies becoming increasingly sophisticated to ensure that generated candidates are not only theoretically optimal but also practically viable. Approaches that decompose molecules into substructures allow for fine-grained control and local optimization, enabling tasks such as scaffold hopping and R-group optimization that were previously difficult for monolithic generative models [398]. Moreover, the balance between exploring novel chemical space and exploiting known high-affinity regions is being managed through disentangled latent spaces in conditional VAEs, which allow for the independent optimization of properties like lipophilicity and synthetic accessibility [399]. The integration of active learning loops further refines this process, where generative models iteratively learn from molecular metrics and simulation data to uncover novel scaffolds significantly dissimilar to known compounds [400]. Collectively, these advancements illustrate a maturing ecosystem where generative models act as practical engines for discovering next-generation therapeutics. Yet, while biological design focuses on functional ensembles, a parallel challenge exists in the inorganic domain: identifying thermodynamically stable crystal structures within vast material spaces, a topic explored in the following section through the lens of high-throughput materials discovery.

7.3 High-Throughput Materials Discovery and Crystal Structure Prediction

7.3 High-Throughput Materials Discovery and Crystal Structure Prediction

While the preceding section highlighted the transformative role of generative models in designing biological entities, a parallel revolution is unfolding in materials science, where Graph Neural Networks (GNNs) and emerging Graph Foundation Models (GFMs) are shifting the paradigm from heuristic-driven experimentation to data-centric, accelerated screening of vast chemical spaces. The primary objective in this domain mirrors the challenges of protein engineering: identifying thermodynamically stable inorganic crystals, reticular materials for specific applications such as carbon capture, and complex substitutional alloys within an almost infinite search space. Traditional density functional theory (DFT) calculations, while accurate, are computationally prohibitive for exploring the billions of potential compounds that exist within the periodic table. Consequently, machine learning models trained on graph representations of crystal structures have become the cornerstone of modern materials informatics, enabling the rapid prediction of formation energies and stability with near-ab initio accuracy.

A critical advancement in this area is the development of universal graph networks capable of generalizing across diverse chemical and structural domains. Early efforts demonstrated that crystal-graph attention networks could predict thermodynamic stability from unrelaxed structures, yet these models often suffered from biases due to underrepresented elements in training data. To address this, researchers have computed additional data to balance chemical and crystal-symmetry spaces, resulting in universal networks that allow for the reliable exploration of the entire space of inorganic compounds [401]. These models have been successfully applied to screen billions of compounds, uncovering thousands of new materials on the convex hull of thermodynamic stability that were previously unknown. Furthermore, the introduction of pre-trained frameworks like CrysGNN has leveraged massive amounts of unlabeled crystal data to distill structural knowledge, which can then be injected into downstream property predictors to significantly enhance accuracy even with limited labeled data [402].

The challenge of predicting stability is not merely a regression problem but requires careful consideration of classification metrics that correlate with actual discovery success. Recent benchmarking efforts, such as Matbench Discovery, have highlighted a disconnect between global regression metrics and task-relevant classification performance, noting that accurate regressors can still yield high false-positive rates near the decision boundary of stability [403]. To mitigate this, advanced architectures like DeeperGATGNN have been developed to overcome the over-smoothing limitations of standard message-passing networks, enabling the training of very deep models (up to 30 layers) that capture complex physicochemical mechanisms essential for robust property prediction across multiple benchmarks [404]. Additionally, innovations in representation learning, such as the Periodic Complete Representation (PerCNet), ensure that global periodic information including dihedral angles is captured, resolving many-to-one mapping issues where distinct crystals previously shared identical representations [405].

Moving beyond simple inorganic crystals, high-throughput screening has been extensively applied to reticular materials, particularly Metal-Organic Frameworks (MOFs) and Covalent Organic Frameworks (COFs), for carbon dioxide capture. Similar to the generative approaches used for small molecules, the chemical space for these porous materials is effectively infinite, necessitating models that can propose novel, stable structures satisfying specific constraints. Generative Flow Networks (GFlowNets) have emerged as a powerful tool in this context, capable of sampling diverse reticular materials with high gravimetric surface areas and superior working capacities compared to existing databases [406]. Similarly, diffusion-based models like UniMat and MatterGen have demonstrated the ability to generate high-fidelity crystal structures across the periodic table, with MatterGen specifically offering fine-tuning capabilities to steer generation toward desired mechanical, electronic, and magnetic properties [407][408]. These generative approaches are often complemented by structure-agnostic models like MOFormer, which utilize text-string representations and self-supervised learning to predict properties without requiring explicit 3D structure optimization, thereby accelerating the screening process for hypothetical MOFs [409].

In the realm of alloy design and disordered systems, GNNs have proven instrumental in predicting the properties of substitutional alloys and high-entropy materials. By encoding atomic, bond, and global state attributes, frameworks like MegNet can predict formation energies and elastic moduli for systems with substitutional defects that were unseen during training, facilitating the identification of promising paths in alloy discovery [410]. The adaptability of foundation models is further exemplified by the fine-tuning of EquiformerV2, originally trained on the Open Catalyst Project, to infer adsorption energies on high-entropy alloys, demonstrating that knowledge learned from ordered intermetallics can be extrapolated to highly disordered solid solutions [411]. Moreover, the introduction of differentiable alchemical degrees of freedom in machine learning interatomic potentials allows for the smooth interpolation between compositional states, enabling the optimization of solid solution compositions toward target macroscopic properties [412].

Despite these successes, challenges remain in ensuring the synthesizability and dynamic stability of predicted materials, a concern that bridges the gap to the dynamic interaction modeling discussed in the subsequent section. Semi-supervised approaches, such as teacher-student deep neural networks, have been proposed to leverage large amounts of unlabeled data to better distinguish between stable and unstable candidates, significantly improving screening accuracy in large-scale generative design [413]. Furthermore, the integration of physical symmetries and constraints directly into generative models, as seen in physics-guided crystal generative models, has drastically increased the validity of generated structures, ensuring that a high percentage of candidates possess negative formation energies and are thermodynamically viable [414]. As the field progresses, the synergy between high-throughput screening, generative design, and foundation models continues to expand the frontiers of materials discovery. However, just as biological function relies on conformational ensembles rather than static structures, the next frontier in materials science involves moving beyond static lattice predictions to understand the dynamic behaviors and temporal evolution of these complex systems.

7.4 Learning from Conformer Ensembles and Dynamic Interactions

7.4 Learning from Conformer Ensembles and Dynamic Interactions

Bridging the gap between the static lattice predictions discussed in high-throughput materials discovery and the continuum simulations of fluid dynamics, the paradigm of molecular representation is undergoing a profound transformation. The field is shifting from static, two-dimensional graph topologies or single energy-minimized structures to dynamic, three-dimensional geometric representations that capture the intrinsic flexibility and temporal evolution of biomolecular systems. While static snapshots suffice for identifying thermodynamically stable crystals, they fail to account for the thermodynamic ensembles that govern biological function. In reality, proteins and small molecules exist as distributions of conformers, where the interplay between entropy and enthalpy dictates binding affinity, catalytic efficiency, and allosteric regulation. Addressing this limitation requires advanced machine learning architectures capable of learning from conformer ensembles and modeling time-dependent interactions with physical fidelity.

A critical advancement in this domain is the development of methods that explicitly encode and aggregate information from multiple 3D conformations rather than relying on a single static snapshot. Recognizing that a molecule's 3D shape and conformation are critical for biomolecular recognition, researchers have proposed end-to-end deep learning approaches that operate directly on small-molecule conformational ensembles. These networks leverage a hierarchical representation learning strategy where individual conformers are first encoded as spatial graphs, and subsequently, the sampled ensembles are represented as sets using attention mechanisms to aggregate over individual instances. This set-based learning allows the model to elucidate key conformational poses that drive molecular geometry and binding events, significantly enhancing virtual screening capabilities compared to methods that ignore conformational diversity [415]. Furthermore, to integrate 2D topological information with 3D geometric data, researchers have proposed E(3)-invariant molecular conformer aggregation networks. These models utilize a novel aggregation mechanism based on differentiable solvers for the Fused Gromov-Wasserstein Barycenter problem, effectively combining a molecule's 2D representation with multiple 3D conformers to outperform state-of-the-art property prediction methods [416].

Beyond static ensembles, capturing the dynamic trajectories of protein-ligand interactions is essential for understanding the kinetics of drug binding. The limitations of the traditional “lock-and-key” theory have spurred the creation of models that learn from molecular dynamics (MD) trajectories to seize flexibility information. For instance, spatial-temporal pre-training methods based on modified Equivariant Graph Matching Networks have been developed to capture the time-dependent geometric mobility of conformations along MD trajectories. By employing self-supervised tasks such as atom-level denoising and conformation-level snapshot ordering, these models grant encoder networks the capacity to predict binding affinity and ligand efficacy with remarkable accuracy, revealing a strong correlation between the magnitude of conformational motion and binding strength [417]. Similarly, graph-based deep learning models like Dynaformer have been curated on large-scale MD datasets to learn geometric characteristics from trajectories rather than static crystal structures. This approach has demonstrated state-of-the-art scoring and ranking power, successfully identifying novel hit compounds in virtual screening campaigns that static models missed [418].

To model these complex biophysical interactions with rigorous physical constraints, recent works have introduced multi-grained symmetric differential equation frameworks. These approaches address the challenge of accurate long-timescale simulation by incorporating physics-informed architectures that satisfy group symmetries. A notable example is the NeuralMD framework, which utilizes a BindingNet model satisfying group symmetry via vector frames and an augmented neural differential equation solver that learns trajectories under Newtonian mechanics. This principled approach facilitates numerical MD surrogates that achieve massive speedups over standard simulations while maintaining stability and accuracy in predicting protein-ligand binding dynamics [419]. Additionally, the need to model interactions across sets of objects, such as protein-drug bindings, has led to the development of deep models that enforce Permutation Equivariance. These models ensure that predictions remain consistent regardless of the ordering of inputs, enabling robust generalization to new objects and domains, a crucial feature for handling the variable sizes and compositions of biological complexes [420].

The integration of equivariance into transformer architectures has further revolutionized the handling of 3D molecular interactions. The Generalist Equivariant Transformer (GET) was proposed to universally represent arbitrary 3D complexes as geometric graphs of sets, allowing a single model to encode diverse molecule types such as proteins, small molecules, and RNA/DNAs. By utilizing bilevel attention modules that are E(3) equivariant, GET effectively captures both domain-specific hierarchies and domain-agnostic interaction physics, retaining fine-grained information across all levels of the molecular structure [421]. Moreover, to overcome the computational bottlenecks of explicit MD simulations while preserving dynamic details, innovative algorithms like the tiered tensor transform (3T) have been developed. Although not strictly a deep learning training method, 3T rapidly generates diverse protein-ligant complex conformations by decoupling structure transformation from system physics, achieving higher accuracy in active ligand classification than traditional ensemble docking [422].

Finally, the frontier of dynamic interaction modeling is expanding into zero-shot and generative realms. Frameworks inspired by simulated annealing, such as Str2Str, leverage amortized denoising score matching to perform zero-shot conformation sampling without reliance on target-specific simulation data, offering orders of magnitude speedup over long MD runs [423]. Complementing this, force-guided SE(3) diffusion models like ConfDiff incorporate physical prior knowledge to generate protein conformations with rich diversity and high fidelity, surpassing existing score-based methods in conformation prediction tasks [394]. Collectively, these advancements signify a decisive move towards dynamic, physics-aware graph learning, where the temporal evolution and ensemble nature of molecular systems are no longer approximations but central components of the learning objective. By establishing robust methods for simulating discrete particle dynamics and interaction forces, this section lays the necessary groundwork for the subsequent exploration of how these differentiable, physics-grounded concepts scale to solve Partial Differential Equations and simulate complex continuous fluid systems.

7.5 Solving Partial Differential Equations and Simulating Complex Fluids

7.5 Solving Partial Differential Equations and Simulating Complex Fluids

Building upon the principles of dynamic interaction modeling and physics-informed architectures introduced in the previous section, the application of Graph Neural Networks (GNNs) extends naturally to the continuum domain, specifically addressing the simulation of fluid dynamics and the solution of Partial Differential Equations (PDEs). These tasks represent some of the most computationally demanding challenges in scientific computing. Traditional numerical methods, such as Computational Fluid Dynamics (CFD) solvers based on Finite Element or Finite Volume methods, provide high fidelity but often incur prohibitive costs for real-time applications, iterative design optimization, or large-scale parameter sweeps. In response, the scientific community has increasingly turned to GNNs and differentiable graph architectures to construct efficient surrogate models. These data-driven approaches leverage the inherent ability of graphs to represent irregular meshes and unstructured particle systems, offering a pathway to accelerate forward simulations and solve complex inverse problems with unprecedented speed, thereby bridging the gap between discrete molecular dynamics and continuous field simulations.

A primary application of these technologies is the development of learned simulators for particle-based and fluid systems. The Graph Network-based Simulator (GNS) framework has emerged as a pivotal architecture in this domain, representing physical states as nodes and local interactions as edges to learn interaction laws through message passing. This approach allows models to generalize from training scenarios involving thousands of particles to test conditions with significantly higher particle counts and different initial configurations [355]. By discretizing the domain into material points and learning the latent interactions, GNS architectures have demonstrated the capability to predict particle kinematics in complex boundary conditions with errors typically within 5% of traditional Material Point Method (MPM) simulations, while achieving speedups of up to 5,000 times [424]. Such efficiency is critical for applications ranging from granular flow prediction in geotechnical engineering to the modeling of deformable materials in robotics, effectively scaling the concepts of interaction learning from biomolecular ensembles to macroscopic physical systems.

Beyond pure surrogate modeling, the integration of differentiable programming has revolutionized the approach to inverse problems in fluid dynamics. Inverse problems, which involve inferring unknown material parameters or boundary conditions from observed data, are notoriously difficult for traditional solvers due to their non-differentiable nature and high computational cost. Differentiable graph networks enable the use of gradient-based optimization to solve these problems efficiently. For instance, differentiable GNS frameworks have been successfully applied to inverse analysis of granular flows, such as landslides, where they identify material properties like friction angles by minimizing the error between predicted and target runout distances [425]. This capability extends to fluid dynamics, where differentiable simulators allow for the identification of spatially varying viscosity and conductivity fields from sparse velocity observations, effectively embedding deep neural networks within the PDE model to recover complex physical fields [426].

To further enhance physical consistency and generalization, hybrid approaches that combine learned surrogates with traditional numerical methods have gained traction. These methods aim to retain the speed of neural networks while enforcing conservation laws and physical constraints inherent in classical solvers. A notable example is the hybrid GNS/MPM approach, which interleaves MPM steps within GNS rollouts to satisfy conservation laws and minimize error accumulation, achieving a 24x speedup compared to pure numerical simulations while maintaining physical fidelity [125]. Similarly, other frameworks have integrated differentiable fluid dynamics simulators directly inside graph networks, combining coarse-resolution CFD simulations with learned corrections to generalize well to novel scenarios that pure data-driven models might fail to capture [427]. These hybrid strategies are particularly effective in addressing the limitations of purely data-driven models, such as error accumulation over long time horizons and poor extrapolation to unseen geometries, echoing the need for rigorous physical constraints seen in dynamic molecular modeling.

The versatility of graph-based methods is also evident in their application to diverse fluid phenomena, including turbulent flows and multi-phase systems. Multi-scale graph neural networks have been developed to infer unsteady continuum mechanics across varying length scales, demonstrating good extrapolation to new domain geometries for atmospheric and oceanic processes [358]. Furthermore, specialized architectures like GraphSplineNets utilize differentiable orthogonal spline collocation methods to forecast dynamical systems governed by the Navier-Stokes equations, improving the accuracy-speedup trade-off in both regular and irregular domains [428]. For problems involving moving boundaries, such as flow past plunging foils, immersed boundary-aware physics-informed neural networks (PINNs) have been formulated to reconstruct velocity and pressure fields without the need for body-attached reference frames, although challenges remain in balancing data efficiency with physical constraint satisfaction [429].

Addressing the specific challenges of mesh-based simulations, researchers have scaled MeshGraphNets to handle meshes with millions of nodes through domain decomposition strategies, enabling the generation of CFD simulations on 3D meshes that were previously intractable for standard GNNs due to hardware limitations [126]. Additionally, physics-informed formulations have been extended to graph networks to solve PDEs on arbitrary meshes without labeled data, utilizing graph exterior calculus to construct differential operators and enforce boundary conditions strictly [121]. These advancements underscore a paradigm shift where differentiable graph networks are not merely replacing traditional solvers but are being intricately woven into the fabric of scientific simulation to create robust, scalable, and physically consistent tools for discovery. As these models mature, they lay the computational groundwork for the next evolution in scientific AI: the integration of such differentiable, physics-grounded simulators with neuro-symbolic reasoning and multimodal data fusion, a frontier explored in the following section.

7.6 Neuro-Symbolic and Multimodal Integration in Scientific Applications

7.6 Neuro-Symbolic and Multimodal Integration in Scientific Applications

Building upon the robust, physics-grounded simulators established in the previous section, the frontier of scientific discovery is now expanding toward the convergence of sub-symbolic graph learning, symbolic reasoning, and multimodal data integration. While differentiable graph networks excel at approximating continuous physical dynamics, traditional deep learning approaches often struggle with the rigorous logical consistency, explicit interpretability, and data efficiency required for high-stakes scientific inference. To address these limitations, emerging applications are shifting toward neuro-symbolic frameworks that fuse the flexibility of neural networks with the structured reasoning capabilities of symbolic systems. By leveraging heterogeneous data sources ranging from textual literature to complex simulation outputs, this integration is proving transformative in automated hypothesis generation, materials discovery, and the interpretation of intricate biological pathways.

A primary application of this paradigm is the automation of hypothesis generation through the synthesis of vast scientific corpora into structured knowledge representations. Recent advancements have demonstrated the efficacy of transforming unstructured scientific literature into ontological knowledge graphs, enabling generative artificial intelligence to perform sophisticated graph reasoning. By leveraging transitive and isomorphic properties within these graphs, researchers can uncover unprecedented interdisciplinary relationships that link dissimilar concepts, such as drawing structural parallels between biological materials and artistic compositions to propose novel material designs [430]. These systems utilize deep node embeddings and path sampling strategies to identify gaps in current knowledge and predict material behaviors that conventional approaches might miss. Furthermore, to handle the temporal dynamics inherent in scientific evolution, variational inference models have been developed to capture the changing relationships between biomedical terms over time, formulating hypothesis generation as a future connectivity prediction task on dynamic attributed graphs [431].

In the realm of biology and medicine, the integration of multimodal data is critical for deciphering complex systems where information resides at varying scales and resolutions. The biological domain is characterized by rich heterogeneity, involving entities such as genes, proteins, and diseases, connected by diverse relational semantics. To effectively mine these complex networks, models like edge2vec have been proposed to incorporate edge semantics into representation learning, significantly outperforming homogeneous graph methods in tasks such as compound-gene bioactivity prediction and drug target identification [432]. Moreover, the challenge of integrating disparate data modalities—such as histology images, genomic sequences, and clinical records—is being addressed through heterogeneous graph neural networks. For instance, the Pathology-Genome Heterogeneous Graph (PGHG) framework successfully integrates whole slide images with bulk RNA-Seq data, utilizing biological prior knowledge to guide feature extraction and attention-based fusion for improved cancer survival analysis [433]. Similarly, multimodal similarity learning graph neural networks have been employed to create unified gene representations from single-cell sequencing and spatial transcriptomic data, capturing functional similarities across different tissues and species [434].

Beyond mere data fusion, the incorporation of symbolic reasoning into graph learning enhances the interpretability and logical rigor of scientific models. Neuro-symbolic approaches are increasingly utilized to bridge the gap between data-driven embeddings and explicit logical constraints. In biological knowledge graphs, methods combining symbolic logic with neural networks generate node embeddings that encode both explicit and implicit information, facilitating tasks like protein-protein interaction prediction and disease gene identification with performance matching or exceeding traditional feature-engineered approaches [435]. These frameworks often employ rule learning modules alongside knowledge graph embedding modules in a closed-loop fashion, where symbolic rules enhance semantic associations and neural embeddings refine rule quality, thereby addressing the incompleteness of biological knowledge bases [436]. Additionally, policy-guided walks based on reinforcement learning have been integrated with logical rules to improve link prediction in drug repurposing tasks, offering a transparent reasoning path that purely neural models cannot provide [437].

The synergy between Large Language Models (LLMs) and graph structures further amplifies these capabilities, creating hybrid systems capable of complex reasoning over scientific data. LLMs are being empowered with graph reasoning abilities through prompt-augmented frameworks that allow them to utilize external graph tools for multi-step logic and precise calculations, essential for analyzing protein molecules and bibliographic networks [438]. Conversely, graph structures are used to ground LLMs, mitigating hallucinations and enhancing their ability to answer complex questions that require reasoning over both situational context and structured world knowledge [439]. Emerging agents, such as the Graph Agent, leverage LLMs alongside inductive-deductive reasoning modules to convert graph structures into textual data, enabling free-of-training adaptation to various graph reasoning tasks with human-interpretable explanations [440]. Furthermore, the concept of "Graph-of-Thought" reasoning models human thought processes as non-linear graphs rather than sequential chains, significantly boosting accuracy in multimodal scientific reasoning tasks [441].

Finally, the integration of physics-informed constraints with these advanced neuro-symbolic and multimodal architectures represents a crucial step toward trustworthy scientific simulation. Differentiable graph network simulators have been developed to accelerate particle and fluid simulations by learning local interaction laws while respecting conservation principles, enabling the solution of inverse problems through automatic differentiation [125]. These physics-informed graph networks (PIGNs) utilize graph exterior calculus to construct differential operators on unstructured data, offering distinct advantages over standard neural networks for complex multiscale systems [120]. By combining these physical priors with neuro-symbolic reasoning, the scientific community is moving toward autonomous discovery systems that are not only accurate but also physically consistent, interpretable, and capable of generating novel, testable hypotheses across diverse scientific frontiers.

8 Open Challenges, Benchmarking Standards, and Future Roadmaps

8.1 Unified Benchmarks and Standardized Evaluation Metrics for Scientific Graphs

8.1 Unified Benchmarks and Standardized Evaluation Metrics for Scientific Graphs

The rapid advancement of Graph Neural Networks (GNNs) and Graph Foundation Models (GFMs) in scientific discovery has exposed a critical disconnect: while model architectures are becoming increasingly sophisticated, the evaluation frameworks used to validate them remain rooted in simplified, static assumptions. Early successes often relied on homogeneous graph datasets that fail to capture the dynamic, heterogeneous, and multimodal nature of real-world physical and biological systems. This reliance on inadequate benchmarks risks producing models that excel in controlled environments but falter when deployed in the complex, shifting contexts of actual scientific inquiry. To ensure that GFMs can truly serve as reliable partners in discovery, the community must coalesce around unified benchmarks that transcend predictive accuracy, incorporating rigorous standards for physical consistency, causal correctness, and robustness against distributional shifts. This foundational step is a prerequisite for the advanced neuro-symbolic and causal reasoning architectures discussed in the following section, as these hybrid models require evaluation protocols capable of measuring logical rigor and adherence to physical laws.

A primary deficiency in current benchmarking practices is the dominance of static and homogeneous data structures, which cannot represent the temporal evolution and structural diversity inherent in scientific domains. Real-world systems, ranging from epidemic spread to academic collaboration networks, are fundamentally dynamic and heterogeneous. Recent efforts have begun to address this by introducing datasets that incorporate time-evolving nodes and edges, such as the Temporal Graph Benchmark (TGB), which spans diverse domains including social, trade, and transportation networks to provide realistic evaluation protocols for temporal tasks [442]. Similarly, the EXPERT initiative highlights the necessity of dynamic heterogeneous academic graphs, demonstrating that models trained on static benchmarks struggle with multi-step forecasting tasks where both node attributes and edge topologies change over time [443]. Furthermore, the generation of synthetic dynamic environments, as seen in the Dynamic Dataset Generator (DDG), offers a controlled mechanism to simulate heterogeneous local and global changes, allowing researchers to systematically evaluate clustering and learning algorithms under varying degrees of temporal severity [444]. These initiatives underscore that future benchmarks must mandate the inclusion of dynamic heterogeneity to ensure models can learn from the fluid nature of scientific data.

Beyond structural dynamics, scientific benchmarks must evolve to handle multimodal data fusion and complex relational structures. Scientific discovery rarely relies on a single data source; instead, it integrates images, text, sensor readings, and graph structures. The lack of standardized solutions for integrating such multimodal data forces practitioners into trial-and-error processes, hindering the development of generalizable models [445]. Emerging frameworks like "Multimodal learning with graphs" propose blueprints for combining diverse modalities while leveraging cross-modal dependencies, yet comprehensive benchmarks that test these capabilities across image-intensive, knowledge-grounded, and language-intensive scenarios remain scarce [446]. In healthcare, for instance, effective phenotyping requires the simultaneous analysis of chest radiographs and physiological time-series data, where dynamic training approaches have shown superior performance over single-modality baselines [447]. A unified benchmark must therefore provide standardized pipelines for ingestible multimodal inputs, ensuring that models are evaluated on their ability to synthesize information across disparate data types rather than optimizing for a single modality in isolation.

Perhaps most critically, the metrics used to evaluate scientific graph models must expand beyond traditional accuracy measures to include physical consistency, causal validity, and out-of-distribution (OOD) generalization. In scientific domains, a model that predicts accurately but violates conservation laws or causal mechanisms is scientifically useless. The concept of "context shift" describes semantically meaningful changes in the data generation process that static benchmarks often miss, leading to models that are brittle in production environments [448]. To combat this, new evaluation paradigms must assess a model's ability to maintain performance under distribution shifts. Benchmarks like Wild-Tab have begun to rigorously test OOD generalization in tabular regression, revealing that many sophisticated methods struggle to outperform simple Empirical Risk Minimization when faced with real-world distribution shifts [449]. Similarly, the DrugOOD dataset challenges models to generate hierarchical semantic environments to handle high heterogeneity in drug discovery tasks, emphasizing the need for metrics that capture invariant learning across diverse scaffolds and sizes [450].

Furthermore, evaluating causal correctness is paramount for scientific discovery, where distinguishing correlation from causation drives hypothesis generation. Tools like SpaCE provide the first realistic benchmark datasets specifically designed to evaluate causal inference methods in the presence of spatial confounding, offering true counterfactuals and confounding scores that are essential for validating causal models [451]. Without such standardized causal benchmarks, there is a risk of propagating spurious correlations that fail to generalize to new experimental conditions. Additionally, metrics must account for uncertainty and reliability. Frameworks like Data-SUITE and Data-IQ emphasize the importance of identifying incongruous regions within the training distribution and stratifying data based on aleatoric uncertainty, providing a more nuanced view of model reliability than aggregate accuracy scores [452; 453]. The “Veridical Data Science” framework further argues for stability as a core principle, suggesting that evaluation metrics must quantify how human judgment calls and data perturbations impact results throughout the lifecycle [454].

Finally, the standardization of metrics themselves is a pressing challenge. The “Metrics Maze” in other fields illustrates the confusion caused by a lack of consensus on score magnitudes and their human interpretability [455]. In scientific graphs, we need metrics that quantify not just error rates but also adherence to physical laws, such as energy conservation in molecular dynamics or mass balance in fluid simulations. The XFlow benchmark suite represents a step in this direction by offering a cohesive platform to assess flow behaviors across diverse domains, addressing the compartmentalization of network research [456]. Moving forward, the community must adopt a holistic evaluation strategy that integrates dynamic structural complexity, multimodal integration, causal robustness, and physical fidelity. Only through such comprehensive and standardized benchmarks can we ensure that Graph Foundation Models truly accelerate scientific discovery rather than merely fitting noise in static snapshots of complex systems, thereby laying the empirical groundwork for the next generation of neuro-symbolic and causally aware scientific AI.

8.2 Bridging the Gap: Neuro-Symbolic Integration and Causal Reasoning

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While Section 8.1 established the critical need for unified benchmarks that evaluate physical consistency, causal validity, and robustness, meeting these rigorous standards requires a fundamental shift in model architecture. Purely data-driven, sub-symbolic approaches, despite their proficiency in pattern recognition within high-dimensional scientific data, often function as “black boxes” that lack the logical rigor and adherence to physical laws demanded by the evaluation metrics proposed previously. To bridge this gap between advanced evaluation protocols and model capability, the field must transition toward hybrid architectures that seamlessly integrate symbolic reasoning and causal inference with the flexible representation learning of Graph Neural Networks (GNNs) and Graph Foundation Models (GFMs). This neuro-symbolic integration is not merely an additive process but a foundational rethinking of how models learn, reason, and generalize in domains governed by strict physical and logical constraints, thereby providing the necessary infrastructure for the dynamic and multimodal systems discussed in the subsequent section.

A primary imperative in this fusion is the direct incorporation of hard physical constraints and domain knowledge into the learning framework to satisfy the physical consistency metrics highlighted in Section 8.1. Purely data-driven models are prone to generating predictions that violate fundamental conservation laws or physical mechanisms, a flaw that is unacceptable in high-stakes scientific applications where benchmark scores must reflect adherence to reality. To address this, emerging frameworks advocate for theory-guided approaches where symbolic knowledge acts as a scaffold for neural computation. For instance, the concept of Theory-Guided Hard Constraint Projection (HCP) illustrates how physical constraints, such as governing equations, can be discretized and enforced through projection operations, ensuring that model predictions strictly conform to physical mechanisms even in data-scarce regimes [457]. Similarly, libraries like DomiKnowS have been developed to facilitate the seamless integration of symbolic domain knowledge into deep learning architectures, allowing logical constraints over outputs or latent variables to be added generically, thereby improving both performance and explainability [458]. By embedding symbolic rules—whether derived from first principles or expert ontologies—into the loss function or architecture, models can achieve higher data efficiency and robustness against noisy observations, effectively preventing the prediction of invalid physical configurations [459].

Beyond enforcing static constraints to meet benchmarks for physical fidelity, the next frontier involves learning structural causal models to distinguish genuine causal mechanisms from spurious correlations, directly addressing the need for causal validity metrics. Scientific discovery demands an understanding of "why" phenomena occur, not just "what" patterns exist. Heterogeneous graphs, which are prevalent in scientific domains ranging from genomics to materials science, often suffer from limited generalizability because standard GNNs do not align with human causal reasoning. The development of frameworks like HG-SCM (Heterogeneous Graph as Structural Causal Model) represents a significant step forward, mimicking human perception by constructing intelligible variables and automatically learning task-level causal relationships [460]. This approach enhances out-of-distribution generalization, a critical capability for scientific models that must operate under varying conditions and pass the rigorous OOD tests described in the previous section. Furthermore, recent advancements propose methods for causal discovery that allow for a two-way interaction between humans and machines, where neural networks expose learnt causal graphs that experts can contest and modify before re-injection, ensuring the final model adheres to verified expert knowledge [461]. Such causal-aware architectures are essential for moving from associative learning to true scientific reasoning, enabling models to infer governing differential equations and interaction laws directly from observational data [462].

The development of hybrid systems that offer both the flexibility of neural networks and the interpretability of symbolic logic is also reshaping the landscape of relational reasoning on graphs, providing the logical backbone required for the complex data structures analyzed in Section 8.3. Traditional neuro-symbolic approaches often struggle with the computational complexity of interfacing sub-symbolic perception with symbolic reasoning. However, novel frameworks are emerging to overcome these bottlenecks. For example, Logical Neural Networks (LNNs) provide an end-to-end differentiable architecture where every neuron corresponds to a component of a logical formula, enabling omnidirectional inference and resilience to inconsistent knowledge [463]. In the context of graph completion and link prediction, models like Graph Collaborative Reasoning (GCR) translate graph structures into logical expressions, converting relational tasks into neural logic reasoning problems that leverage neighbor information more effectively than associative baselines [464]. Additionally, approaches that combine rule learning with knowledge graph embeddings in a closed-loop fashion, such as EngineKG, demonstrate how symbolic rules can enhance semantic associations while neural embeddings refine rule quality, achieving superior performance in inference tasks [436].

Moreover, the extraction of symbolic laws from trained neural networks offers a pathway to discovering new scientific principles, transforming GNNs from mere predictors into tools for hypothesis generation that can be validated against the standardized benchmarks advocated earlier. Techniques involving symbolic regression applied to the message functions of GNNs have successfully recovered known physical laws, such as Newton's law of gravitation, and even discovered new analytic formulas for complex systems like dark matter distributions [462] [145]. The integration of retrieval-based causal learning with Graph Information Bottleneck theory further advances this by compressing explanatory subgraphs to improve both prediction accuracy and interpretability, ensuring that the learned explanations are causally relevant rather than coincidental [465].

Ultimately, the roadmap to autonomous scientific discovery requires a paradigm shift where neuro-symbolic integration becomes the standard rather than the exception, serving as the logical bridge between the rigorous evaluation standards of Section 8.1 and the dynamic, multimodal architectural challenges of Section 8.3. By combining the data-driven adaptability of deep learning with the logical rigor of symbolic AI, future Graph Foundation Models will be capable of not only predicting outcomes but also explaining them, adhering to physical laws, and reasoning about counterfactuals. This synthesis is vital for building AI scientists that can operate with the trustworthiness and logical consistency demanded by the scientific community, ensuring that AI-driven discoveries are both innovative and fundamentally sound, while providing the stable reasoning core necessary to navigate the messy, evolving reality of dynamic scientific systems.

8.3 Handling Dynamic, Heterogeneous, and Multimodal Scientific Systems

8.3 Handling Dynamic, Heterogeneous, and Multimodal Scientific Systems

While the integration of neuro-symbolic reasoning and causal inference detailed in the previous section provides the logical backbone for trustworthy AI, the practical realization of these capabilities depends on architectures capable of ingesting the messy, evolving reality of scientific data. Real-world scientific systems—from climate modeling and protein folding to nuclear nonproliferation analysis—are inherently dynamic, heterogeneous, and multimodal, standing in stark contrast to the static, homogeneous graph structures often assumed in traditional benchmarks. Consequently, bridging the gap between rigorous causal reasoning and autonomous discovery requires models that can not only reason logically but also adapt to continuous temporal evolution and fuse disparate data sources. A primary challenge in this domain is capturing long-range temporal dependencies while simultaneously adapting to structural changes in the underlying graph. Recent advancements have demonstrated that re-conceptualizing dynamic graphs as sequence modeling problems allows for the effective application of Transformer architectures, which excel at handling long-term dependencies often missed by recurrent or discrete-time models [466]. By introducing novel temporal alignment techniques, these simplified Transformer models can capture inherent evolution patterns without the need for overly complex module designs, offering a robust pathway for modeling systems where interaction topologies shift unpredictably.

Beyond temporal dynamics, scientific data is rarely complete or uniformly distributed, posing significant hurdles for model generalization and the reliable application of causal rules. The issue of missing data is pervasive in spatiotemporal scientific tasks, whether due to sensor failures in environmental monitoring or incomplete observations in biological assays. Addressing this requires moving beyond standard imputation techniques to models that balance strong inductive biases with high expressivity, ensuring that the "scaffold" for symbolic reasoning remains intact even in data-scarce regimes. For instance, low rankness-induced Transformers have been proposed to leverage the inherent structural priors of spatiotemporal data, achieving a balance that allows for generalizable imputation across heterogeneous datasets ranging from air quality monitoring to solar energy prediction [467]. Similarly, advanced variational autoencoders equipped with position-aware graph embeddings have shown efficacy in simultaneously imputing missing node features and reconstructing evolving graph structures, a capability critical for networked time series where the underlying connectivity is not fixed [468]. These approaches highlight the necessity of integrating uncertainty quantification and adaptive mechanisms to handle the sparse and noisy nature of scientific observations, thereby providing a stable foundation for the causal discovery mechanisms discussed previously.

The heterogeneity of scientific systems further demands sophisticated strategies for integrating multimodal data to support the comprehensive world models required for autonomy. Scientific inquiry often involves fusing information from text (literature, experimental logs), images (microscopy, satellite imagery), simulations (physics engines), and structured graphs (molecular structures, social networks). A unified framework must be able to merge complex intra- and inter-relations across these modalities without losing valuable information, effectively creating the rich representational space needed for hypothesis generation. Hypergraph-based models have emerged as a powerful solution for this, enabling the representation of high-order relations among items from different modalities and facilitating scalable, distributed learning across multiple GPUs [469]. In the context of biomedical research, such as oncology, the integration of radiology, pathology, genomics, and clinical records into a cohesive graph representation is essential for accurate diagnosis and treatment planning. Deep neural networks, particularly Graph Neural Networks (GNNs) and Transformers, are increasingly being employed to extract and fuse these diverse data types, overcoming the limitations of traditional unimodal methods [470].

Moreover, the dynamic nature of multimodal interactions requires models that can adapt to changing data distributions and modalities over time, a prerequisite for the self-improving agents envisioned in autonomous laboratories. In fields like single-cell biology, static interaction graphs are often suboptimal because they fail to incorporate downstream task information or capture the fluid interactions between different omics layers. Frameworks like scMoFormer demonstrate the potential of leveraging Transformers in an end-to-end manner to model both intra- and cross-modality interactions, significantly outperforming static GNN approaches [471]. Similarly, in urban computing and smart city applications, Transformer-based systems have been designed with interchangeable input embeddings and output heads to handle virtually any combination of IoT data sources, supporting multi-task learning across diverse domains such as traffic forecasting and visual disease classification [472]. This flexibility is crucial for transitioning from passive analysis to active experimentation, where the AI must interpret new sensory inputs in real-time.

Looking forward, the integration of automated attribute completion and dynamic structure learning will be paramount to enabling the closed-loop systems described in the subsequent section. Methods that employ automated spatiotemporal augmentation and contrastive learning can adapt to heterogeneous region graphs generated from multi-view data sources, improving robustness against noise and distribution shifts [473]. Additionally, the ability to anticipate capability evolution in research communities using dynamic heterogeneous graph representations underscores the importance of forecasting not just node states but also the emergence of new collaboration patterns and technical expertise [474]. As scientific datasets grow in complexity, future architectures must prioritize flexibility, allowing for the seamless addition of new modalities and the continuous adaptation to evolving system dynamics. This entails a shift towards models that can learn from streaming tensor decompositions and handle multi-aspect graphs without requiring full re-computation, ensuring scalability in the face of ever-expanding scientific data [475]. Ultimately, mastering these dynamic, heterogeneous, and multimodal challenges provides the essential perceptual and representational infrastructure upon which the autonomous, self-driving laboratories of the future will be built.

8.4 The Roadmap to Autonomous Scientific Discovery and Closed-Loop Systems

8.4 The Roadmap to Autonomous Scientific Discovery and Closed-Loop Systems

Building upon the robust perceptual and representational infrastructure for dynamic, heterogeneous, and multimodal systems established in the previous section, the field now faces its ultimate frontier: the transition from AI-assisted analysis to fully autonomous "AI Scientists." This roadmap envisions a paradigm shift where intelligent systems do not merely augment human researchers but operate as independent agents capable of formulating hypotheses, designing experiments, executing physical protocols, and refining theoretical models in a continuous, closed-loop fashion. By leveraging the adaptive graph architectures discussed previously, these agents aim to achieve Nobel-tier scientific breakthroughs with minimal human intervention, effectively closing the gap between data ingestion and knowledge generation.

The foundational layer of this roadmap involves the evolution of AI from passive pattern recognizers to active hypothesis generators. While current large language models have demonstrated the capacity to structure vast scientific knowledge and propose testable hypotheses, they often suffer from high error rates that necessitate human oversight [476]. The next generation of autonomous systems must transcend these limitations by integrating the rigorous logical constraints and domain-specific ontologies enabled by neuro-symbolic reasoning. This ensures that generated hypotheses are not only novel but also physically plausible and experimentally viable. Consequently, the focus must shift from static knowledge retrieval to dynamic reasoning systems that can simulate the "Observation–Hypothesis–Prediction–Experimentation" loop historically practiced by human scientists, utilizing explainable AI to derive interpretable insights from the complex, multimodal data streams handled by the architectures in Section 8.3 [477].

Central to the realization of autonomous discovery is the deployment of closed-loop systems that tightly couple computational inference with physical experimentation, directly addressing the challenge of adapting to changing data distributions highlighted earlier. In domains such as materials science and chemistry, autonomous experimentation (AE) platforms are already beginning to map complex phase diagrams and synthesize metastable materials by iteratively selecting experiments based on active learning strategies [478]. These systems utilize hierarchical AI agents, such as the Scientific Autonomous Reasoning Agent (SARA), which coordinate robotic synthesis and characterization to explore vast parameter spaces orders of magnitude faster than traditional methods. The roadmap dictates that future systems must generalize this capability across diverse scientific disciplines, integrating the multi-modal sensing and real-time decision-making techniques discussed in Section 8.3 to handle the stochasticity and latency inherent in physical experiments [479].

A critical component of this vision is the development of "self-driving laboratories" or "science factories" that operate with a high degree of modularity and scalability, relying on the flexible graph representations required for heterogeneous systems. These facilities will rely on reconfigurable robotic workcells linked by distributed computing infrastructures, enabling the concurrent execution of thousands of distinct experimental workflows [480]. To manage the complexity of such operations, future architectures must incorporate robust regulatory frameworks where high-level policies are represented as structured graph data, allowing for automated safety validation before robotic execution [481]. This ensures that while the generative components of the AI explore novel scientific frontiers, the deterministic control layers maintain strict adherence to safety and ethical standards, thereby mitigating the risks of uncontrolled autonomy.

Furthermore, the roadmap emphasizes the necessity of "self-improving" capabilities, where AI agents learn from their own interactions with the environment to refine their world models and experimental strategies autonomously. Inspired by developmental robotics, these agents should possess intrinsic motivations and curiosity-driven exploration mechanisms that allow them to identify and investigate anomalies without explicit human instruction [482]. By adopting frameworks such as Self-Initiated Open World Learning (SOL), these systems can detect novelties, characterize unknown phenomena, and incrementally update their internal models, effectively functioning as lifelong learners in open-ended scientific environments [483]. This contrasts sharply with current deep learning systems that require offline retraining and human-defined objective functions for every new task, a limitation that the dynamic graph models of the previous section aim to overcome

[484].

Achieving Level 5 autonomy, defined as the ability to produce scientific knowledge with no human intervention, remains a grand challenge akin to the development of fully autonomous vehicles [485]. Reaching this level requires overcoming significant hurdles in causal reasoning, uncertainty quantification, and the integration of symbolic and sub-symbolic AI, building directly on the foundations laid in Section 8.2 and the data handling capabilities of Section 8.3. Future systems must be capable of distinguishing between spurious correlations and causal mechanisms, ensuring that discovered laws are robust and generalizable. The vision includes the creation of "AI Research Associates" that can navigate the space of parsimonious hypotheses, compiling them into trainable computation graphs to learn phenomenological relations from sparse and noisy data [486].

Ultimately, the successful deployment of autonomous scientific discovery systems will rely on a synergistic ecosystem where human scientists act as shepherds rather than operators, guiding high-level goals while AI agents manage the intricate details of discovery. This collaboration promises to democratize science, accelerate the resolution of global challenges such as climate change and pandemics, and potentially unlock fundamental truths about the universe that have remained beyond human reach [487]. However, as we advance along this roadmap toward fully autonomous agents, it is imperative to recognize that the realization of Level 5 autonomy is critically contingent upon overcoming the current "epistemic monoculture" in model development. The risks of outcome homogenization and the neglect of diverse scientific perspectives, if left unaddressed, could undermine the very innovation these systems seek to automate. Therefore, the transition to autonomy must be paralleled by a concerted effort to diversify models and data-centric approaches, a crucial pivot detailed in the following section to ensure the resilience and ethical integrity of future scientific AI.

8.5 Overcoming the Epistemic Monoculture: Diversity in Models and Data-Centric Approaches

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While Section 8.4 envisions a future of fully autonomous "AI Scientists" operating in closed-loop systems, the realization of such Level 5 autonomy is critically contingent upon overcoming the current "epistemic monoculture" that pervades Graph Foundation Models (GFMs). The rapid ascent of deep learning has been characterized by an overwhelming reliance on scaling laws, where progress is equated with increasing model parameters and dataset sizes. While this trajectory has unlocked unprecedented capabilities, it has simultaneously fostered a research environment that converges on a narrow set of benchmarks, architectures, and evaluation metrics, thereby disincentivizing exploration beyond the dominant scaling paradigm. Historical analysis reveals that this shift began when AI research moved from autonomous, theory-driven exploration to a benchmark-centric model focused exclusively on predictive accuracy, a transition that streamlined progress but entrenched conservatism and hindered the development of explainable and efficient systems [488]. In the context of scientific discovery, where physical consistency, causal correctness, and data scarcity are paramount, adhering strictly to scaling-only paradigms poses significant risks, including outcome homogenization and the neglect of critical environmental and social dimensions. Without addressing these systemic fragilities, the autonomous agents described in the previous section risk propagating narrow, biased, or environmentally unsustainable discovery pathways.

To overcome this monoculture and build the robust foundation required for autonomous science, the field must pivot toward a diversified research agenda that prioritizes data-centric graph learning over mere model expansion. The current obsession with state-of-the-art accuracy on static benchmarks often obscures the fundamental disparities between artificial and natural intelligence, particularly regarding the ability to learn implicitly from contextual environments and generalize beyond training distributions [489]. A data-centric approach emphasizes the quality, diversity, and representativeness of scientific graph data rather than the sheer volume of parameters. This involves rigorous scrutiny of data collection practices to ensure reliability and fairness, as biased or non-representative datasets can lead to unfair outcomes and reinforce systemic inequalities in scientific applications [490]. Furthermore, the component-sharing hypothesis suggests that when multiple decision-makers or scientific domains rely on the same foundational models or training data, it exacerbates outcome homogenization, potentially institutionalizing exclusion and limiting the discovery of novel scientific phenomena [491]. Therefore, fostering diversity in data sources and curation strategies is essential to prevent the convergence of scientific insights into a single, potentially flawed perspective, ensuring that autonomous systems have access to a rich, multi-faceted view of the scientific landscape.

Parallel to data-centricity, the integration of green computing principles is imperative to ensure the environmental sustainability of graph-based scientific AI, a prerequisite for the scalable "science factories" envisioned in the roadmap to autonomy. The super-linear growth in computational demands for training large-scale models has resulted in a substantial carbon footprint, raising concerns about the ecological viability of current AI trajectories [492]. The prevailing focus on accuracy often comes at the expense of energy efficiency, creating a rebound effect where efficiency gains are consumed by increased model complexity rather than reducing overall emissions [493]. To counter this, the community must adopt "Green AI" methodologies that treat energy consumption and carbon footprint as first-class metrics alongside accuracy. This includes exploring algorithmic innovations such as tensor networks, which can significantly reduce computational requirements without compromising performance, thereby enabling more sustainable AI development [494]. Additionally, the concept of "Green Federated Learning" offers a pathway to minimize carbon emissions by optimizing training parameters and leveraging decentralized computational resources, although careful design is required to balance efficiency with performance [495].

Beyond data and sustainability, breaking the epistemic monoculture requires the active exploration of non-standard architectures that deviate from the dominant transformer and message-passing paradigms. The current landscape is dominated by models that prioritize scaling, often ignoring alternative computational substrates that may offer superior efficiency or interpretability for specific scientific tasks—capabilities that are crucial for the trust and accountability discussed in the following section. For instance, nature-inspired approaches such as Spiking Neural Networks (SNNs) mirror the energy-efficient processing of the mammalian brain and present a viable alternative to traditional Artificial Neural Networks, yet they remain underutilized due to hardware gaps and a lack of comparative research [496]. Similarly, the pursuit of "Green Learning" paradigms characterized by small model sizes, low computational complexity, and logical transparency offers a compelling alternative to opaque, energy-intensive deep learning models [497]. By diversifying the architectural toolkit to include these non-standard approaches, the scientific community can develop models that are not only more efficient but also better suited to the constraints of edge devices and the specific inductive biases of physical systems.

Moreover, addressing the epistemic monoculture necessitates a broader engagement with social and ethical dimensions, recognizing that technical choices are inherently sociotechnical. The lack of diversity within the computing community itself contributes to the narrowness of problem formulations and the types of biases encountered in datasets, underscoring the need for inclusive teams to drive innovation [498]. A diversified research agenda must also consider the societal impacts of AI, moving beyond technical metrics to evaluate fairness, accountability, and the potential for reinforcing structural inequalities [499]. By integrating insights from social sciences and humanities, researchers can develop frameworks that promote equitable access to AI tools and ensure that scientific discoveries benefit a broad spectrum of society [500]. This sociotechnical awareness serves as a critical bridge to the challenges of trust and human-centric interpretability, ensuring that the drive for diversity directly supports the ethical deployment of AI in high-stakes domains.

Ultimately, overcoming the epistemic monoculture in Graph Neural Networks and Graph Foundation Models requires a concerted shift away from the singular pursuit of scaling. It demands a holistic strategy that values data quality, environmental stewardship, architectural diversity, and social responsibility. By embracing these diverse pillars, the field can foster a more robust, equitable, and sustainable ecosystem for scientific discovery, one that is resilient to the limitations of current paradigms and capable of addressing the complex, multifaceted challenges of the future. This transition is not merely a technical adjustment but a fundamental reorientation of scientific values, ensuring that AI serves as a versatile and responsible partner in the quest for knowledge [501]. Only by diversifying the underlying models and data can we ensure that the autonomous systems of tomorrow are capable of genuine innovation rather than mere optimization of existing biases, thereby laying the necessary groundwork for the trust and accountability mechanisms detailed in the subsequent section.

8.6 Trust, Accountability, and Human-Centric Interpretability in High-Stakes Domains

8.6 Trust, Accountability, and Human-Centric Interpretability in High-Stakes Domains

Building upon the diversified ecosystem advocated in Section 8.5, the deployment of Graph Foundation Models (GFMs) in scientific discovery necessitates a rigorous framework for trust, accountability, and human-centric interpretability. While overcoming epistemic monoculture through architectural and data diversity lays the groundwork for robust systems, the transition of these models from experimental tools to integral components of high-stakes workflows—such as drug design, materials science, and climate modeling—introduces profound societal and ethical imperatives. The inherent opacity of deep graph learning architectures poses significant risks; without rigorous validation protocols and explainable frameworks grounded in the diverse paradigms previously discussed, the adoption of GFMs could propagate errors, reinforce biases, and erode human agency in critical decision-making processes. Therefore, the future roadmap for scientific AI must prioritize systems that are not only accurate but also transparent, accountable, and aligned with human values, ensuring that the move away from scaling-only paradigms directly supports ethical deployment.

A central challenge in establishing trust within this diversified landscape is bridging the gap between technical performance metrics and the actual utility of explanations for human stakeholders. While post-hoc interpretability methods have gained traction, they often fail to provide justifications grounded in the reality of data distributions or underlying physical laws, a deficiency exacerbated when models lack causal grounding. For instance, counterfactual explanations, which suggest minimal changes to input graphs to alter predictions, are frequently generated without ensuring connection to ground-truth data, leading to "unjustified counterfactual explanations" that may mislead scientists [211]. To mitigate this, future frameworks must integrate causal reasoning to ensure that explanations reflect genuine mechanistic relationships rather than spurious correlations. This aligns with the broader push for combining sub-symbolic graph learning with symbolic methods to incorporate domain knowledge and causality, thereby producing explanations that are faithful to the model's internal decision-making process and understandable to domain experts [217]. Such integration is particularly vital when leveraging the non-standard, logic-transparent architectures highlighted in the previous section to enhance interpretability by design.

Furthermore, the concept of accountability in GFMs extends beyond mere explainability to encompass the entire lifecycle of the model, from data curation to deployment, reflecting the data-centric principles essential for breaking monocultures. The "Foundation Model Transparency Index" highlights a critical deficit in the current ecosystem: major developers often fail to disclose downstream impacts, usage policies, or mechanisms for redress, creating an environment where accountability is obscured by opacity [301]. In scientific contexts, where models may influence public health or environmental policy, this lack of transparency is unacceptable. We must advocate for the creation of "Ecosystem Graphs" that document the dependencies between datasets, models, and applications, providing a transparent map of the sociotechnical landscape surrounding scientific GFMs [502]. Such documentation is essential for auditing models and ensuring that they adhere to ethical standards and regulatory requirements, effectively operationalizing the social responsibility dimensions discussed earlier.

Human-centric interpretability also requires addressing the dynamic nature of scientific knowledge and the potential for "unfortunate counterfactual events," where recommendations based on model explanations become obsolete or harmful due to model retraining or shifting environmental conditions [503]. In high-stakes scientific domains, providing actionable recourse is insufficient if that recourse does not account for temporal dependencies and the evolving state of the system. Consequently, explanation frameworks must evolve to include temporal reasoning and continuous validation, ensuring that the guidance provided to scientists remains valid and safe over time. This necessitates a shift from static, one-off explanations to dynamic, interactive systems that allow users to explore the "explanatory multiverse" of possible counterfactual paths, granting them agency in selecting the most appropriate course of action [504].

The tension between competing ethical pillars—such as fairness, privacy, and transparency—further complicates the deployment of trustworthy GFMs, echoing the trade-offs inherent in balancing accuracy with sustainability and diversity. Attempts to optimize for one pillar in isolation can inadvertently undermine others, leading to unintended consequences in real-world applications [505]. For example, enhancing transparency through detailed model disclosures might conflict with privacy-preserving techniques required for sensitive biomedical data. Addressing these trade-offs requires a holistic, socio-ethical perspective that considers the interconnectedness of these values and their impact on vulnerable populations [506]. It also demands the development of unified evaluation metrics that assess not just predictive accuracy but also the ethical robustness and social responsibility of the model [507].

Moreover, the integration of Graph Foundation Models into autonomous scientific discovery systems raises questions about the preservation of human self-determination, a critical consideration as we envision the "AI Scientists" of the future. As AI systems become more capable of generating hypotheses and designing experiments, there is a risk that humans may become overly reliant on centralized, opaque algorithms, eroding their ability to critically evaluate and direct scientific inquiry [508]. To counter this, future research must focus on developing neuro-symbolic architectures that keep humans in the loop, leveraging knowledge graphs to provide context and semantics that empower researchers to maintain control over the discovery process. This includes creating interfaces that facilitate "algorithmic recourse" in multi-agent environments, ensuring that scientists can effectively challenge and modify model decisions when necessary [509], thus preventing the emergence of new forms of algorithmic dependency.

Finally, the path toward trustworthy GFMs in science requires a fundamental rethinking of how uncertainty is quantified and communicated, serving as a cornerstone for reliable autonomous operation. Ethical AI practice demands that models explicitly acknowledge the statistical foundations of their predictions and the limits of their knowledge, rather than presenting outputs as absolute truths [510]. By adopting frameworks that enhance the quality and meaning of uncertainty, such as conformal prediction and Bayesian belief assessment, we can build systems that are more responsive to feedback and transparent to evaluation. Ultimately, achieving trust and accountability in high-stakes scientific domains is not merely a technical challenge but a sociotechnical imperative that requires interdisciplinary collaboration, rigorous governance, and an unwavering commitment to human-centered design principles [511]. Only through such a comprehensive approach can we ensure that Graph Foundation Models serve as reliable partners in the pursuit of scientific truth, rather than sources of unverified and potentially harmful assertions, thereby securing the foundation for the next generation of scientific AI.

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[496] From Convolutions towards Spikes The Environmental Metric that the Community currently Misses

[497] Green Learning Introduction, Examples and Outlook

[498] No computation without representation Avoiding data and algorithm biases through diversity

[499] Rethinking Fairness An Interdisciplinary Survey of Critiques of Hegemonic ML Fairness Approaches

[500] Catalyzing Equity in STEM Teams Harnessing Generative AI for Inclusion and Diversity

[501] Efficiency is Not Enough A Critical Perspective of Environmentally Sustainable AI

[502] Ecosystem Graphs The Social Footprint of Foundation Models

[503] A Series of Unfortunate Counterfactual Events the Role of Time in Counterfactual Explanations

[504] Navigating Explanatory Multiverse Through Counterfactual Path Geometry

[505] Tensions Between the Proxies of Human Values in AI

[506] Are You Worthy of My Trust A Socioethical Perspective on the Impacts of Trustworthy AI Systems on the Environment and Human Society

[507] A Survey of Trustworthy Graph Learning Reliability, Explainability, and Privacy Protection

[508] Trust, Accountability, and Autonomy in Knowledge Graph-based AI for Self-determination

[509] Multi-Agent Algorithmic Recourse

[510] On the meaning of uncertainty for ethical AI philosophy and practice

[511] On the Opportunities and Risks of Foundation Models